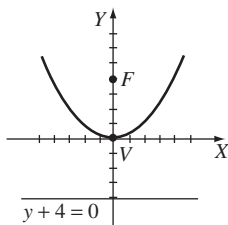
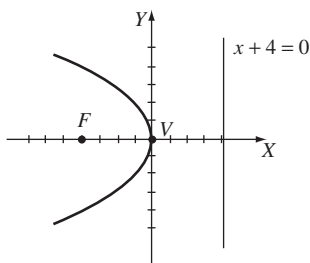


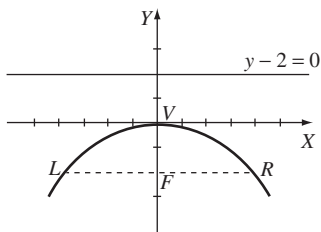
4. Foco: $F(0, 4)$, Directriz: $y + 4 = 0$, $LR = 16$,
Eje: $x = 0$



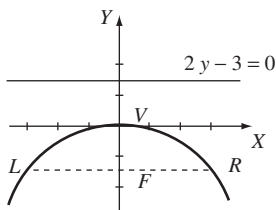
5. Foco: $F(-4, 0)$, Directriz: $x - 4 = 0$, $LR = 16$,
Eje: $y = 0$



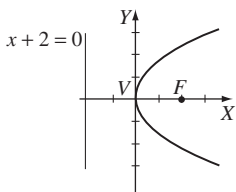
6. Foco: $F(0, -2)$, Directriz: $y - 2 = 0$, $LR = 8$,
Eje: $x = 0$



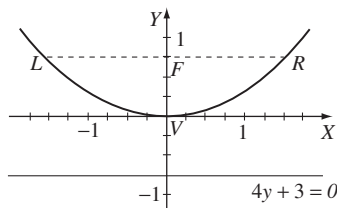
7. Foco: $F\left(0, -\frac{3}{2}\right)$, Directriz: $2y - 3 = 0$, $LR = 6$,
Eje: $x = 0$



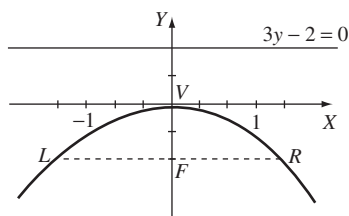
8. Foco: $F(2, 0)$, Directriz: $x + 2 = 0$, $LR = 8$,
Eje: $y = 0$



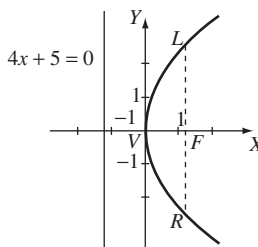
9. Foco: $F\left(0, \frac{3}{4}\right)$, Directriz: $4y + 3 = 0$, $LR = 3$,
Eje: $x = 0$



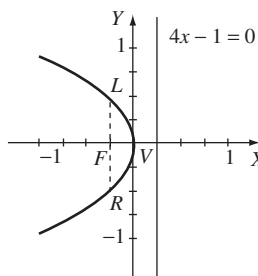
10. Foco: $F\left(0, -\frac{2}{3}\right)$, Directriz: $3y - 2 = 0$, $LR = \frac{8}{3}$,
Eje: $x = 0$



11. Foco: $F\left(\frac{5}{4}, 0\right)$, Directriz: $4x + 5 = 0$, $LR = 5$,
Eje: $y = 0$

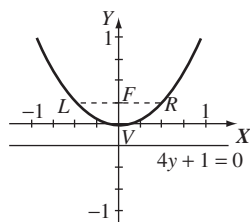


12. Foco: $F\left(-\frac{1}{4}, 0\right)$, Directriz: $4x - 1 = 0$, $LR = 1$,
Eje: $y = 0$



13. Foco: $F\left(0, \frac{1}{4}\right)$, Directriz: $4y + 1 = 0$, $LR = 1$

Eje: $x = 0$



- | | |
|----------------------|-------------------------------------|
| 14. $y^2 + 20x = 0$ | 25. $x^2 - y = 0$ |
| 15. $x^2 - 24y = 0$ | 26. $y^2 + 18x = 0$ |
| 16. $y^2 - 8x = 0$ | 27. $x^2 + 4y = 0$ |
| 17. $x^2 + 4y = 0$ | 28. $3y^2 - 16x = 0$ |
| 18. $y^2 + 2x = 0$ | 29. $3x^2 + 16y = 0$ |
| 19. $3x^2 + 28y = 0$ | 30. $3\sqrt{13}$ unidades |
| 20. $x^2 - 8y = 0$ | 31. $5\sqrt{2}$ unidades |
| 21. $y^2 + 24x = 0$ | 32. $2x^2 - y = 0$, $y^2 - 4x = 0$ |
| 22. $x^2 + 10y = 0$ | 33. $y^2 + 3x = 0$, $y^2 - 3x = 0$ |
| 23. $y^2 + 6x = 0$ | 34. $y^2 - 12x = 0$ |
| 24. $3y^2 - 16x = 0$ | 35. $x^2 - 8y = 0$ |

EJERCICIO 27

V: Vértice, F: Foco, LR: Lado recto, D: Directriz

- $V(1, 5)$, $F(4, 5)$, $LR = 12$, $D: x + 2 = 0$, Eje: $y = 5$
- $V(6, -2)$, $F(6, -6)$, $LR = 16$, $D: y - 2 = 0$, Eje: $x = 6$
- $V(-2, -4)$, $F(-7, -4)$, $LR = 20$, $D: x - 3 = 0$, Eje: $y = -4$
- $V(-1, 5)$, $F(-1, 4)$, $LR = 4$, $D: y - 6 = 0$, Eje: $x = -1$
- $V(-2, 0)$, $F(0, 0)$, $LR = 8$, $D: x + 4 = 0$, Eje: $y = 0$
- $V(0, 2)$, $F(0, 8)$, $LR = 24$, $D: y + 4 = 0$, Eje: $x = 0$
- $V(-4, 2)$, $F\left(-4, \frac{7}{2}\right)$, $LR = 6$, $D: 2y - 1 = 0$, Eje: $x = -4$
- $V\left(\frac{4}{5}, -3\right)$, $F\left(\frac{41}{20}, -3\right)$, $LR = 5$, $D: 20x + 9 = 0$, Eje: $y = -3$
- $V\left(\frac{3}{2}, 2\right)$, $F\left(\frac{3}{2}, 3\right)$, $LR = 4$, $D: y - 1 = 0$, $x = \frac{3}{2}$
- $V\left(2, -\frac{1}{4}\right)$, $F\left(\frac{19}{8}, -\frac{1}{4}\right)$, $LR = \frac{3}{2}$, $D: x = \frac{13}{8}$, Eje: $y = -\frac{1}{4}$
- $V\left(\frac{1}{2}, -\frac{3}{2}\right)$, $F\left(\frac{1}{2}, -\frac{1}{2}\right)$, $LR = 4$, $D: 2y + 5 = 0$, Eje: $x = \frac{1}{2}$
- $V(3, -1)$, $F\left(\frac{10}{3}, -1\right)$, $LR = \frac{4}{3}$, $D: 3x - 8 = 0$, Eje: $y = -1$
- $V\left(0, \frac{1}{4}\right)$, $F\left(0, \frac{3}{4}\right)$, $LR = 2$, $D: 4y + 1 = 0$, Eje: $x = 0$

14. $V(1, 0)$, $F\left(\frac{21}{16}, 0\right)$, $LR = \frac{5}{4}$, $D: 16x - 11 = 0$, $E: y = 0$

- $y^2 + 20x - 8y - 24 = 0$
- $x^2 - 6x + 16y + 25 = 0$
- $y^2 - 8x - 4y + 28 = 0$
- $x^2 + 10x - 12y + 49 = 0$
- $y^2 - 2x + 8y + 20 = 0$
- $3x^2 + 18x - 28y - 29 = 0$
- $y^2 + 24x - 12y - 60 = 0$
- $x^2 - 8x - 16y + 32 = 0$
- $y^2 + 8y - 20x + 36 = 0$
- $x^2 + 28y - 28 = 0$
- $y^2 + 14x - 4y + 25 = 0$
- $x^2 - 14x - 10y + 54 = 0$
- $x^2 - 2x - 8y - 23 = 0$
- $x^2 + 6x - 24y + 129 = 0$
 $x^2 + 6x + 24y - 111 = 0$
- $y^2 + 24x - 4y - 116 = 0$
- $x^2 - 6x + 24y - 87 = 0$
- $(4, 8)$, $(7, -4)$

EJERCICIO 28

- | | |
|---------------------------|----------------------------|
| 1. $y^2 - x + 2y + 1 = 0$ | 5. $x^2 - 4x - y + 3 = 0$ |
| 2. $y^2 - 4x = 0$ | 6. $x^2 - y + 1 = 0$ |
| 3. $3y^2 - x + 7 = 0$ | 7. $2x^2 + 3x - y + 1 = 0$ |
| 4. $y^2 - x - 4 = 0$ | 8. $x^2 + 6x - 2y + 5 = 0$ |

EJERCICIO 29

- | | |
|----------------------|------------------------|
| 1. 8 m | 6. $x - 4y + 24 = 0$ |
| 2. 62.5 cm | 7. $2x - y - 9 = 0$ |
| 3. 18.75 cm | 8. $x - 2y + 2 = 0$ |
| 4. 27.71 cm | 9. $x - 3y - 3 = 0$ |
| 5. $x - y - 2 = 0$ | 10. $2x - 3y - 16 = 0$ |

CAPÍTULO 9

EJERCICIO 30

- $5x^2 + 9y^2 - 180 = 0$
- $5x^2 + 9y^2 - 45 = 0$
- $49x^2 + 24y^2 - 1176 = 0$
- $25x^2 + 16y^2 + 100x - 128y - 44 = 0$
- $9x^2 + 25y^2 - 18x + 100y - 791 = 0$

EJERCICIO 31

1. $V(\pm 2, 0), F(\pm 1, 0), B(0, \pm\sqrt{3}), LR = 3, e = \frac{1}{2},$
 $\overline{V_1V_2} = 4, \overline{F_1F_2} = 2 \text{ y } \overline{B_1B_2} = 2\sqrt{3}$
2. $V(0, \pm 3), F(0, \pm 2), B(\pm\sqrt{5}, 0), LR = \frac{10}{3}, e = \frac{2}{3},$
 $\overline{V_1V_2} = 6, \overline{F_1F_2} = 4 \text{ y } \overline{B_1B_2} = 2\sqrt{5}$
3. $V(0, \pm 2\sqrt{3}), F(0, \pm\sqrt{7}), B(\pm\sqrt{5}, 0), LR = \frac{5\sqrt{3}}{3},$
 $e = \frac{\sqrt{21}}{6}, \overline{V_1V_2} = 4\sqrt{3}, \overline{F_1F_2} = 2\sqrt{7} \text{ y } \overline{B_1B_2} = 2\sqrt{5}$
4. $V(\pm 8, 0), F(\pm 2\sqrt{15}, 0), B(0, \pm 2), LR = 1, e = \frac{\sqrt{15}}{4},$
 $\overline{V_1V_2} = 16, \overline{F_1F_2} = 4\sqrt{15} \text{ y } \overline{B_1B_2} = 4$
5. $V(\pm 5, 0), F(\pm 4, 0), B(0, \pm 3), LR = \frac{18}{5}, e = \frac{4}{5},$
 $\overline{V_1V_2} = 10, \overline{F_1F_2} = 8 \text{ y } \overline{B_1B_2} = 6$
6. $V(0, \pm 4), F(0, \pm 2\sqrt{3}), B(\pm 2, 0), LR = 2, e = \frac{\sqrt{3}}{2},$
 $\overline{V_1V_2} = 8, \overline{F_1F_2} = 4\sqrt{3} \text{ y } \overline{B_1B_2} = 4$
7. $V(0, \pm \frac{5}{2}), F(0, \pm \frac{5\sqrt{5}}{6}), B(\pm \frac{5}{3}, 0), LR = \frac{20}{9}, e = \frac{\sqrt{5}}{3},$
 $\overline{V_1V_2} = 5, \overline{F_1F_2} = \frac{5\sqrt{5}}{3} \text{ y } \overline{B_1B_2} = \frac{10}{3}$
8. $V(0, \pm 1), F(0, \pm \frac{\sqrt{3}}{2}), B(\pm \frac{1}{2}, 0), LR = \frac{1}{2}, e = \frac{\sqrt{3}}{2},$
 $\overline{V_1V_2} = 2, \overline{F_1F_2} = \sqrt{3}, \overline{B_1B_2} = 1$
9. $V(0, \pm\sqrt{3}), F(0, \pm 1), B(\pm\sqrt{2}, 0), LR = \frac{4\sqrt{3}}{3},$
 $e = \frac{\sqrt{3}}{3}, \overline{V_1V_2} = 2\sqrt{3}, \overline{F_1F_2} = 2 \text{ y } \overline{B_1B_2} = 2\sqrt{2}$
10. $V(0, \pm \frac{1}{3}), F(0, \pm \frac{\sqrt{7}}{12}), B(\pm \frac{1}{4}, 0), LR = \frac{3}{8},$
 $e = \frac{\sqrt{7}}{4}, \overline{V_1V_2} = \frac{2}{3}, \overline{F_1F_2} = \frac{\sqrt{7}}{6} \text{ y } \overline{B_1B_2} = \frac{1}{2}$

11. $V(\pm 4, 0), F(\pm 3, 0), B(0, \pm\sqrt{7}), LR = \frac{7}{2}, e = \frac{3}{4},$
 $\overline{V_1V_2} = 8, \overline{F_1F_2} = 6, \overline{B_1B_2} = 2\sqrt{7}$
12. $V(\pm 1, 0), F(\pm \frac{\sqrt{2}}{2}, 0), B(0, \pm \frac{\sqrt{2}}{2}), LR = 1,$
 $e = \frac{\sqrt{2}}{2}, \overline{V_1V_2} = 2, \overline{F_1F_2} = \sqrt{2}, \overline{B_1B_2} = \sqrt{2}$
13. $V(0, \pm\sqrt{5}), F(0, \pm\sqrt{3}), B(\pm\sqrt{2}, 0), LR = \frac{4\sqrt{5}}{5},$
 $e = \frac{\sqrt{15}}{5}, \overline{V_1V_2} = 2\sqrt{5}, \overline{F_1F_2} = 2\sqrt{3} \text{ y } \overline{B_1B_2} = 2\sqrt{2}$
14. $V(0, \pm 2\sqrt{2}), F(0, \pm\sqrt{6}), B(\pm\sqrt{2}, 0), LR = \sqrt{2},$
 $e = \frac{\sqrt{3}}{2}, \overline{V_1V_2} = 4\sqrt{2}, \overline{F_1F_2} = 2\sqrt{6} \text{ y } \overline{B_1B_2} = 2\sqrt{2}$
15. $V(\pm 3, 0), F(\pm\sqrt{6}, 0), B(0, \pm\sqrt{3}), LR = 2, e = \frac{\sqrt{6}}{3},$
 $\overline{V_1V_2} = 6, \overline{F_1F_2} = 2\sqrt{6} \text{ y } \overline{B_1B_2} = 2\sqrt{3}$
16. $V(0, \pm 2\sqrt{3}), F(0, \pm 2\sqrt{2}), B(\pm 2, 0), LR = \frac{4\sqrt{3}}{3},$
 $e = \frac{\sqrt{6}}{3}, \overline{V_1V_2} = 4\sqrt{3}, \overline{F_1F_2} = 4\sqrt{2} \text{ y } \overline{B_1B_2} = 4$

EJERCICIO 32

1. $5x^2 + 9y^2 - 180 = 0$
2. $7x^2 + 9y^2 - 63 = 0$
3. $x^2 + 5y^2 - 5 = 0$
4. $49x^2 + 24y^2 - 1176 = 0$
5. $3x^2 + y^2 - 3 = 0$
6. $16x^2 + 25y^2 - 400 = 0$
7. $7x^2 + 16y^2 - 112 = 0$
8. $4x^2 + 13y^2 - 52 = 0$
9. $9x^2 + 14y^2 - 126 = 0$
10. $3x^2 + 2y^2 - 12 = 0$
11. $5x^2 + y^2 - 5 = 0$
12. $65x^2 + 16y^2 - 1040 = 0$
13. $9x^2 + 5y^2 - 45 = 0$
14. $9x^2 + 25y^2 - 225 = 0$
15. $16x^2 + 7y^2 - 448 = 0$

16. $12x^2 + 16y^2 - 3 = 0$
 17. $8x^2 + 9y^2 - 72 = 0$; $9x^2 + 8y^2 - 72 = 0$
 18. $9x^2 + 4y^2 - 36 = 0$
 19. $x^2 + 4y^2 - 16 = 0$
 20. $9x^2 + 13y^2 - 117 = 0$
 21. $x^2 + 4y^2 - 9 = 0$
 22. $x^2 + 2y^2 - 2 = 0$; $2x^2 + y^2 - 2 = 0$

EJERCICIO 33

1. $C(2, 1)$, $V_1(2, 5)$, $V_2(2, -3)$, $F_1(2, 1 + \sqrt{7})$, $F_2(2, 1 - \sqrt{7})$

$$B_1(5, 1), B_2(-1, 1), LR = \frac{9}{2}, e = \frac{\sqrt{7}}{4}, \overline{V_1V_2} = 8,$$

$$\overline{F_1F_2} = 2\sqrt{7}, \overline{B_1B_2} = 6$$

2. $C\left(\frac{2}{3}, 1\right)$, $V_1\left(\frac{8}{3}, 1\right)$, $V_2\left(-\frac{4}{3}, 1\right)$, $F_1\left(\frac{2+3\sqrt{3}}{3}, 1\right)$,

$$F_2\left(\frac{2-3\sqrt{3}}{3}, 1\right), B_1\left(\frac{2}{3}, 2\right), B_2\left(\frac{2}{3}, 0\right), LR = 1, e = \frac{\sqrt{3}}{2},$$

$$\overline{V_1V_2} = 4, \overline{F_1F_2} = 2\sqrt{3}, \overline{B_1B_2} = 2$$

3. $C(-5, 1)$, $V_1(-2, 1)$, $V_2(-8, 1)$, $F_1(-5 + \sqrt{6}, 1)$,

$$F_2(-5 - \sqrt{6}, 1), B_1(-5, 1 + \sqrt{3}), B_2(-5, 1 - \sqrt{3}),$$

$$LR = 2, e = \frac{\sqrt{6}}{3}, \overline{V_1V_2} = 6, \overline{F_1F_2} = 2\sqrt{6}, \overline{B_1B_2} = 2\sqrt{3}$$

4. $C(0, 2)$, $V_1(0, 7)$, $V_2(0, -3)$, $F_1(0, 5)$, $F_2(0, -1)$, $B_1(4, 2)$,

$$B_2(-4, 2), LR = \frac{32}{5}, e = \frac{3}{5}, \overline{V_1V_2} = 10, \overline{F_1F_2} = 6, \overline{B_1B_2} = 8$$

5. $C(5, -2)$, $V_1(9, -2)$, $V_2(1, -2)$, $F_1(5 + \sqrt{15}, -2)$,

$$F_2(5 - \sqrt{15}, -2), B_1(5, -1), B_2(5, -3), LR = \frac{1}{2},$$

$$e = \frac{\sqrt{15}}{4}, \overline{V_1V_2} = 8, \overline{F_1F_2} = 2\sqrt{15}, \overline{B_1B_2} = 2$$

6. $C(2, 3)$, $V_1(2, 9)$, $V_2(2, -3)$, $F_1(2, 3 + 3\sqrt{3})$,

$$F_2(2, 3 - 3\sqrt{3}), B_1(5, 3), B_2(-1, 3), LR = 3, e = \frac{\sqrt{3}}{2},$$

$$\overline{V_1V_2} = 12, \overline{F_1F_2} = 6\sqrt{3}, \overline{B_1B_2} = 6$$

7. $C\left(-\frac{5}{2}, \frac{3}{4}\right)$, $V_1\left(-\frac{5}{2}, \frac{15}{4}\right)$, $V_2\left(-\frac{5}{2}, -\frac{9}{4}\right)$,

$$F_1\left(-\frac{5}{2}, \frac{3}{4} + \sqrt{5}\right), F_2\left(-\frac{5}{2}, \frac{3}{4} - \sqrt{5}\right), B_1\left(-\frac{1}{2}, \frac{3}{4}\right),$$

$$B_2\left(-\frac{9}{2}, \frac{3}{4}\right), LR = \frac{8}{3}, e = \frac{\sqrt{5}}{3}, \overline{V_1V_2} = 6,$$

$$\overline{F_1F_2} = 2\sqrt{5}, \overline{B_1B_2} = 4$$

8. $C(1, 2)$, $V_1(4, 2)$, $V_2(-2, 2)$, $F_1(1 + \sqrt{5}, 2)$, $F_2(1 - \sqrt{5}, 2)$,

$$B_1(1, 4), B_2(1, 0), LR = \frac{8}{3}, e = \frac{\sqrt{5}}{3}, \overline{V_1V_2} = 6,$$

$$\overline{F_1F_2} = 2\sqrt{5}, \overline{B_1B_2} = 4$$

9. $C\left(-\frac{7}{3}, \frac{3}{4}\right)$, $V_1\left(-2, \frac{3}{4}\right)$, $V_2\left(-\frac{8}{3}, \frac{3}{4}\right)$,

$$F_1\left(\frac{-28 + \sqrt{7}}{12}, \frac{3}{4}\right), F_2\left(\frac{-28 - \sqrt{7}}{12}, \frac{3}{4}\right), B_1\left(-\frac{7}{3}, 1\right),$$

$$B_2\left(-\frac{7}{3}, \frac{1}{2}\right), LR = \frac{3}{8}, e = \frac{\sqrt{7}}{4}, \overline{V_1V_2} = \frac{2}{3},$$

$$\overline{F_1F_2} = \frac{\sqrt{7}}{6}, \overline{B_1B_2} = \frac{1}{2}$$

10. $C\left(-\frac{5}{3}, -\frac{7}{2}\right)$, $V_1\left(-\frac{5}{3}, \frac{-7 + 2\sqrt{3}}{2}\right)$,

$$V_2\left(-\frac{5}{3}, \frac{-7 - 2\sqrt{3}}{2}\right), F_1\left(-\frac{5}{3}, -\frac{5}{2}\right), F_2\left(-\frac{5}{3}, -\frac{9}{2}\right),$$

$$B_1\left(\frac{-5 + 3\sqrt{2}}{3}, -\frac{7}{2}\right), B_2\left(\frac{-5 - 3\sqrt{2}}{3}, -\frac{7}{2}\right),$$

$$LR = \frac{4\sqrt{3}}{3}, e = \frac{\sqrt{3}}{3}, \overline{V_1V_2} = 2\sqrt{3}, \overline{F_1F_2} = 2,$$

$$\overline{B_1B_2} = 2\sqrt{2}$$

11. $C(-3, 2)$, $V_1(0, 2)$, $V_2(-6, 2)$, $F_1(-1, 2)$, $F_2(-5, 2)$,

$$B_1(-3, 2 + \sqrt{5}), B_2(-3, 2 - \sqrt{5}), LR = \frac{10}{3}, e = \frac{2}{3},$$

$$\overline{V_1V_2} = 6, \overline{F_1F_2} = 4, \overline{B_1B_2} = 2\sqrt{5}$$

$$\begin{aligned}
 12. \quad & C\left(-\frac{5}{2}, \frac{4}{3}\right), V_1\left(\frac{1}{2}, \frac{4}{3}\right), V_2\left(-\frac{11}{2}, \frac{4}{3}\right), \\
 & F_1\left(\frac{-5+2\sqrt{5}}{2}, \frac{4}{3}\right), F_2\left(\frac{-5-2\sqrt{5}}{2}, \frac{4}{3}\right), \\
 & B_1\left(-\frac{5}{2}, \frac{10}{3}\right), B_2\left(-\frac{5}{2}, -\frac{2}{3}\right), LR = \frac{8}{3}, e = \frac{\sqrt{5}}{3}, \\
 & \overline{V_1V_2} = 6, \overline{F_1F_2} = 2\sqrt{5}, \overline{B_1B_2} = 4
 \end{aligned}$$

$$\begin{aligned}
 13. \quad & C\left(-\frac{1}{2}, \frac{12}{5}\right), V_1\left(\frac{9}{2}, \frac{12}{5}\right), V_2\left(-\frac{11}{2}, \frac{12}{5}\right) \\
 & F_1\left(\frac{-1+2\sqrt{21}}{2}, \frac{12}{5}\right), F_2\left(\frac{-1-2\sqrt{21}}{2}, \frac{12}{5}\right), \\
 & B_1\left(-\frac{1}{2}, \frac{22}{5}\right), B_2\left(-\frac{1}{2}, \frac{2}{5}\right), LR = \frac{8}{5}, e = \frac{\sqrt{21}}{5}, \\
 & \overline{V_1V_2} = 10, \overline{F_1F_2} = 2\sqrt{21}, \overline{B_1B_2} = 4
 \end{aligned}$$

$$\begin{aligned}
 14. \quad & C(0, -1), V_1(1, -1), V_2(-1, -1), F_1\left(\frac{\sqrt{3}}{2}, -1\right), \\
 & F_2\left(-\frac{\sqrt{3}}{2}, -1\right), B_1\left(0, -\frac{1}{2}\right), B_2\left(0, -\frac{3}{2}\right), LR = \frac{1}{2}, \\
 & e = \frac{\sqrt{3}}{2}, \overline{V_1V_2} = 2, \overline{F_1F_2} = \sqrt{3}, \overline{B_1B_2} = 1
 \end{aligned}$$

$$\begin{aligned}
 15. \quad & C(-2, 0), V_1(-2, 2), V_2(-2, -2), F_1(-2, 1), F_2(-2, -1), \\
 & B_1(-2+\sqrt{3}, 0), B_2(-2-\sqrt{3}, 0), LR = 3, \\
 & e = \frac{1}{2}, \overline{V_1V_2} = 4, \overline{F_1F_2} = 2, \overline{B_1B_2} = 2\sqrt{3}
 \end{aligned}$$

$$\begin{aligned}
 16. \quad & C\left(-\frac{3}{2}, \frac{1}{3}\right), V_1\left(-\frac{3}{2}, \frac{13}{3}\right), V_2\left(-\frac{3}{2}, -\frac{11}{3}\right), \\
 & F_1\left(-\frac{3}{2}, \frac{1}{3}+\sqrt{7}\right), F_2\left(-\frac{3}{2}, \frac{1}{3}-\sqrt{7}\right), B_1\left(\frac{3}{2}, \frac{1}{3}\right), \\
 & B_2\left(-\frac{9}{2}, \frac{1}{3}\right), LR = \frac{9}{2}, e = \frac{\sqrt{7}}{4}, \overline{V_1V_2} = 8, \\
 & \overline{F_1F_2} = 2\sqrt{7}, \overline{B_1B_2} = 6
 \end{aligned}$$

$$\begin{aligned}
 17. \quad & C(-1, 2), V_1\left(-\frac{1}{2}, 2\right), V_2\left(-\frac{3}{2}, 2\right), F_1\left(-1+\frac{\sqrt{5}}{6}, 2\right), \\
 & F_2\left(-1-\frac{\sqrt{5}}{6}, 2\right), B_1\left(-1, \frac{7}{3}\right), B_2\left(-1, \frac{5}{3}\right), LR = \frac{4}{9}, \\
 & e = \frac{\sqrt{5}}{3}, \overline{V_1V_2} = 1, \overline{F_1F_2} = \frac{\sqrt{5}}{3}, \overline{B_1B_2} = \frac{2}{3}
 \end{aligned}$$

EJERCICIO 34

$$\begin{aligned}
 1. \quad & \frac{(x-7)^2}{16} + \frac{(y+2)^2}{4} = 1 \\
 & x^2 + 4y^2 - 14x + 16y + 49 = 0 \\
 2. \quad & \frac{(x-3)^2}{25} + \frac{(y-3)^2}{9} = 1 \\
 & 9x^2 + 25y^2 - 54x - 150y + 81 = 0 \\
 3. \quad & \frac{(x+2)^2}{7} + \frac{(y+1)^2}{16} = 1 \\
 & 16x^2 + 7y^2 + 64x + 14y - 41 = 0 \\
 4. \quad & \frac{(x-4)^2}{16} + \frac{y^2}{9} = 1 \\
 & 9x^2 + 16y^2 - 72x = 0 \\
 5. \quad & \frac{(x-3)^2}{25} + \frac{(y-4)^2}{4} = 1 \\
 & 4x^2 + 25y^2 - 24x - 200y + 336 = 0 \\
 6. \quad & \frac{(x-6)^2}{100} + \frac{(y-5)^2}{36} = 1 \\
 & 9x^2 + 25y^2 - 108x - 250y + 49 = 0 \\
 7. \quad & \frac{x^2}{5} + \frac{(y+2)^2}{9} = 1 \\
 & 9x^2 + 5y^2 + 20y - 25 = 0 \\
 8. \quad & \frac{(x-3)^2}{4} + \frac{(y+2)^2}{36} = 1 \\
 & 9x^2 + y^2 - 54x + 4y + 49 = 0 \\
 9. \quad & \frac{(x+4)^2}{9} + \frac{(y-1)^2}{25} = 1 \\
 & 25x^2 + 9y^2 + 200x - 18y + 184 = 0 \\
 10. \quad & \frac{(x+7)^2}{36} + \frac{(y-5)^2}{4} = 1 \\
 & x^2 + 9y^2 + 14x - 90y + 238 = 0
 \end{aligned}$$

CAPÍTULO 10

$$11. \frac{(x+6)^2}{25} + \frac{(y+2)^2}{16} = 1$$

$$16x^2 + 25y^2 + 192x + 100y + 276 = 0$$

$$12. \frac{\left(x - \frac{8}{3}\right)^2}{36} + \frac{\left(y + \frac{11}{2}\right)^2}{16} = 1$$

$$144x^2 + 324y^2 - 768x + 3\,564y + 5\,641 = 0$$

$$13. \frac{(x-5)^2}{9} + \frac{(y-7)^2}{1} = 1$$

$$x^2 + 9y^2 - 10x - 126y + 457 = 0$$

$$14. \frac{(x+4)^2}{16} + \frac{y^2}{7} = 1$$

$$7x^2 + 16y^2 + 56x = 0$$

$$15. \frac{(x+1)^2}{25} + \frac{(y-2)^2}{9} = 1$$

$$9x^2 + 25y^2 + 18x - 100y - 116 = 0$$

$$16. \frac{(x-3)^2}{64} + \frac{(y-6)^2}{48} = 1$$

$$3x^2 + 4y^2 - 18x - 48y - 21 = 0$$

$$17. \frac{(x+5)^2}{36} + \frac{(y-3)^2}{11} = 1$$

$$11x^2 + 36y^2 + 110x - 216y + 203 = 0$$

$$18. \frac{(x-2)^2}{4} + \frac{(y-1)^2}{12} = 1$$

$$3x^2 + y^2 - 12x - 2y + 1 = 0$$

EJERCICIO 35

- | | |
|-------------------|-------------------|
| 1. Conjunto vacío | 6. Conjunto vacío |
| 2. Punto | 7. Un punto |
| 3. Elipse | 8. Un punto |
| 4. Elipse | 9. Conjunto vacío |
| 5. Elipse | 10. Elipse |

EJERCICIO 36

- | | |
|---------------------------------------|--|
| 1. $3x^2 + 8y^2 + 18x + 8y - 21 = 0$ | 6. $x^2 + 4y^2 - 16 = 0$ |
| 2. $9x^2 + 4y^2 - 36x - 16y + 16 = 0$ | 7. $3x^2 + y^2 - 3 = 0$ |
| 3. $4x^2 + y^2 - 32x - 4y + 64 = 0$ | 8. $9x^2 + 25y^2 - 225 = 0$ |
| 4. $x^2 + 9y^2 - 18y = 0$ | 9. $9x^2 + 4y^2 + 18x + 24y + 9 = 0$ |
| 5. $4x^2 + 25y^2 - 16x - 84 = 0$ | 10. $25x^2 + 4y^2 - 100x + 8y + 4 = 0$ |

EJERCICIO 37

- | | |
|----------------|-----------------------------|
| 1. 30.0588 UA | 4. 11.8578 años |
| 2. 0.72298 UA | 5. $3x - \sqrt{3}y + 6 = 0$ |
| 3. 1.8739 años | 6. $3x + 5y - 44 = 0$ |

EJERCICIO 38

- | | |
|--------------------------------|---|
| 1. $5x^2 - 4y^2 - 20 = 0$ | 5. $3x^2 - 4y^2 - 12 = 0$ |
| 2. $16x^2 - 9y^2 - 144 = 0$ | 6. $7x^2 - 9y^2 - 70x + 72y - 32 = 0$ |
| 3. $13y^2 - 36x^2 - 468 = 0$ | 7. $9y^2 - 16x^2 - 96x - 36y - 252 = 0$ |
| 4. $156y^2 - 100x^2 - 975 = 0$ | |

EJERCICIO 39

- $V(\pm 9, 0)$, $F(\pm 3\sqrt{10}, 0)$, $B(0, \pm 3)$, $\overline{V_1V_2} = 18$,
 $\overline{F_1F_2} = 6\sqrt{10}$, $\overline{B_1B_2} = 6$, $LR = 2$, $e = \frac{\sqrt{10}}{3}$.
 Asíntotas: $y = \pm \frac{1}{3}x$
- $V(\pm 1, 0)$, $F(\pm \sqrt{5}, 0)$, $B(0, \pm 2)$, $\overline{V_1V_2} = 2$,
 $\overline{F_1F_2} = 2\sqrt{5}$, $\overline{B_1B_2} = 4$, $LR = 8$, $e = \sqrt{5}$.
 Asíntotas: $y = \pm 2x$
- $V(0, \pm 2\sqrt{2})$, $F(0, \pm \sqrt{13})$, $B(\pm \sqrt{5}, 0)$, $\overline{V_1V_2} = 4\sqrt{2}$,
 $\overline{F_1F_2} = 2\sqrt{13}$, $\overline{B_1B_2} = 2\sqrt{5}$, $LR = \frac{5\sqrt{2}}{2}$, $e = \frac{\sqrt{26}}{4}$.
 Asíntotas: $y = \pm \frac{2\sqrt{10}}{5}x$
- $V(0, \pm 2a)$, $F(0, \pm \sqrt{5}a)$, $B(\pm a, 0)$, $\overline{V_1V_2} = 4a$,
 $\overline{F_1F_2} = 2\sqrt{5}a$, $\overline{B_1B_2} = 2a$, $LR = a$, $e = \frac{\sqrt{5}}{2}$.
 Asíntotas: $y = \pm 2x$
- $V(\pm \sqrt{5}, 0)$, $F(\pm 3, 0)$, $B(0, \pm 2)$, $\overline{V_1V_2} = 2\sqrt{5}$,
 $\overline{F_1F_2} = 6$, $\overline{B_1B_2} = 4$, $LR = \frac{8\sqrt{5}}{5}$, $e = \frac{3\sqrt{5}}{5}$.
 Asíntotas: $y = \pm \frac{2}{\sqrt{5}}x$
- $V(\pm 3, 0)$, $F(\pm 5, 0)$, $B(0, \pm 4)$, $\overline{V_1V_2} = 6$, $\overline{F_1F_2} = 10$,
 $\overline{B_1B_2} = 8$, $LR = \frac{32}{3}$, $e = \frac{5}{3}$. Asíntotas: $y = \pm \frac{4}{3}x$
- $V(0, \pm 1)$, $F(0, \pm \sqrt{5})$, $B(\pm 2, 0)$, $\overline{V_1V_2} = 2$,
 $\overline{F_1F_2} = 2\sqrt{5}$, $\overline{B_1B_2} = 4$, $LR = 8$; $e = \sqrt{5}$.
 Asíntotas: $y = \pm \frac{1}{2}x$
- $V(\pm 5, 0)$, $F(\pm \sqrt{105}, 0)$, $B(0, \pm 4\sqrt{5})$, $\overline{V_1V_2} = 10$,
 $\overline{F_1F_2} = 2\sqrt{105}$, $\overline{B_1B_2} = 8\sqrt{5}$, $LR = 32$; $e = \frac{\sqrt{105}}{5}$.
 Asíntotas: $y = \pm \frac{4\sqrt{5}}{5}x$
- $V(0, \pm 4)$, $F(0, \pm 2\sqrt{13})$, $B(\pm 6, 0)$, $\overline{V_1V_2} = 8$,
 $\overline{F_1F_2} = 4\sqrt{13}$, $\overline{B_1B_2} = 12$, $LR = 18$; $e = \frac{\sqrt{13}}{2}$.
 Asíntotas: $y = \pm \frac{2}{3}x$
- $V(0, \pm 2)$, $F(0, \pm 2\sqrt{2})$, $B(\pm 2, 0)$, $\overline{V_1V_2} = 4$,
 $\overline{F_1F_2} = 4\sqrt{2}$, $\overline{B_1B_2} = 4$, $LR = 4$; $e = \sqrt{2}$.
 Asíntotas: $y = \pm x$

11. $V(0, \pm\sqrt{5}), F(0, \pm\sqrt{11}), B(\pm\sqrt{6}, 0), \overline{V_1V_2} = 2\sqrt{5},$

$$\overline{F_1F_2} = 2\sqrt{11}, \overline{B_1B_2} = 2\sqrt{6}, LR = \frac{12\sqrt{5}}{5},$$

$$e = \frac{\sqrt{55}}{5}. \text{ Asíntotas: } y = \pm \frac{\sqrt{30}}{6}x$$

12. $V(\pm\sqrt{5}, 0), F(\pm\sqrt{17}, 0), B(0, \pm 2\sqrt{3}), \overline{V_1V_2} = 2\sqrt{5},$

$$\overline{F_1F_2} = 2\sqrt{17}, \overline{B_1B_2} = 4\sqrt{3}, LR = \frac{24\sqrt{5}}{5},$$

$$e = \frac{\sqrt{85}}{5}. \text{ Asíntotas: } y = \pm \frac{2\sqrt{15}}{5}x$$

EJERCICIO 40

- | | |
|------------------------------|------------------------------|
| 1. $7y^2 - 9x^2 - 63 = 0$ | 11. $5y^2 - 6x^2 - 30 = 0$ |
| 2. $9x^2 - 16y^2 - 144 = 0$ | 12. $3x^2 - 4y^2 - 12 = 0$ |
| 3. $3x^2 - 2y^2 + 12 = 0$ | 13. $x^2 - 2y^2 - 12 = 0$ |
| 4. $x^2 - 2y^2 - 8 = 0$ | 14. $5y^2 - 6x^2 - 30 = 0$ |
| 5. $4x^2 - y^2 - 4 = 0$ | 15. $9y^2 - 16x^2 - 256 = 0$ |
| 6. $5x^2 - 2y^2 - 40 = 0$ | 16. $x^2 - 9y^2 - 9 = 0$ |
| 7. $4x^2 - 9y^2 - 36 = 0$ | 17. $5x^2 - 6y^2 - 30 = 0$ |
| 8. $25y^2 - 16x^2 - 400 = 0$ | 18. $9x^2 - 4y^2 - 36 = 0$ |
| 9. $x^2 - 4y^2 - 36 = 0$ | 19. $4x^2 - 9y^2 - 36 = 0$ |
| 10. $x^2 - 3y^2 - 12 = 0$ | |

EJERCICIO 41

1. $C(-3, 4), V(-3 \pm 5, 4), F(-3 \pm \sqrt{34}, 4), B(-3, 4 \pm 3),$

$$\overline{V_1V_2} = 10, \overline{F_1F_2} = 2\sqrt{34}, \overline{B_1B_2} = 6, LR = \frac{18}{5},$$

$$e = \frac{\sqrt{34}}{5}. \text{ Asíntotas: } y - 4 = \pm \frac{3}{5}(x + 3)$$

2. $C(-1, 0), V(-1, 0 \pm 2), F(-1, 0 \pm \sqrt{5}), B(-1 \pm 1, 0),$

$$\overline{V_1V_2} = 4, \overline{F_1F_2} = 2\sqrt{5}, \overline{B_1B_2} = 2, LR = 1,$$

$$e = \frac{\sqrt{5}}{2}. \text{ Asíntotas: } y = \pm 2(x + 1)$$

3. $C(0, -2), V(0 \pm 3, -2), F(0 \pm \sqrt{13}, -2), B(0, -2 \pm 2),$

$$\overline{V_1V_2} = 6, \overline{F_1F_2} = 2\sqrt{13}, \overline{B_1B_2} = 4, LR = \frac{8}{3},$$

$$e = \frac{\sqrt{13}}{3}. \text{ Asíntotas: } y + 2 = \pm \frac{2}{3}x$$

4. $C(1, 2), V(1, 2 \pm \sqrt{2}), F(1, 2 \pm \sqrt{10}), B(1 \pm 2\sqrt{2}, 2),$

$$\overline{V_1V_2} = 2\sqrt{2}, \overline{F_1F_2} = 2\sqrt{10}, \overline{B_1B_2} = 4\sqrt{2},$$

$$LR = 8\sqrt{2}, e = \sqrt{5}. \text{ Asíntotas: } y - 2 = \pm \frac{1}{2}(x - 1)$$

5. $C(-1, -3), V(-1, -3 \pm 3), F(-1, -3 \pm \sqrt{13}),$

$$B(-1 \pm 2, -3), \overline{V_1V_2} = 6, \overline{F_1F_2} = 2\sqrt{13}, \overline{B_1B_2} = 4,$$

$$LR = \frac{8}{3}, e = \frac{\sqrt{13}}{3}. \text{ Asíntotas: } y + 3 = \pm \frac{3}{2}(x + 1)$$

6. $C(-2, 1), V(-2 \pm 4, 1), F(-2 \pm 5, 1), B(-2, 1 \pm 3),$

$$\overline{V_1V_2} = 8, \overline{F_1F_2} = 10, \overline{B_1B_2} = 6, LR = \frac{9}{2}, e = \frac{5}{4}.$$

$$\text{Asíntotas: } y - 1 = \pm \frac{3}{4}(x + 2)$$

7. $C\left(\frac{1}{2}, 1\right), V\left(\frac{1}{2} \pm 3, 1\right), F\left(\frac{1}{2} \pm \sqrt{13}, 1\right),$

$$B\left(\frac{1}{2}, 1 \pm 2\right), \overline{V_1V_2} = 6, \overline{F_1F_2} = 2\sqrt{13}, \overline{B_1B_2} = 4,$$

$$LR = \frac{8}{3}, e = \frac{\sqrt{13}}{3}. \text{ Asíntotas: } y - 1 = \pm \frac{2}{3}\left(x - \frac{1}{2}\right)$$

8. $C(-3, 0), V(-3, 0 \pm 2), F(-3, 0 \pm \sqrt{5}), B(-3 \pm 1, 0),$

$$\overline{V_1V_2} = 4, \overline{F_1F_2} = 2\sqrt{5}, \overline{B_1B_2} = 2, LR = 1, e = \frac{\sqrt{5}}{2}.$$

$$\text{Asíntotas: } y = \pm 2(x + 3)$$

9. $C\left(\frac{1}{2}, \frac{1}{2}\right), V\left(\frac{1}{2}, \frac{1}{2} \pm 2\right), F\left(\frac{1}{2}, \frac{1}{2} \pm 2\sqrt{2}\right),$

$$B\left(\frac{1}{2} \pm 2, \frac{1}{2}\right), \overline{V_1V_2} = 4, \overline{F_1F_2} = 4\sqrt{2}, \overline{B_1B_2} = 4,$$

$$LR = 4, e = \sqrt{2}. \text{ Asíntotas: } y - \frac{1}{2} = \pm \left(x - \frac{1}{2}\right)$$

10. $C(0, -2), V(0 \pm 3, -2), F(0 \pm 3\sqrt{5}, -2),$

$$B(0, -2 \pm 6), \overline{V_1V_2} = 6, \overline{F_1F_2} = 6\sqrt{5}, \overline{B_1B_2} = 12,$$

$$LR = 24, e = \sqrt{5}. \text{ Asíntotas: } y + 2 = \pm 2x$$

11. $C(3, -2), V(3 \pm 1, -2), F(3 \pm \sqrt{2}, -2),$

$$B(3, -2 \pm 1), \overline{V_1V_2} = 2, \overline{F_1F_2} = 2\sqrt{2}, \overline{B_1B_2} = 2,$$

$$LR = 2, e = \sqrt{2}. \text{ Asíntotas: } y + 2 = \pm(x - 3)$$

12. $C(2, -2), V(2, -2 \pm 3), F(2, -2 \pm \sqrt{10}),$

$$B(2 \pm 1, -2), \overline{V_1V_2} = 6, \overline{F_1F_2} = 2\sqrt{10}, \overline{B_1B_2} = 2,$$

$$LR = \frac{2}{3}, e = \frac{\sqrt{10}}{3}. \text{ Asíntotas: } y + 2 = \pm 3(x - 2)$$

$$13. \quad C\left(\frac{1}{2}, \frac{1}{3}\right), V\left(\frac{1}{2} \pm 3, \frac{1}{3}\right), F\left(\frac{1}{2} \pm \sqrt{13}, \frac{1}{3}\right), \\ B\left(\frac{1}{2}, \frac{1}{3} \pm 2\right), \overline{V_1 V_2} = 6, \overline{F_1 F_2} = 2\sqrt{13}, \\ \overline{B_1 B_2} = 4, LR = \frac{8}{3}, e = \frac{\sqrt{13}}{3}.$$

$$\text{Asíntotas: } y - \frac{1}{3} = \pm \frac{2}{3}\left(x - \frac{1}{2}\right)$$

$$14. \quad C(4, 3), V(4 \pm 2\sqrt{2}, 3), F(4 \pm 2\sqrt{3}, 3), \\ B(4, 3 \pm 2), \overline{V_1 V_2} = 4\sqrt{2}, \overline{F_1 F_2} = 4\sqrt{3}, \\ \overline{B_1 B_2} = 4, LR = 2\sqrt{2}, e = \frac{\sqrt{6}}{2}.$$

$$\text{Asíntotas: } y - 3 = \pm \frac{\sqrt{2}}{2}(x - 4)$$

$$15. \quad C(-1, -3), V(-1, -3 \pm \sqrt{6}), F(-1, -3 \pm \sqrt{11}), \\ B(-1 \pm \sqrt{5}, -3), \overline{V_1 V_2} = 2\sqrt{6}, \overline{F_1 F_2} = 2\sqrt{11}, \\ \overline{B_1 B_2} = 2\sqrt{5}, LR = \frac{5}{3}\sqrt{6}, e = \frac{\sqrt{66}}{6}.$$

$$\text{Asíntotas: } y + 3 = \pm \frac{\sqrt{30}}{5}(x + 1)$$

$$16. \quad C(-4, -1), V(-4 \pm 2, -1), F(-4 \pm \sqrt{7}, -1), \\ B(-4, -1 \pm \sqrt{3}), \overline{V_1 V_2} = 4, \overline{F_1 F_2} = 2\sqrt{7}, \\ \overline{B_1 B_2} = 2\sqrt{3}, LR = 3, e = \frac{\sqrt{7}}{2}.$$

$$\text{Asíntotas: } y + 1 = \pm \frac{\sqrt{3}}{2}(x + 4)$$

$$17. \quad C(2, 5), V(2 \pm 2\sqrt{3}, 5), F(2 \pm 3\sqrt{2}, 5), \\ B(2, 5 \pm \sqrt{6}), \overline{V_1 V_2} = 4\sqrt{3}, \overline{F_1 F_2} = 6\sqrt{2}, \\ \overline{B_1 B_2} = 2\sqrt{6}, LR = 2\sqrt{3}, e = \frac{\sqrt{6}}{2}.$$

$$\text{Asíntotas: } y - 5 = \pm \frac{\sqrt{2}}{2}(x - 2)$$

$$18. \quad C(-7, -1), V(-7 \pm \sqrt{2}, -1), F(-7 \pm 2, -1), \\ B(-7, -1 \pm \sqrt{2}), \overline{V_1 V_2} = 2\sqrt{2}, \overline{F_1 F_2} = 4, \\ \overline{B_1 B_2} = 2\sqrt{2}, LR = 2\sqrt{2}, e = \sqrt{2}.$$

$$\text{Asíntotas: } y + 1 = \pm 1(x + 7)$$

$$19. \quad C(-7, -1), V(-7 \pm 1, -1), F(-7 \pm \sqrt{3}, -1), \\ B(-7, -1 \pm \sqrt{2}), \overline{V_1 V_2} = 2, \overline{F_1 F_2} = 2\sqrt{3}, \\ \overline{B_1 B_2} = 2\sqrt{2}, LR = 4, e = \sqrt{3}.$$

$$\text{Asíntotas: } y + 1 = \pm \sqrt{2}(x + 7)$$

$$20. \quad C(-1, 5), V(-1 \pm \sqrt{3}, 5), F(-1 \pm \sqrt{7}, 5), \\ B(-1, 5 \pm 2), \overline{V_1 V_2} = 2\sqrt{3}, \overline{F_1 F_2} = 2\sqrt{7}, \\ \overline{B_1 B_2} = 4, LR = \frac{8}{3}\sqrt{3}, e = \frac{\sqrt{21}}{3}.$$

$$\text{Asíntotas: } y - 5 = \pm \frac{2\sqrt{3}}{3}(x + 1)$$

EJERCICIO 42

- $16x^2 - 9y^2 + 18y - 153 = 0$
- $11y^2 - 25x^2 + 22y - 200x - 664 = 0$
- $9x^2 - 16y^2 - 36x - 64y - 172 = 0$
- $5y^2 - 4x^2 - 30y + 8x + 21 = 0$
- $9x^2 - 16y^2 - 54x + 64y - 127 = 0$
- $5x^2 - 4y^2 + 60x + 24y + 124 = 0$
- $3x^2 - y^2 - 12x + 6y - 9 = 0$
- $9x^2 - 7y^2 + 18x + 42y + 9 = 0$
- $x^2 - 2y^2 - 8x + 12y - 10 = 0$
- $6x^2 - 5y^2 + 24x - 30y + 9 = 0$
- $3x^2 - 4y^2 + 24x - 8y + 32 = 0$
- $9x^2 - 16y^2 - 18x + 96y - 279 = 0$
- $4y^2 - 5x^2 - 8y - 10x - 21 = 0$
- $9x^2 - 4y^2 + 36x + 8y - 4 = 0$

EJERCICIO 43

- | | | |
|-----------------------|--------------|---------------|
| 1. $k = 5$ | 5. $k = 1$ | 9. $k = 0$ |
| 2. $k = -2$ | 6. $k = 1$ | 10. $k = -7$ |
| 3. $k = 77$ | 7. $k = 64$ | 11. $k = -32$ |
| 4. $k = -\frac{1}{6}$ | 8. $k = -35$ | |

EJERCICIO 44

- $5x - 4y + 16 = 0$
- $5x + \sqrt{34}y + 25 = 0$
- $5x - 4y + 26 = 0$
- $2x - 5y - 30 = 0$
- $x + 4y - 9 = 0$

CAPÍTULO 11

EJERCICIO 45

- $y'^2 - \sqrt{2}x = 0$
- $3x'^2 + 4y'^2 - 12 = 0$
- $x'^2 - y'^2 + 4 = 0$
- $x'^2 + y' = 0$
- $x'^2 + 2y'^2 + 4x' + 4y' - 3 = 0$

EJERCICIO 46

- $x'^2 - y'^2 - 1 = 0$
- $y'^2 - 8x' = 0$
- $x'^2 + 4y'^2 - 4x' - 8y' + 4 = 0$
- $x'^2 - \sqrt{3}y' = 0$
- $x'^2 - y'^2 - 2 = 0$
- $4x'^2 + 9y'^2 + 8\sqrt{2}x' - 28 = 0$
- $3x'^2 + y'^2 - 2\sqrt{2}x' - 10\sqrt{2}y' + 10 = 0$
- $6x'^2 + y'^2 - 6\sqrt{5}x' + 8\sqrt{5}y' = 0$
- $2y'^2 - 5\sqrt{2}x' + 15\sqrt{2}y' = 0$
- $3x'^2 - y'^2 - 24\sqrt{2}x + 104 = 0$

EJERCICIO 47

- $y''^2 - 4x'' = 0$
- $4x''^2 + y''^2 - 4 = 0$
- $4x''^2 + 9y''^2 - 36 = 0$
- $4x''^2 - 9y''^2 - 36 = 0$
- $x''^2 + 8y'' = 0$
- $y''^2 + 6x'' = 0$
- $x''^2 - y''^2 + 9 = 0$
- $x''^2 - 12y'' = 0$
- $4x''^2 + y''^2 - 4 = 0$

EJERCICIO 48

- Parábola
- Elipse
- Hipérbola
- Parábola
- Elipse
- Parábola
- Elipse
- Elipse
- Parábola
- Hipérbola

EJERCICIO 49

- $x - y + 3 = 0$;
 $x + 2y = 0$
- $3x + y + 1 = 0$
- $x - y + 1 = 0$
- $x + y + 3 = 0$;
 $4x - y + 7 = 0$
- $3x - 2y = 0$;
 $x + y = 0$
- $x + 5y - 2 = 0$
- $P(-1, -1)$
- $3x + 4y - 2 = 0$
- $x + 3y - 2 = 0$;
 $x - 2y + 2 = 0$
- $2x + 3y - 5 = 0$

EJERCICIO 50

- $5x^2 + 9y^2 + 48x - 54y - 63 = 0$
- $4x^2 + 3y^2 - 8x - 12y + 4 = 0$
- $y^2 + 14x - 6y - 12 = 0$
- $16x^2 - 9y^2 + 200y - 400 = 0$
- $x^2 - 16xy + y^2 - 36x + 18y + 45 = 0$
- $3x^2 + 4xy - 18x - 2y - 4 = 0$
- $144x^2 + 432xy + 324y^2 + 984x - 1332y - 1127 = 0$
- $2xy + a^2 = 0$
- $7y^2 - 24xy - 6y + 144x - 225 = 0$
- $x^2 + 4y^2 - 6\sqrt{3}ax + 11a^2 = 0$

EJERCICIO 51

- $x = \pm \frac{25}{3}$
- $y = \pm \frac{16}{\sqrt{7}}$
- $x = \pm \frac{9}{\sqrt{5}}$
- $x = \pm \frac{4}{\sqrt{29}}$
- $y = \pm \frac{16}{5}$
- $y = \pm \frac{25}{\sqrt{34}}$
- $y = -2 \pm \frac{9}{\sqrt{5}}$
- $x = -3 \pm \frac{25}{\sqrt{41}}$
- $x = \pm \frac{\sqrt{2}}{2}$
- $y = 1 \pm \frac{16}{\sqrt{7}}$

EJERCICIO 52

- $3x + 4y - 25 = 0$
- $9x + 4y - 31 = 0$
- $5x - 4y - 21 = 0$
- $6x - 5y + 31 = 0$
- $12x - 5y - 38 = 0$
- $x - 4y + 17 = 0$
- $3x - 5y + 18 = 0$
- $y - 2 = 0$
- $x - y - 7 = 0$
- $16x + 25y + 41 = 0$

EJERCICIO 53

- $3x - 2y + 13 = 0$; $3x - 2y - 13 = 0$
- $2x - 3y + 3\sqrt{5} = 0$; $2x - 3y - 3\sqrt{5} = 0$
- $x - 6y + 14 = 0$
- $x + 2y + 1 = 0$
- $x + 4y - 3 = 0$; $x + 4y + 9 = 0$

EJERCICIO 54

- $x - y + 2 = 0$; $x + 2y + 8 = 0$
- $y = 0$; $2x + y + 8 = 0$
- $\sqrt{3}x - 3y + 6 = 0$; $\sqrt{3}x + 3y - 6 = 0$
- $3x - 5y + 18 = 0$; $5x + 3y - 38 = 0$
- $2x - y + 1 = 0$; $x + y - 4 = 0$

CAPÍTULO 12

EJERCICIO 55

1. $A(3\sqrt{2}, 3\sqrt{2})$
2. $R(2, -2\sqrt{3})$
3. $P(-4, 4)$
4. $A(4\sqrt{3}, 4)$
5. $B(5, -5\sqrt{3})$
6. $C(0, -4)$
7. $Q\left(\frac{5}{2}, \frac{5}{2}\sqrt{3}\right)$
15. $B\left(-\frac{3}{8}\sqrt{2+\sqrt{3}}, -\frac{3}{8}\sqrt{2-\sqrt{3}}\right)$
16. $A(13, 67^\circ 22' 48'') = (-13, 247^\circ 22' 48'')$
17. $P(2\sqrt{13}, 213^\circ 41' 24'') = (-2\sqrt{13}, 33^\circ 41' 24'')$
18. $C(5, 323^\circ 7' 48'') = (-5, 143^\circ 7' 48'')$
19. $B(15, 306^\circ 52' 11'') = (-15, 126^\circ 52' 11'')$
20. $C(4, 0^\circ) = (-4, 180^\circ)$
21. $W(6, 270^\circ) = (-6, 90^\circ)$
22. $M(5, 306^\circ 52' 11'') = (-5, 126^\circ 52' 11'')$
23. $Q(13, 157^\circ 22' 48'') = (-13, 337^\circ 22' 48'')$
24. $D(-1, 135^\circ) = (1, 315^\circ)$
25. $F(25, 16^\circ 15' 36'') = (-25, 196^\circ 15' 36'')$
26. $Z(1, 150^\circ) = (-1, 330^\circ)$
27. $Q(\sqrt{34}, 329^\circ 2' 10'') = (-\sqrt{34}, 149^\circ 2' 10'')$
28. $L(3, 180^\circ) = (-3, 0^\circ)$
29. $J\left(\frac{\sqrt{17}}{2}, 284^\circ 2' 10''\right) = \left(-\frac{\sqrt{17}}{2}, 104^\circ 2' 10''\right)$
30. $K(5, 90^\circ) = (-5, 270^\circ)$

8. $M\left(-\frac{7}{2}\sqrt{2}, \frac{7}{2}\sqrt{2}\right)$

9. $N(5\sqrt{2}, 5\sqrt{2})$

10. $S\left(-\frac{15}{2}\sqrt{3}, \frac{15}{2}\right)$

11. $T\left(\frac{3}{2}, -\frac{3}{2}\sqrt{3}\right)$

12. $A(-1, \sqrt{3})$

13. $S\left(-\frac{\sqrt{3}}{4}, \frac{1}{4}\right)$

14. $C\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$

EJERCICIO 56

1. $d_{AB} = \sqrt{5} u$
2. $d_{CD} = 3\sqrt{5} u$
3. $d_{EF} = 17 u$
4. $d_{GH} = 2\sqrt{7} u$
5. $d_U = \sqrt{89 - 40\sqrt{3}} u$
6. $A = 36u^2$
7. $A = 6u^2$
8. $A = 32u^2$

EJERCICIO 57

1. $r \sin \theta + 3 = 0$
2. $r \cos \theta - 5 = 0$
3. $\theta = 60^\circ$
4. $r = \frac{6}{2\cos \theta - 3\sin \theta}$
5. $r = \frac{2}{\sin \theta + \cos \theta}$
6. $r = \frac{p}{\cos(\theta - w)}$
13. $r = \sin 2\theta$
14. $r^2 \cos^2 \theta - 2r \cos \theta - 4r \sin \theta - 3 = 0$
15. $r = \frac{\pm 6}{\sqrt{5\cos^2 \theta + 4}}$
16. $r = \frac{\pm 20}{\sqrt{16 + 9\sin^2 \theta}}$
17. $9r^2 \cos^2 \theta + 25r^2 \sin^2 \theta - 72r \sin \theta - 81 = 0$
18. $r = \frac{-36\sin \theta}{4 + 5\sin^2 \theta}$
19. $r = \frac{\pm 3}{\sqrt{\cos 2\theta}}$
20. $r = \frac{\pm 12}{\sqrt{25\cos^2 \theta - 9}}$
21. $9r^2 \cos^2 \theta - 4r^2 \sin^2 \theta + 8r \sin \theta - 40 = 0$
22. $r = \cos \theta \pm 1$
23. $r = \pm \sqrt{\frac{-8}{\sin 2\theta}}$
24. $r = \pm \sqrt{\cos 2\theta}$
25. $r = \frac{1}{2} \sin 2\theta$
26. $r \cos^2 \theta (r \sin \theta - 2) = 16 \sin \theta$
27. $r = 12 \operatorname{ctg} \theta \csc \theta$
28. $r = \frac{12}{3\sin \theta - 4\cos \theta}$
29. $r = \frac{\pm 4}{\sqrt{\cos^2 \theta - 4\sin^2 \theta}}$
7. $r = \pm 4$
8. $r + 4 \cos \theta = 0$
9. $r - 2 \sin \theta = 0$
10. $r^2 - 4r \cos \theta - 6r \sin \theta - 12 = 0$
11. $r = \frac{-8\cos \theta}{\sin^2 \theta} = -8\cot \theta \csc \theta$
12. $r^2 \sin^2 \theta - 12r \cos \theta - 36 = 0$

30. $r^2 \cos^2 \theta + 4r \sin \theta - 8 = 0$

31. $r = -3 \sin \theta \pm 2$

32. $r = \frac{\pm 6}{\sqrt{4 + 5 \sin^2 \theta}}$

33. $r^2 - 2r \cos \theta = 8$

34. $r^2 (3 + \sin^2 \theta) - 6r \cos \theta - 9 = 0$

35. $r^2 (4 - 9 \cos^2 \theta) - 8r \sin \theta - 6 = 0$

36. $r^2 \cos^2 \theta - 5r \sin \theta + 15 = 0$

37. $r = \frac{2 - 4 \operatorname{ctg} \theta}{3 \sin \theta}$

38. $r = \frac{\pm 2\sqrt{2}}{\sqrt{2 \cos 2\theta + 3 \sin 2\theta}}$

39. $r = \frac{1 \pm \sqrt{1 + \tan \theta}}{\cos \theta}$

40. $\frac{1}{2} r^2 \sin 2\theta - r \sin \theta - 3r \cos \theta + 2 = 0$

EJERCICIO 58

1. $y - 5 = 0$
2. $x + 8 = 0$
3. $x^2 + 2y^2 - 4x = 0$
4. $x^2 + y^2 - 4y = 0$
5. $9x^2 + 5y^2 + 20y - 25 = 0$
6. $(x^2 + y^2)^{\frac{3}{2}} - 2xy = 0$
7. $x^2 + 2y^2 - 256 = 0$
8. $y^2 - 10x - 25 = 0$
9. $4(x^2 + y^2) = (x^2 + y^2 + x)^2$
10. $x^4 + y^4 - 15x^2 - 16y^2 + 2x^2y^2 - 2x^3 - 2xy^2 = 0$
11. $16(x^2 + y^2) = (x^2 + y^2 + 4x)^2$
12. $2xy - 9 = 0$
13. $x^2 - 6y - 9 = 0$
14. $y^2 + 8x - 16 = 0$
15. $(x^2 + y^2)^{\frac{3}{2}} = 4(x^2 - y^2)$
16. $(x^2 + y^2)^{\frac{3}{2}} = y$
17. $9x^2 + 8y^2 + 12y - 36 = 0$
18. $3x^2 + 4y^2 - 4x - 4 = 0$
19. $3x^2 - y^2 + 12x + 9 = 0$
20. $3x^2 + 4y^2 + 2x - 1 = 0$

21. $3x^2 + 4y^2 - 8x - 16 = 0$

22. $(x^2 + y^2)^2 = 16(x^2 - y^2)$

23. $5x^2 - 4y^2 - 36x + 36 = 0$

24. $x^2 + 4y - 4 = 0$

25. $4x^3 - y^2 = 0$

26. $x^2 + y^2 - 5x + 3y - 8 = 0$

27. $x^2 + 3x - 2y + 4 = 0$

28. $y^2 - 12x = 0$

29. $x^2 - y^2 = 1$

30. $x + \sqrt{3}y + 8 = 0$

31. $x - \sqrt{3}y = 0$

32. $(x^2 + y^2)^2 = x^3 - 3xy^2$

33. $(x^2 + y^2)^2 = 2x^3 - 6xy^2$

34. $(x^2 + y^2)^2 \left[\sqrt{x^2 + y^2} - 5 \right] = -40x^2y^2$

35. $x \tan \frac{\sqrt{x^2 + y^2}}{3} - y = 0$

36. $2(x^2 + y^2)^{\frac{3}{2}} - 9(x^2 + y^2)^{\frac{1}{2}} + 9x = 0$

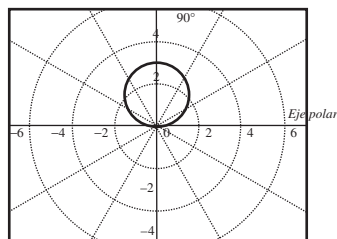
37. $2(x^2 + y^2)^{\frac{3}{2}} - 4(x^2 + y^2) + \sqrt{x^2 + y^2} - x = 0$

EJERCICIO 59

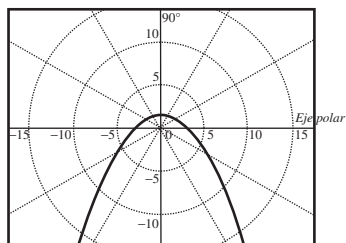
- | | |
|--------------------------|--------------------------|
| 1. Parábola horizontal | 11. Hipérbola vertical |
| 2. Parábola horizontal | 12. Parábola vertical |
| 3. Parábola vertical | 13. Hipérbola horizontal |
| 4. Parábola vertical | 14. Elipse horizontal |
| 5. Elipse vertical | 15. Parábola horizontal |
| 6. Hipérbola vertical | 16. Elipse horizontal |
| 7. Elipse horizontal | 17. Elipse horizontal |
| 8. Elipse horizontal | 18. Elipse vertical |
| 9. Hipérbola horizontal | 19. Parábola vertical |
| 10. Hipérbola horizontal | 20. Elipse vertical |

EJERCICIO 60

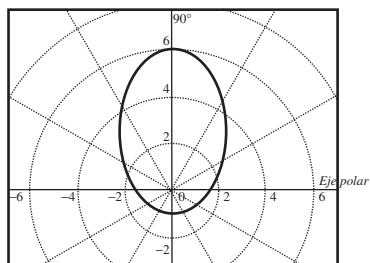
1. $r = 3 \sin \theta$



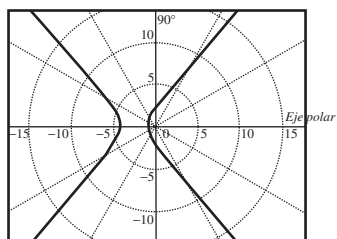
2. $r = \frac{3}{1 + \sin \theta}$



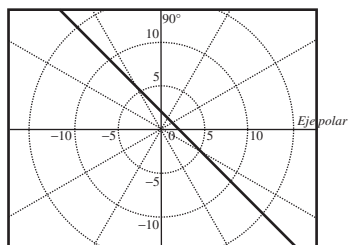
3. Elipse



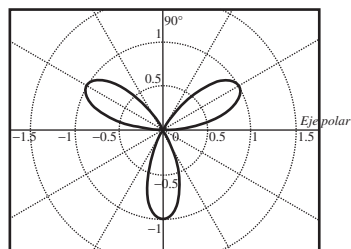
4. $r = \frac{4}{2 - 3 \cos \theta}$



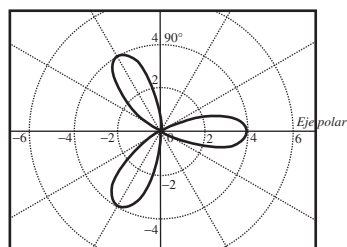
5. $r = \frac{2}{\sin \theta + \cos \theta}$



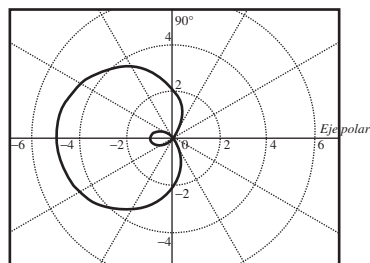
6. $r = \sin 3\theta$



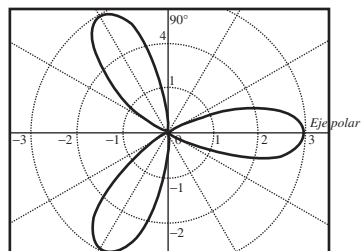
7. $r = 4 \cos 3\theta$



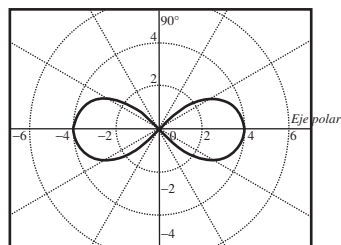
8. $r = 2 - 3 \cos \theta$



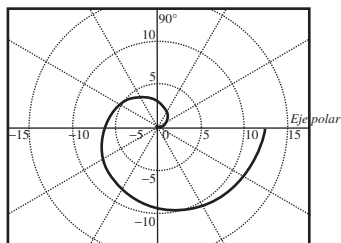
9. $r = 3 \cos 3\theta$



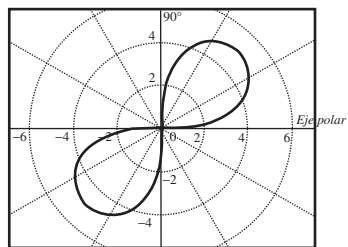
10. $r^2 = 16 \cos 2\theta$



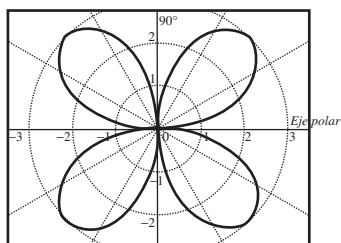
11. $r = 2\theta$



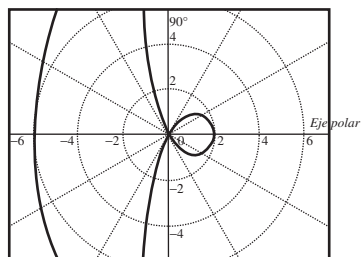
16. $r^2 = 25 \operatorname{sen} 2\theta$



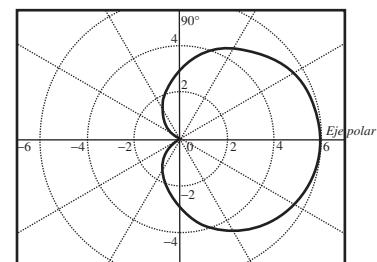
12. $r = 3 \operatorname{sen} 2\theta$



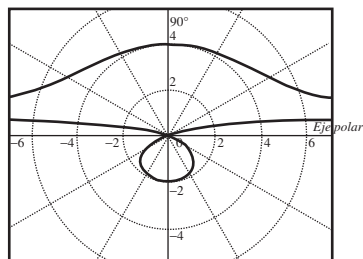
17. $r = 4 - 2 \sec \theta$



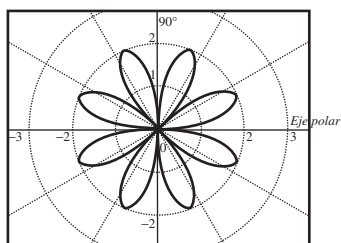
13. $r = 3(1 + \cos \theta)$



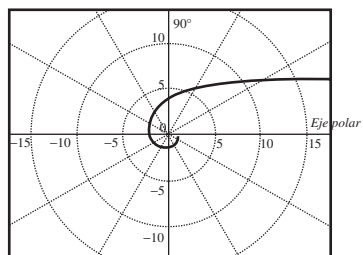
18. $r = 3 + \csc \theta$



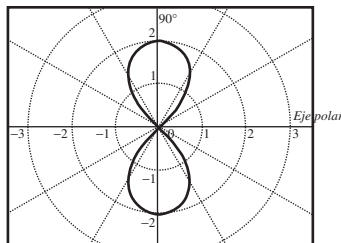
14. $r = 2 \operatorname{sen} 4\theta$



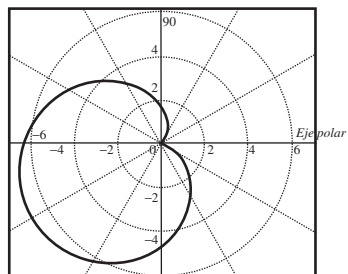
19. $r = \frac{2\pi}{\theta}$



15. $r^2 = -4 \cos 2\theta$



20. $r = \theta(1 - \cos \theta)$



EJERCICIO 61

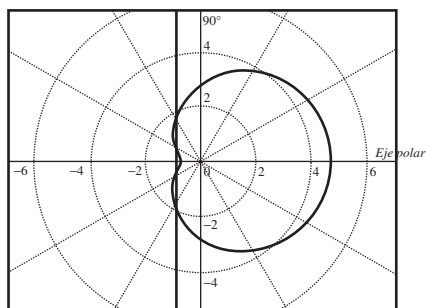
1. $r \cos \theta - 5 = 0$
2. $r \operatorname{sen} \theta + 7 = 0$
3. $r \operatorname{sen} \theta - 5 = 0$
4. $r \cos \theta + 1 = 0$
5. $r \cos(\theta - 60^\circ) = 5 \sqrt{3}$
6. $r \cos(\theta - 75^\circ) = 4 \sqrt{2}$
7. $r \cos(\theta - 150^\circ) = 2$
8. $r \cos(\theta - 135^\circ) = 5$

EJERCICIO 62

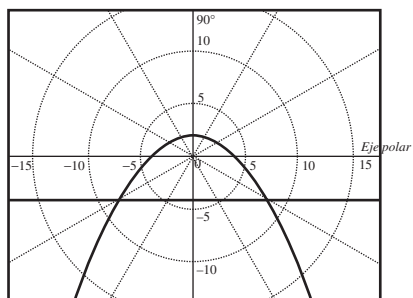
1. $r^2 - 6r \cos(\theta - 30^\circ) - 72 = 0$
2. $r^2 - 10r \cos(\theta - 120^\circ) + 24 = 0$
3. $r^2 - 20r \cos(\theta - 45^\circ) + 84 = 0$
4. $r - 14 \operatorname{sen} \theta = 0$
5. $r - 6 = 0$

EJERCICIO 63

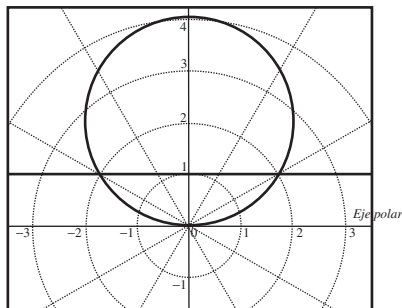
1. $(1, 30^\circ), (-\sqrt{3}, 60^\circ), (\sqrt{3}, 300^\circ), (-1, 330^\circ)$



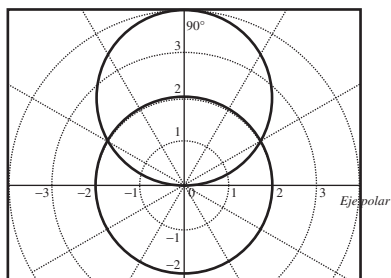
2. $(8, 210^\circ), (8, 330^\circ)$



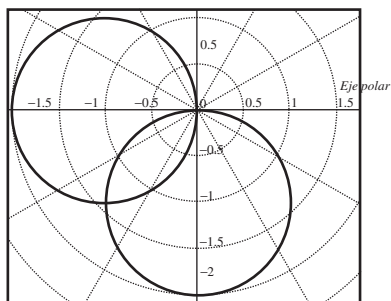
3. $(2, 30^\circ), (2, 150^\circ)$



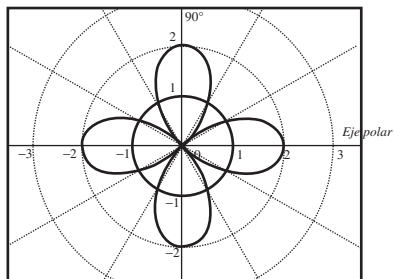
4. $(2, 30^\circ), (2, 150^\circ)$



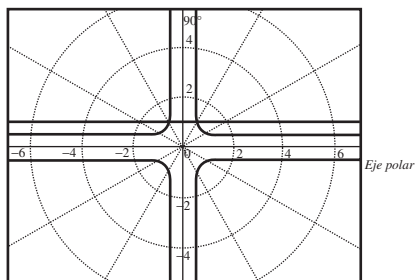
5. $(-\sqrt{2}, 45^\circ), (\sqrt{2}, 225^\circ)$



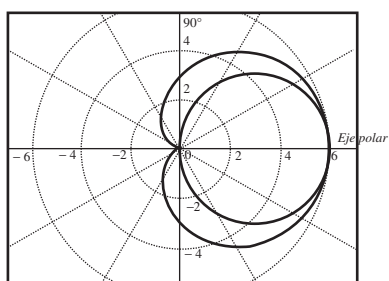
6. $(1, 30^\circ), (1, 330^\circ), (1, 150^\circ), (1, 210^\circ)$



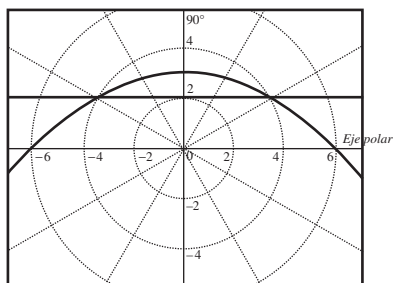
7. $\left(\frac{2}{\sqrt{3}}, 60^\circ\right), \left(-\frac{2}{\sqrt{3}}, 300^\circ\right)$



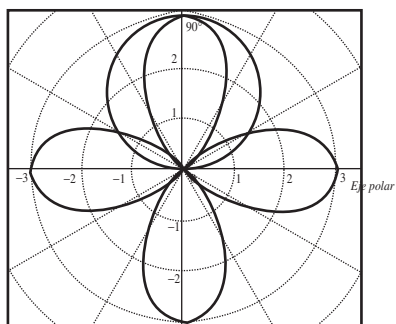
8. $(6, 0^\circ)$



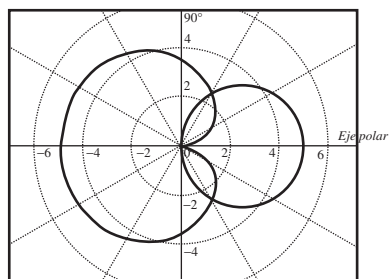
9. $(4, 30^\circ), (4, 150^\circ)$



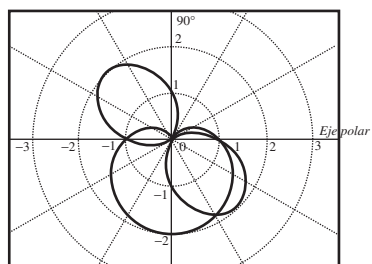
10. $\left(\frac{3}{2}, 30^\circ\right), \left(\frac{3}{2}, 150^\circ\right)$



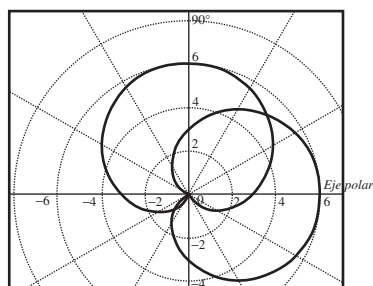
11. $\left(\frac{5}{2}, 60^\circ\right), \left(\frac{5}{2}, 300^\circ\right)$



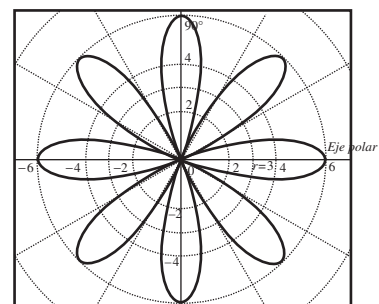
12. $(1, 0^\circ), (1, 180^\circ), (1.8, 300^\circ), (0.13, 60^\circ)$



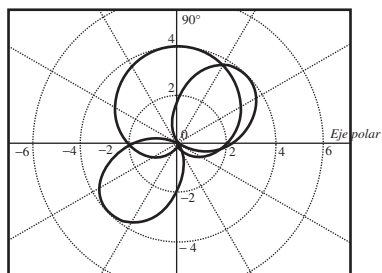
13. $\left(\frac{3}{2}(2+\sqrt{2}), 45^\circ\right), \left(\frac{3}{2}(2-\sqrt{2}), 225^\circ\right)$



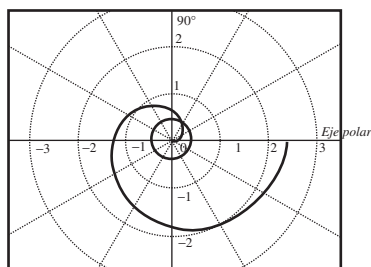
14. $(3, 15^\circ), (3, 75^\circ), (3, 105^\circ), (3, 165^\circ), (3, 195^\circ), (3, 255^\circ), (3, 285^\circ), (3, 345^\circ)$



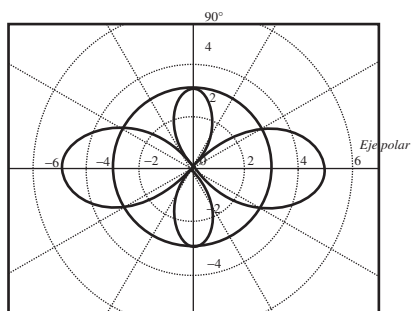
- 15.
- $(2, 0^\circ)$
- ,
- $(2, 180^\circ)$
- ,
- $(3.7, 60^\circ)$
- ,
- $(0.26, 300^\circ)$



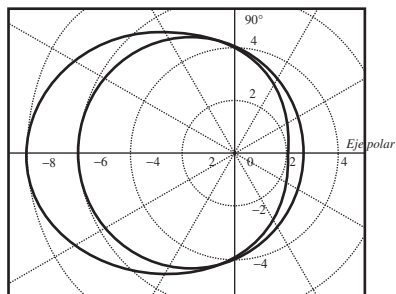
- 16.
- $(2, 60^\circ)$



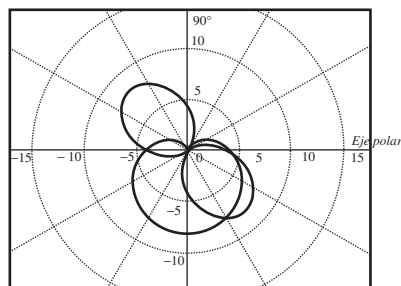
- 17.
- $(3, 30^\circ)$
- ,
- $(3, 150^\circ)$
- ,
- $(3, 210^\circ)$
- ,
- $(3, 330^\circ)$



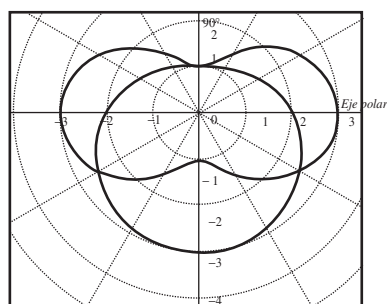
- 18.
- $(4, 90^\circ)$
- ,
- $(4, 270^\circ)$



- 19.
- $(4, 0^\circ)$
- ,
- $(4, 180^\circ)$
- ,
- $(4 - 2\sqrt{3}, 60^\circ)$
- ,
- $(4 + 2\sqrt{3}, 300^\circ)$



- 20.
- $(1, 90^\circ)$
- ,
- $(2.5, 210^\circ)$
- ,
- $(2.5, 330^\circ)$



CAPÍTULO 13

EJERCICIO 64

1. $x - 4y = 0$
2. $\frac{x-b}{a} = \frac{y+d}{c}$
3. $2bx - ay - 2ab = 0$
4. $3x + 4y - 2 = 0$
5. $2x - y - 2 = 0$
6. $y^2 - x + 4y + 3 = 0$
7. $x^2 - 4x - y + 8 = 0$
8. $2x - y - 1 = 0$
9. $2x^3 - y + 20 = 0$
10. $y^2 - x - 1 = 0$
11. $y^2 - 16x - 2y - 31 = 0$
12. $xy^2 - 2xy + x - y = 0$
13. $x^3 - 2y - 2 = 0$
14. $3x^2 - 21xy + 18y^2 - 25 = 0$
15. $4x^2 + y^2 + x = 0$
16. $y^2 - x^2 - 4y = 0$
17. $x^2 + y^2 = 16$
18. $x^2y + x^2 - y - 4 = 0$
19. $xy = 1$, $xy = -1$
20. $(3y^2 + 2)^2(2 - x) = 1$
21. $y = \frac{1}{1-3x}$
22. $y = (x-1)^{\frac{2}{3}}$
23. $xy = 1$

EJERCICIO 65

1. $\frac{x^2}{16} + \frac{y^2}{49} = 1$

2. $\frac{x^2}{4} + \frac{y^2}{16} = 1$

3. $x^2 + y^2 = 4$

4. $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$

5. $xy = 128$

6. $x^2 - y^2 + 1 = 0$

7. $4x - y + 4 = 0$

8. $2y^2 + x - 1 = 0$

9. $x^4 - 4x^2 + y^2 = 0$

10. $y = 3x - 4x^3$

11. $\frac{(x-1)^2}{4} + \frac{(y-2)^2}{9} = 1$

12. $x^2 + y^2 - 6x - 6y + 17 = 0$

13. $4x^2 + 9y^2 - 16x + 18y - 11 = 0$

14. $4x^2 + y^2 - 32x - 6y + 69 = 0$

15. $x = (y-1)(xy-x+2)$

16. $(x-1)^2 - y^2 = 4$

17. $5x^2 + 4xy + 8y^2 - 36 = 0$

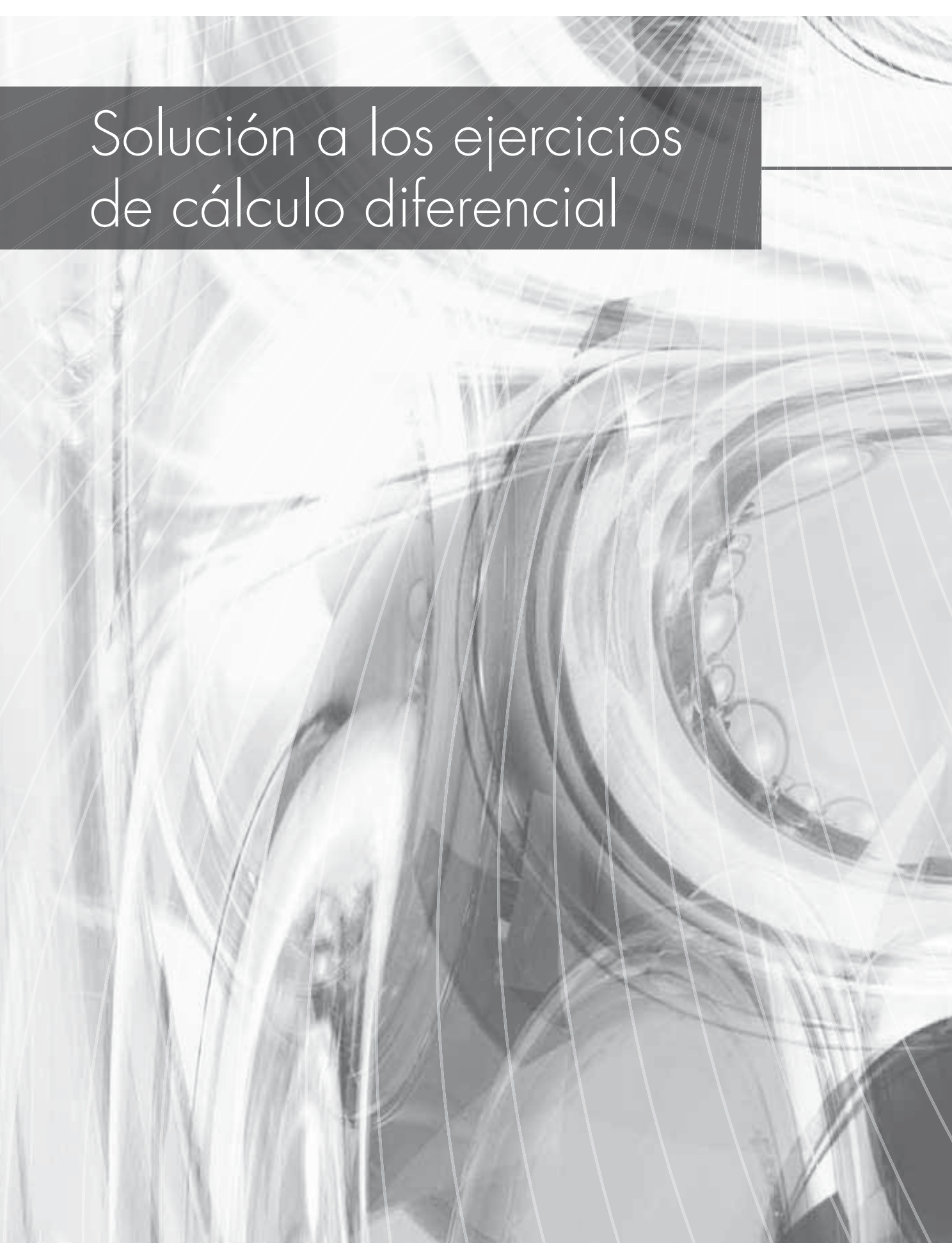
18. $x^2 - 8xy + 17y^2 - 2 = 0$

19. $x - y + 5 = 0$

20. $9x^2 - 4y^2 + 16y - 52 = 0$

21. $y^2 - 2xy = 8$

Solución a los ejercicios de cálculo diferencial



CAPÍTULO 1

EJERCICIO 1

- | | | |
|-------------|--------------|--------------|
| 1. Función | 6. Relación | 11. Función |
| 2. Relación | 7. Función | 12. Relación |
| 3. Función | 8. Relación | 13. Función |
| 4. Relación | 9. Función | 14. Función |
| 5. Función | 10. Relación | 15. Relación |

EJERCICIO 2

- $f\left(-\frac{1}{2}\right) = -\frac{5}{2} f(3) = 15, f(0) = -3$
- $f(a) = a^2 - 5a + 6,$
 $f(a+b) = a^2 + 2ab + b^2 - 5a - 5b + 6$
 $f(x+h) = 3x^2 + 6hx + 3h^2 + 4x + 4h - 2$
- $\frac{f(x+h)-f(x)}{h} = 6x + 3h + 4$
- $f\left(\frac{1}{3}\right) = -\frac{1}{5}, f\left(-\frac{1}{2}\right) = \text{No existe}$

$$f(x+h) - f(x) = \frac{4h}{(2x+2h+1)(2x+1)}$$
- $f(5) = 3, f(4) = 0, f(6) = \sqrt{20} = 2\sqrt{5},$
 $f(3) = \text{No está definida}$
- $f(x+h) = \sqrt{x^2 + 2xh + h^2 - 3}$

$$\frac{f(x+h)-f(x)}{h} = \frac{2x+h}{\sqrt{(x+h)^2 - 3} + \sqrt{x^2 - 3}}$$
- $\frac{f(x+b)-f(x)}{b} = -\frac{1}{(x+b+1)(x+1)}$
- $\frac{f(x+h)-f(x)}{h} = -\frac{1}{\sqrt{1-(x+h)} + \sqrt{1-x}}$
- $f(1) = \frac{4}{3}, f(0) = \frac{5}{2}, f(x+5) = \frac{|x|}{x+7}$
- $f(-1) = 2, f\left(\frac{1}{x}\right) = -\frac{3}{x^2} + 2x^2 - 3x$

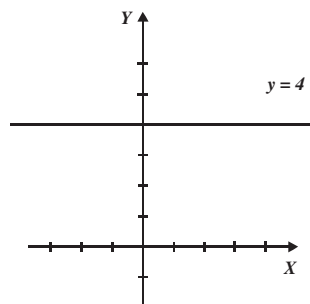
Las demostraciones de los ejercicios 11, 12, 13, 14, 15, 16, 17 y 18, se dejan al estudiante.

EJERCICIO 3

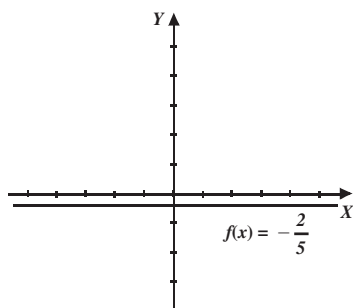
- $(-\infty, \infty) = \{x \in \mathbb{R}\}$
- $(-\infty, \infty) = \{x \in \mathbb{R}\}$
- $(-\infty, -3) \cup (-3, \infty) = \{x \in \mathbb{R} | x \neq -3\}$
- $(-\infty, 5) \cup (5, \infty) = \{x \in \mathbb{R} | x \neq 5\}$
- $(-\infty, -4) \cup (-4, 4) \cup (4, \infty) = \{x \in \mathbb{R} | x \neq -4, x \neq 4\}$
- $(-\infty, 0) \cup (0, 5) \cup (5, \infty) = \{x \in \mathbb{R} | x \neq 0, x \neq 5\}$
- $(-\infty, 2) \cup (2, 5) \cup (5, \infty) = \{x \in \mathbb{R} | x \neq 2, x \neq 5\}$
- $(-\infty, -5) \cup (-5, 5) \cup (5, \infty) = \{x \in \mathbb{R} | x \neq -5, x \neq 5\}$
- $(-\infty, \infty) = \{x \in \mathbb{R}\}$
- $(-\infty, -5) \cup (-5, 0) \cup (0, \infty) = \{x \in \mathbb{R} | x \neq -5, x \neq 0\}$
- $(-\infty, -1) \cup (-1, 0) \cup (0, 1) \cup (1, \infty) = \{x \in \mathbb{R} | x \neq -1, x \neq 0, x \neq 1\}$
- $[-1, \infty) = \{x \geq -1\}$
- $[6, \infty) = \{x \geq 6\}$
- $(-\infty, 2] = \{x \leq 2\}$
- $(-\infty, 4] = \{x \leq 4\}$
- $(-\infty, -5] \cup [5, \infty) = \{x \in \mathbb{R} | x \leq -5 \text{ o } x \geq 5\}$
- $(-\infty, -1] \cup [6, \infty) = \{x \in \mathbb{R} | x \leq -1 \text{ o } x \geq 6\}$
- $[-6, 6] = \{x \in \mathbb{R} | -6 \leq x \leq 6\}$
- $(-\infty, \infty) = \{x \in \mathbb{R}\}$
- $(-\infty, \infty) = \{x \in \mathbb{R}\}$
- $[5, \infty) = \{x \in \mathbb{R} | x \geq 5\}$
- $(2, \infty) = \{x \in \mathbb{R} | x > 2\}$
- $(-\infty, 3) = \{x \in \mathbb{R} | x < 3\}$
- $(-\infty, -2) \cup (-2, \infty) = \{x \in \mathbb{R} | x \neq -2\}$
- $(-\infty, -4] \cup (3, \infty) = \{x \in \mathbb{R} | x \leq -4 \text{ o } x > 3\}$
- $\left[1, \frac{3}{2}\right] = \left\{x \in \mathbb{R} \mid 1 \leq x < \frac{3}{2}\right\}$
- $(-2, \infty) = \{x \in \mathbb{R} | x \geq -2\}$
- $\left(-\infty, \frac{5}{2}\right) = \left\{x \in \mathbb{R} \mid x < \frac{5}{2}\right\}$
- $(0, \infty) = \{x \in \mathbb{R} | x > 0\}$
- $(-1, 3) = \{x \in \mathbb{R} | -1 < x < 3\}$
- $[1, \infty) = \{y \in \mathbb{R} | y \geq 1\}$
- $[-4, \infty) = \{y \in \mathbb{R} | y \geq -4\}$
- $(-\infty, 9] = \{y \in \mathbb{R} | y \leq 9\}$
- $\left(-\infty, \frac{9}{4}\right] = \left\{y \in \mathbb{R} \mid y \leq \frac{9}{4}\right\}$
- $(-\infty, -2) \cup (-2, \infty) = \{y \in \mathbb{R} | y \neq -2\}$
- $\left(-\infty, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right) = \left\{y \in \mathbb{R} \mid y \neq \frac{1}{2}\right\}$
- $[1, \infty) = \{y \in \mathbb{R} | y \geq 1\}$
- $(-\infty, 0] = \{y \in \mathbb{R} | y \leq 0\}$
- $[0, 2] = \{y \in \mathbb{R} | 0 \leq y \leq 2\}$
- $(0, 1] = \{y \in \mathbb{R} | 0 < y \leq 1\}$
- $[0, 1) \cup (1, \infty) = \{x \in \mathbb{R} | 0 \leq y < 1 \text{ o } y > 1\}$
- $[0, \infty) = \{y \in \mathbb{R} | y \geq 0\}$

EJERCICIO 4

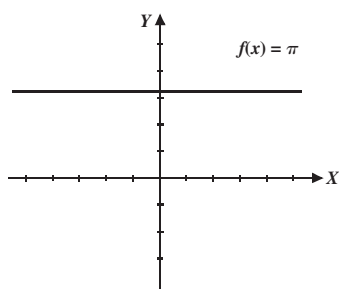
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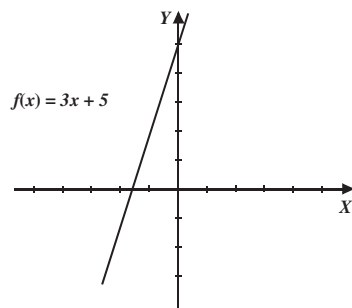
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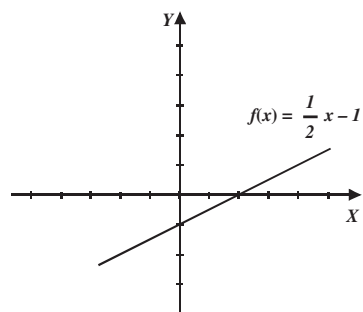
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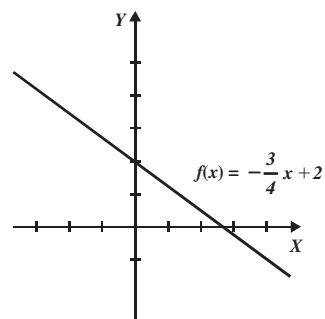
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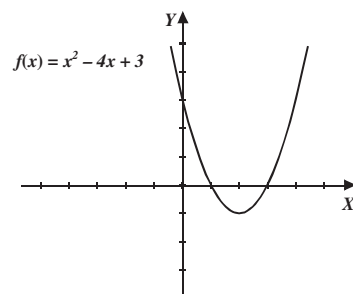
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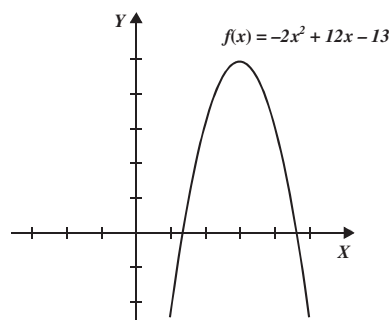
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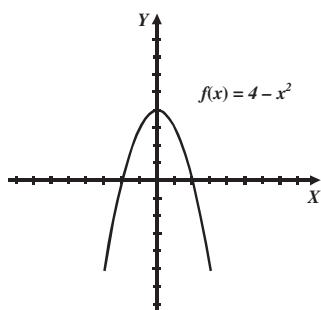
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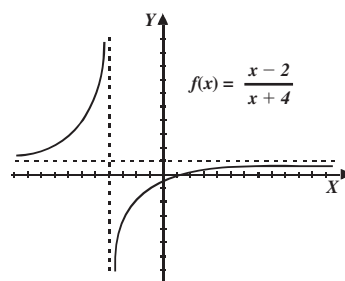
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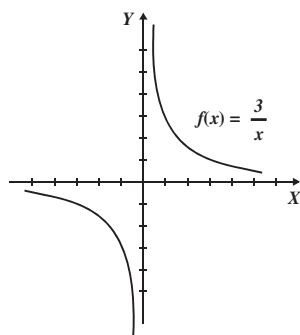
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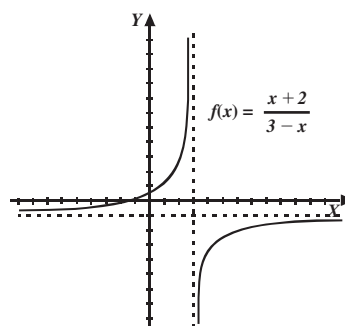
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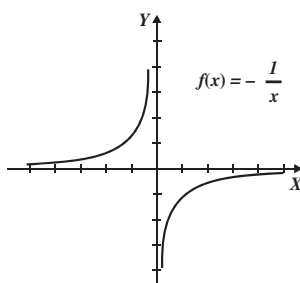
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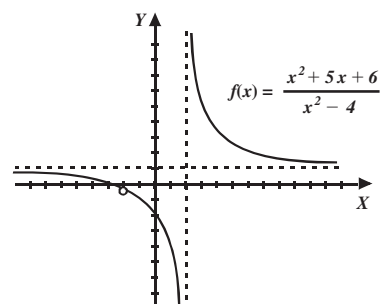
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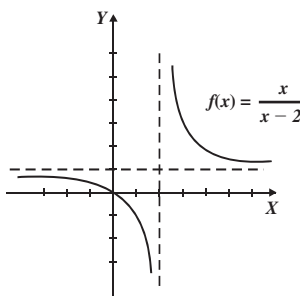
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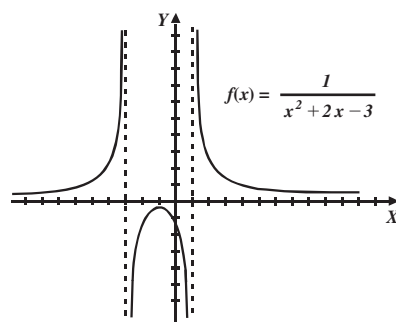
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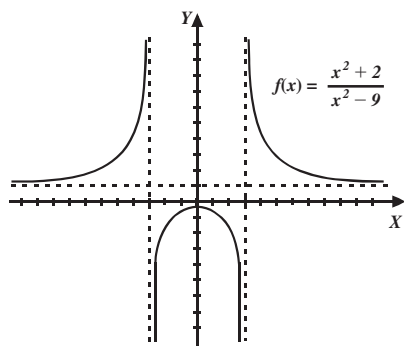
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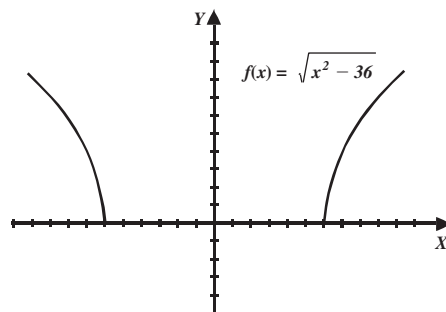
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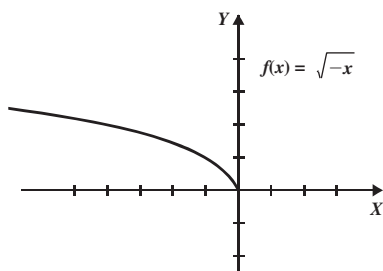
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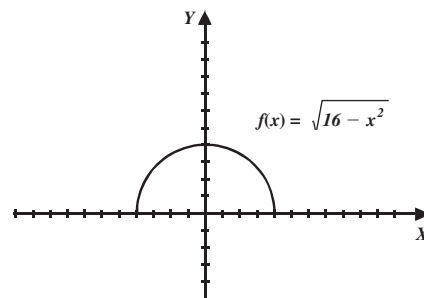
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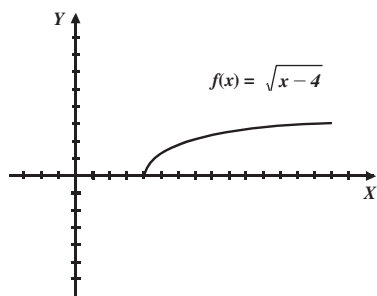
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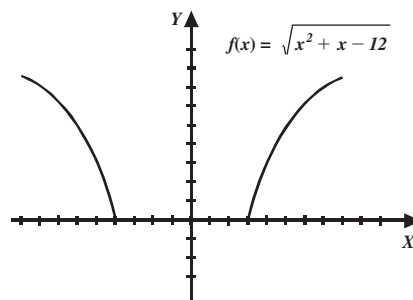
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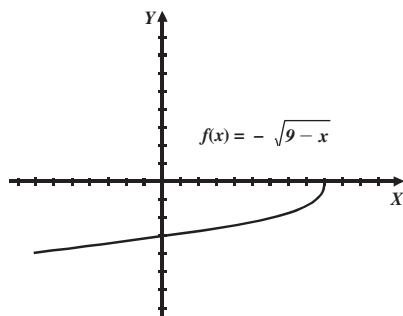
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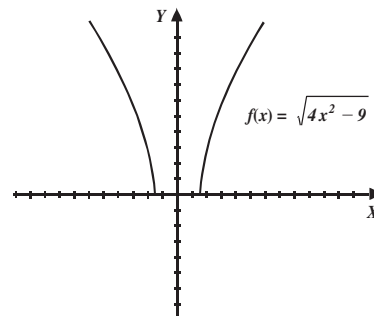
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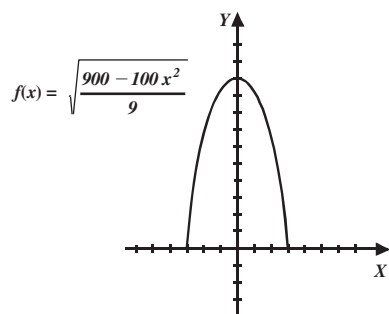
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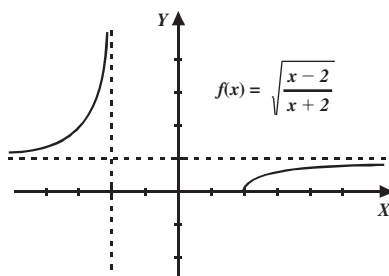
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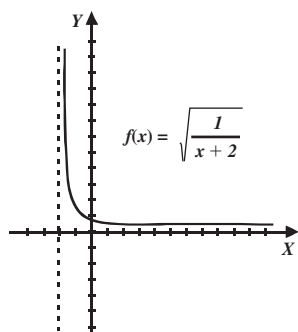
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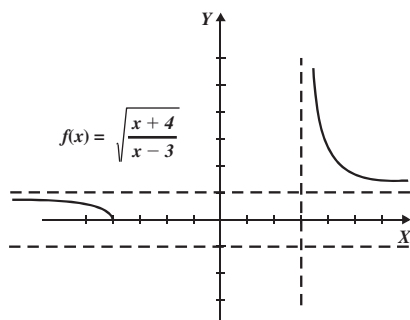
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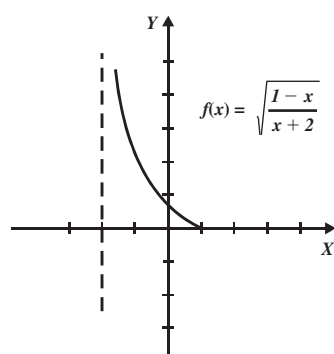
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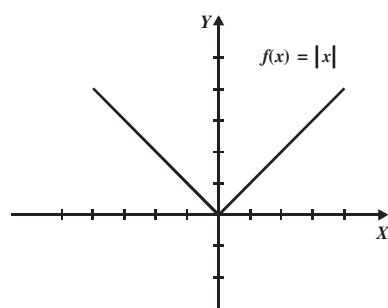
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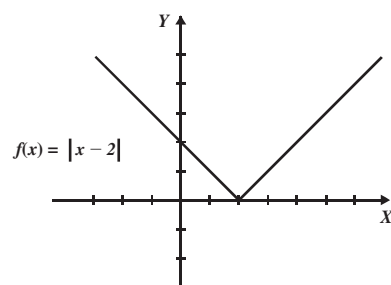
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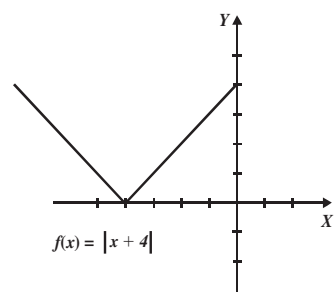
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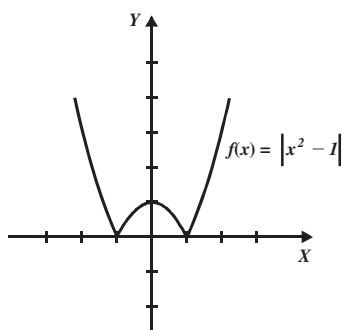
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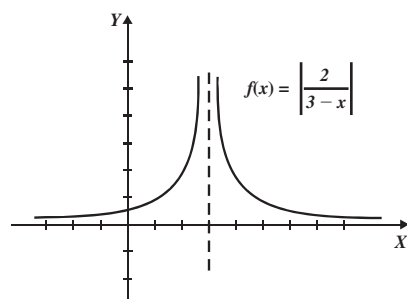
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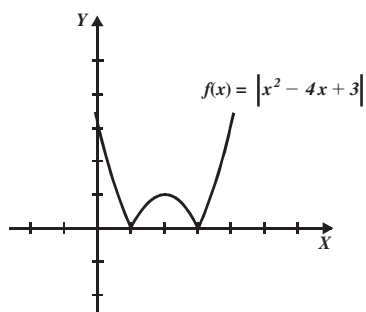
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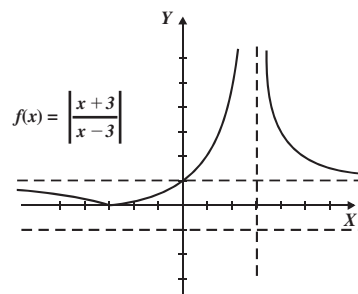
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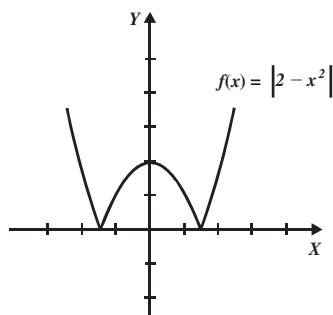
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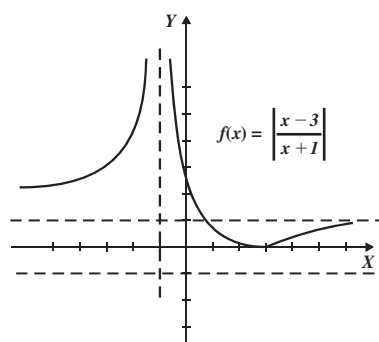
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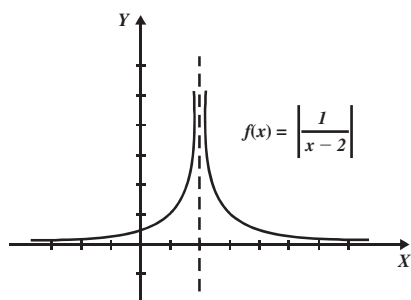
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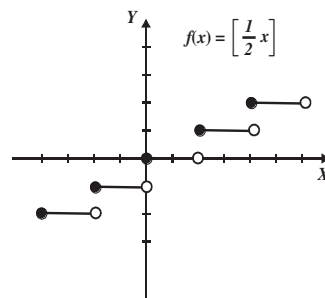
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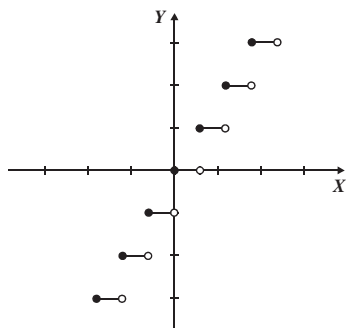
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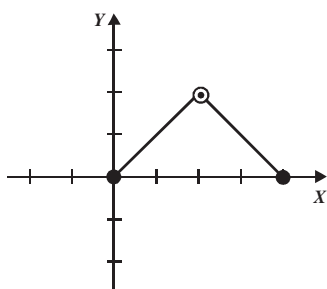
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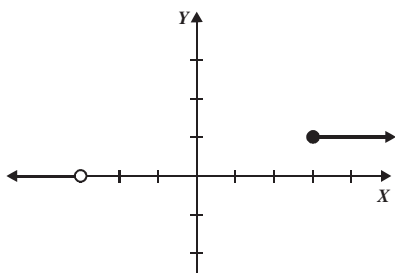
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**EJERCICIO 5**

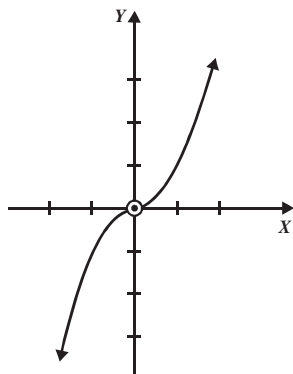
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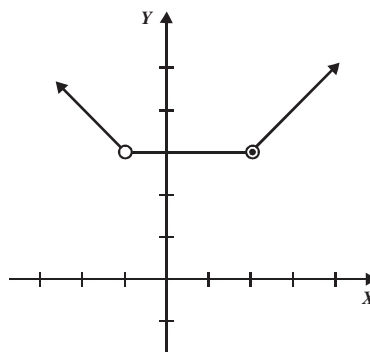
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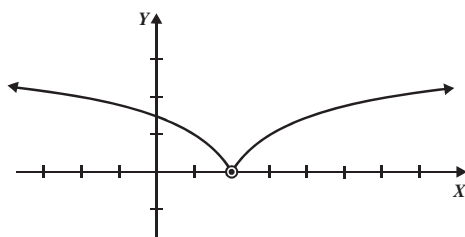
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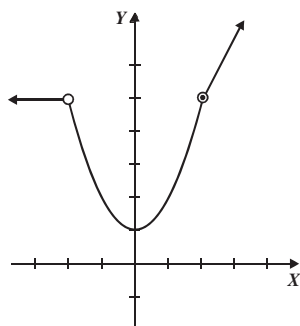
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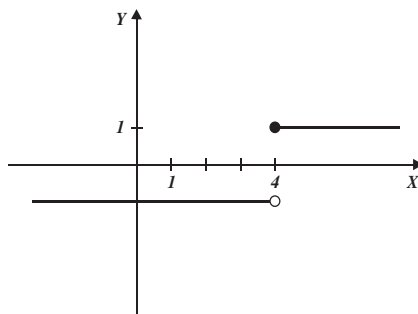
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6.

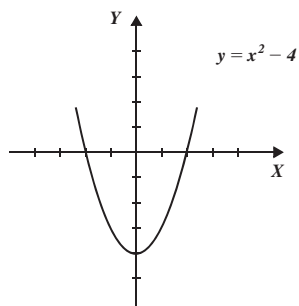


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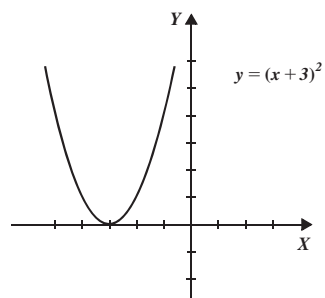


EJERCICIO 6

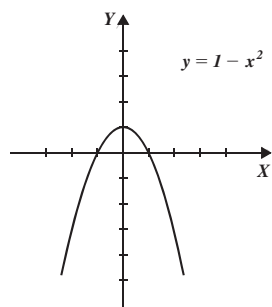
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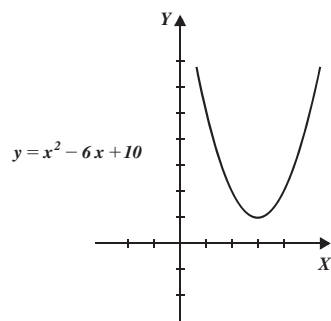
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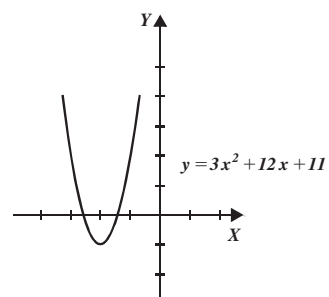
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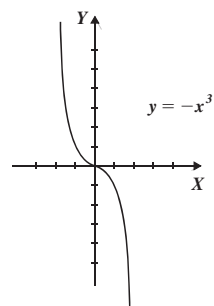
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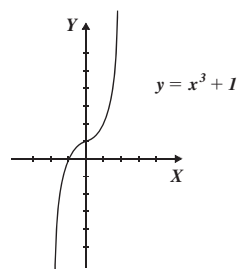
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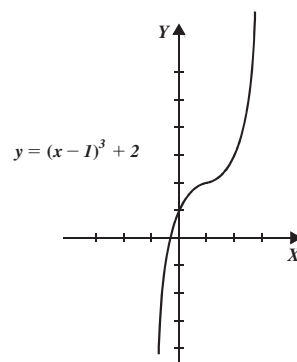
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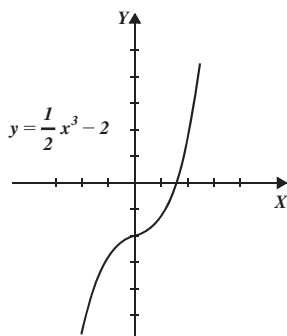
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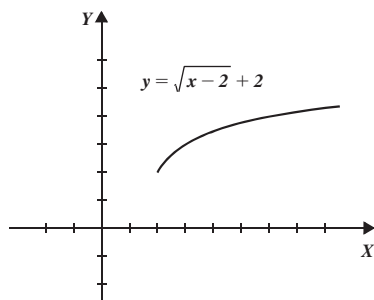
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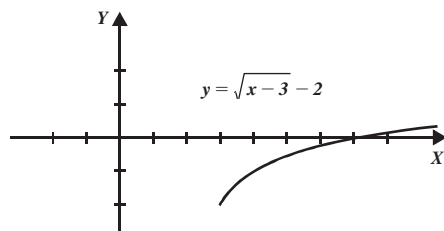
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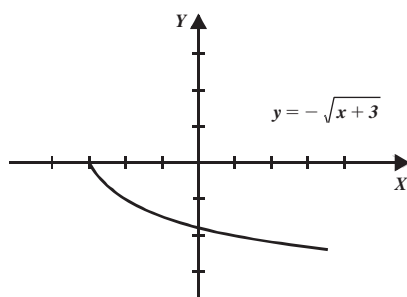
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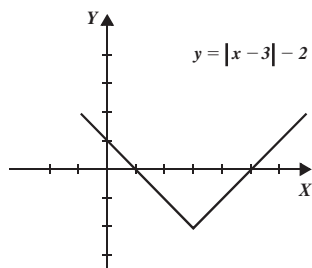
11.



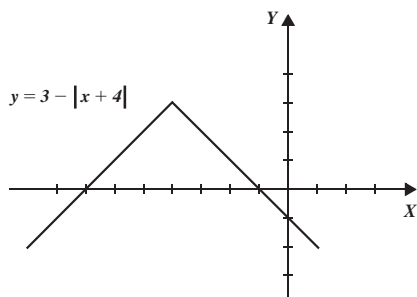
12.



13.



14.

**EJERCICIO 7**

1. Crece: $(0, \infty)$
2. Decrece: $(-\infty, 0)$
Crece: $(0, \infty)$
3. Crece: $(-\infty, +\infty)$
4. Decrece: $(-\infty, 0)$
Crece: $(0, \infty)$
5. Crece: $(2, \infty)$
6. Decrece: $(-\infty, -3)$
7. Decrece: $(0, \infty)$
Crece: $(-\infty, 0)$
8. Decrece: $(-\infty, 3)$
Crece: $(3, \infty)$
9. Decrece: $(0, 3)$
Crece: $(-3, 0)$
10. No crece ni decrece, permanece constante

EJERCICIO 8

- | | |
|--------------|-----------------|
| 1. Biyectiva | 6. Ninguna |
| 2. Ninguna | 7. Biyectiva |
| 3. Ninguna | 8. Inyectiva |
| 4. Biyectiva | 9. Suprayectiva |
| 5. Inyectiva | 10. Biyectiva |

EJERCICIO 9

1. $f(x) + g(x) = 3$
 $f(x) - g(x) = 7$
 $f(x) \cdot g(x) = -10$
 $\frac{f(x)}{g(x)} = -\frac{5}{2}$
2. $f(x) + g(x) = 4x$
 $f(x) - g(x) = -10$
 $f(x) \cdot g(x) = 4x^2 - 25$
 $\frac{f(x)}{g(x)} = \frac{2x-5}{2x+5}$

$$3. f(x) + g(x) = 2x^2 - x - 3$$

$$f(x) - g(x) = -7(x + 1)$$

$$f(x) \cdot g(x) = x^4 - x^3 - 15x^2 - 23x - 10$$

$$\frac{f(x)}{g(x)} = \frac{x-5}{x+2}$$

$$4. f(x) + g(x) = \frac{8x+1}{6}$$

$$f(x) - g(x) = \frac{4x-7}{6}$$

$$f(x) \cdot g(x) = \frac{2x^2 + 3x - 2}{6}$$

$$\frac{f(x)}{g(x)} = \frac{6x-3}{2x+4}$$

$$5. f(x) + g(x) = \sqrt{x-3} + \sqrt{x+4}$$

$$f(x) - g(x) = \sqrt{x-3} - \sqrt{x+4}$$

$$f(x) \cdot g(x) = \sqrt{x^2 + x - 12}$$

$$\frac{f(x)}{g(x)} = \frac{\sqrt{x-3}}{\sqrt{x+4}} = \frac{\sqrt{x^2 + x - 12}}{x+4}$$

$$6. f(x) + g(x) = x + 2\sqrt{x}$$

$$f(x) - g(x) = x$$

$$f(x) \cdot g(x) = x + x\sqrt{x}$$

$$\frac{f(x)}{g(x)} = \sqrt{x} + 1$$

$$7. f(x) + g(x) = \operatorname{sen}^2 x + \cos^2 x = 1$$

$$f(x) - g(x) = \operatorname{sen}^2 x - \cos^2 x = -\cos 2x$$

$$f(x) \cdot g(x) = \operatorname{sen}^2 x \cdot \cos^2 x = \frac{1}{4} \operatorname{sen}^2 2x$$

$$\frac{f(x)}{g(x)} = \frac{\operatorname{sen}^2 x}{\cos^2 x} = \tan^2 x$$

$$8. D_f \cap D_g = \{-1, 3, 5\}$$

$$f + g = \{(-1, 12), (3, 20), (5, 23)\}$$

$$f - g = \{(-1, -8), (3, -8), (5, -9)\}$$

$$f \cdot g = \{(-1, 20), (3, 84), (5, 112)\}$$

$$\frac{f}{g} = \left\{ \left(-1, \frac{1}{5}\right), \left(3, \frac{3}{7}\right), \left(5, \frac{7}{16}\right) \right\}$$

$$9. D_f \cap D_g = \{-2, -1, 0\}$$

$$f + g = \{(-2, 0), (-1, 1), (0, 2)\}$$

$$f - g = \{(-2, -10), (-1, -7), (0, -4)\}$$

$$f \cdot g = \{(-2, -25), (-1, -12), (0, -3)\}$$

$$\frac{f}{g} = \left\{ (-2, -1), \left(-1, -\frac{3}{4}\right), \left(0, -\frac{1}{3}\right) \right\}$$

$$10. D_f \cap D_g = \{-1, 1, 2\}$$

$$f + g = \{(-1, 1), (1, 1), (2, 1)\}$$

$$f - g = \{(-1, -3), (1, 1), (2, 0)\}$$

$$f \cdot g = \left\{ (-1, -2), (1, 0), \left(2, \frac{1}{4}\right) \right\}$$

$$\frac{f}{g} = \left\{ \left(-1, -\frac{1}{2}\right), (2, 1) \right\}$$

$$11. f(x) + r(x) = 2x + 5$$

$$12. f(x) - s(x) = -x^2 + 4x + 13$$

$$13. g(x) \cdot s(x) = x^4 + 2x^3 - 19x^2 - 68x - 60$$

$$14. \frac{g(x)}{r(x)} = x + 3$$

$$15. \frac{s(x)}{r(x)} = x - 5$$

$$16. g(x) - s(x) = 8(x + 2)$$

$$17. f(x) \cdot r(x) = x^2 + 5x + 6$$

$$18. \frac{f(x)}{r(x)} = \frac{x+3}{x+2}$$

$$19. \frac{g(x)}{s(x)} = \frac{x+3}{x-5}$$

$$20. \frac{g(x)}{f(x)} + \frac{s(x)}{r(x)} = 2x - 3$$

$$21. f(x) + g(x) = \frac{x^2 + 2}{x(x+2)}$$

$$22. \frac{f(x)}{g(x)} = \frac{x^2 - x}{x+2}$$

$$23. f(x) \cdot g(x) = \frac{x-1}{x^2 + 2x}$$

$$24. f(x) - h(x) = \frac{5-5x}{(x+2)(x-3)}$$

$$25. g(x) \cdot h(x) = \frac{x-1}{x^2 - 3x}$$

$$26. \frac{f(x)}{g(x)} + h(x) = \frac{x^3 - 3x^2 + 4x - 2}{(x+2)(x-3)}$$

$$27. \frac{h(x)}{f(x)} - g(x) = \frac{x^2 + x + 3}{x(x-3)}$$

$$28. \frac{h(2) - f(1)}{g(3)} = -3$$

$$29. f(x+1) \cdot \frac{1}{h(x+1)} = \frac{x-2}{x+3}$$

$$30. h(x) - g(x) = \frac{x^2 - 2x + 3}{x(x-3)}$$

$$31. \frac{h(x)}{g(x)} - \frac{g(x)}{f(x)} = \frac{x^4 - 2x^3 + x + 6}{x(x-1)(x-3)}$$

$$32. f(x) \cdot h(x) - g(x) = \frac{x^3 - 3x^2 + 2x + 6}{x(x+2)(x-3)}$$

$$33. \frac{f(x) + h(x)}{g(x)} = \frac{x(x-1)(2x-1)}{(x+2)(x-3)}$$

$$34. \frac{1}{g(x) + h(x)} = \frac{x(x-3)}{x^2 - 3}$$

$$35. \frac{1}{1-h(x)} = \frac{3-x}{2}$$

EJERCICIO 10

$$1. (f \circ g)(x) = 12x^2 - 46x + 40, (g \circ f)(x) = 6x^2 - 10x - 7,$$

$$(f \circ f)(x) = 27x^4 - 90x^3 + 24x^2 + 85x + 20,$$

$$(g \circ g)(x) = 4x - 9$$

$$2. (f \circ g)(x) = x, (g \circ f)(x) = x, (f \circ f)(x) = \sqrt[3]{x}, (g \circ g)(x) = x^4$$

$$3. (f \circ g)(x) = 4, (g \circ f)(x) = 2, (f \circ f)(x) = 4, (g \circ g)(x) = 2$$

$$4. (f \circ g)(x) = x, (g \circ f)(x) = x, (f \circ f)(x) = \sqrt{x^2 - 10}$$

$$(g \circ g)(x) = \sqrt{x^2 + 10}$$

$$5. (f \circ g)(x) = x + 2\sqrt{x-1}, (g \circ f)(x) = \sqrt{x^2 + 2x}$$

$$(f \circ f)(x) = (x^2 + 2x + 2)^2, (g \circ g)(x) = \sqrt{\sqrt{x-1} - 1}$$

$$6. (f \circ g)(x) = \frac{1-x}{1+3x}, (g \circ f)(x) = \frac{x+3}{x-1}$$

$$(f \circ f)(x) = -\frac{1}{2x+1}, (g \circ g)(x) = x$$

$$7. (f \circ g)(x) = \log(x-4), (g \circ f)(x) = \log(x-2) - 2$$

$$(f \circ f)(x) = \log[\log(x-2) - 2], (g \circ g)(x) = x - 4$$

$$8. (f \circ g)(x) = \frac{1}{\sqrt{x}}, (g \circ f)(x) = \sqrt{-x^2 - \sqrt{x^4 - 1}}$$

$$(f \circ f)(x) = \sqrt{-\frac{1}{x^2}} \text{ no está definida}$$

$$(g \circ g)(x) = \sqrt{x + \sqrt{x^2 - 1}}$$

$$9. (f \circ g)(x) = \{(1, 5), (2, 6), (3, 7), (4, 8)\}$$

$$(g \circ f)(x) = \text{No está definida}, (f \circ f)(x) = \{(2, 8)\}$$

$$(g \circ g)(x) = \{(1, 3), (2, 4), (3, 5)\}$$

$$10. (f \circ g)(x) = \{(-2, 1), (-1, 4), (0, 9), (1, 16)\}$$

$$(g \circ f)(x) = (1, 4), (f \circ f)(x) = \{(1, 1), (2, 16)\}$$

$$(g \circ g)(x) = \{(-2, 4)\}$$

$$11. (f \circ g)(x) = \{(3, 1), (-2, -3), (1, -1)\}$$

$$(g \circ f)(x) = \{(0, -1), (1, 0)\}, (f \circ f)(x) = \{(0, 3), (-1, -1)\}$$

$$(g \circ g)(x) = \{(-2, -2)\}$$

$$12. f(x) = \frac{1}{x}$$

$$13. f(x) = \sqrt{x}$$

$$14. f(x) = mx + b$$

$$15. f(x) = x^2$$

$$16. f(x) = \sqrt{x-2}$$

$$17. f \circ g \circ h = 81x^2 - 54x + 9$$

$$18. f \circ g \circ h = 1 - 12x^2 + 48x^4 + 64x^6$$

$$19. f \circ g \circ h = \sqrt{2x-9}$$

$$20. f \circ g \circ h = \sin^2(x-2)$$

$$21. f \circ g \circ h = \sec^2 x$$

$$22. f \circ g \circ h = \sin x$$

EJERCICIO 11

$$1. \text{ Ninguna}$$

$$6. \text{ Ninguna}$$

$$11. \text{ Par}$$

$$2. \text{ Par}$$

$$7. \text{ Par}$$

$$12. \text{ Par}$$

$$3. \text{ Impar}$$

$$8. \text{ Par}$$

$$13. \text{ Impar}$$

$$4. \text{ Ninguna}$$

$$9. \text{ Ninguna}$$

$$14. \text{ Par}$$

$$5. \text{ Ninguna}$$

$$10. \text{ Impar}$$

$$15. \text{ Par}$$

EJERCICIO 12

$$1. f^{-1}(x) = x$$

$$2. f^{-1}(x) = \frac{x+5}{2}$$

$$3. f^{-1}(x) = \sqrt{x+9}$$

$$4. \text{ No tiene inversa}$$

$$5. f^{-1}(x) = \sqrt[3]{x}$$

$$6. f^{-1}(x) = \sqrt[5]{x}$$

$$7. f^{-1}(x) = \sqrt[4]{x}$$

$$8. f^{-1}(x) = 3 - x^2$$

$$9. \text{ No tiene inversa}$$

$$10. f^{-1}(x) = \sqrt{4-x^2}$$

$$11. f^{-1}(x) = x^3 - 9$$

$$12. f^{-1}(x) = \frac{1-3x}{2x}$$

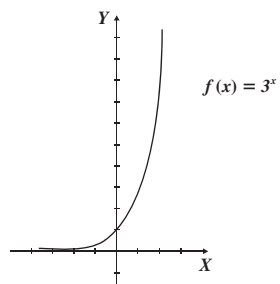
$$13. f^{-1}(x) = \sqrt{x^2 + 1}$$

$$14. f^{-1}(x) = \frac{1+x}{1-x}$$

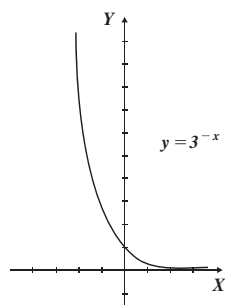
$$15. f^{-1}(x) = \frac{x^2}{x^2 - 1}$$

EJERCICIO 13

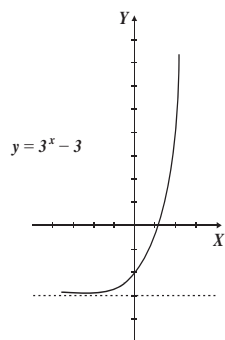
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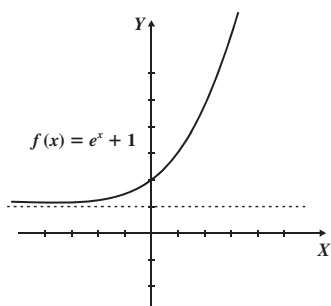
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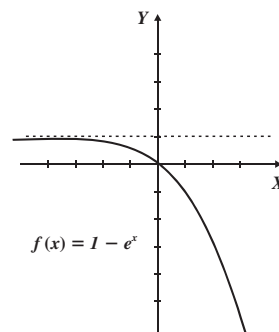
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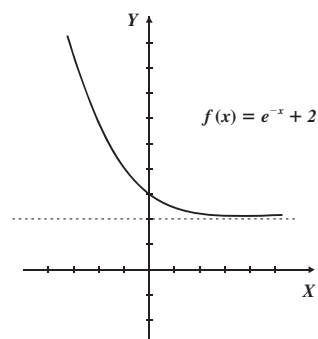
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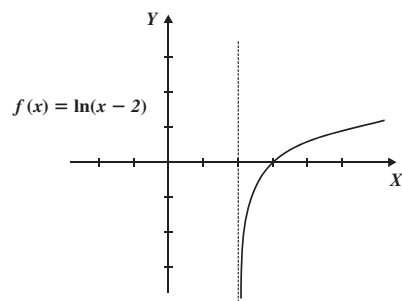
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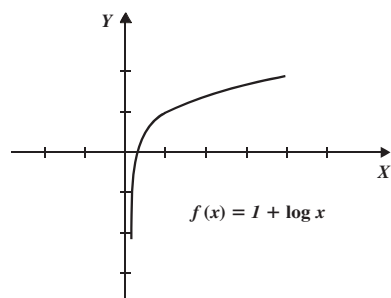
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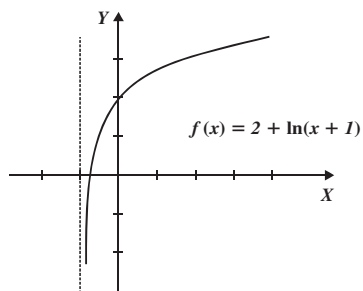
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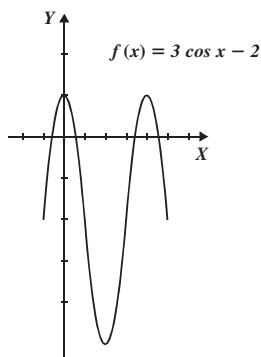
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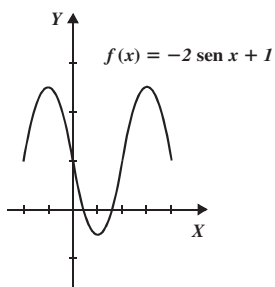
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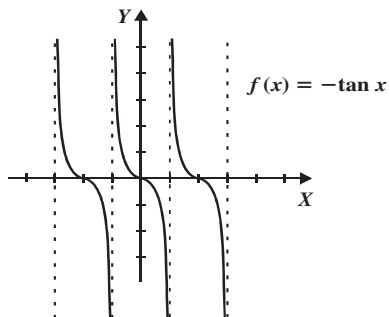
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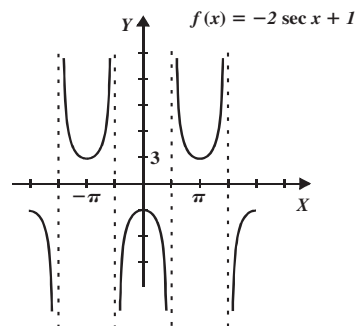
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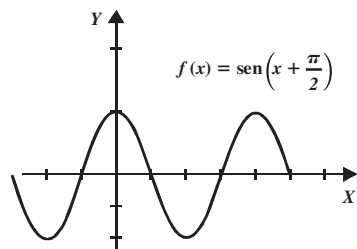
12.



13.



14.

**EJERCICIO 14**

1. $V(h) = 40h$
2. $V(h) = \frac{4}{75} \pi h^3$
3. $P(A) = \frac{12\sqrt{5A}}{5}$
4. $A(d) = \frac{\pi d^2}{4}$
5. $V(x) = \frac{\sqrt{3}}{2} \pi x^3$
6. $A(x) = \frac{3}{4} (x + 2)^2$
7. $V(x) = \frac{\pi r^2}{6} (8r + 15)$
8. $A(x) = \frac{\pi x^2}{3}$
9. $A(x) = 3x \sqrt{16 - x^2}$
10. $A(x) = (x - 4) \left(\frac{540}{x} - 3 \right)$
11. $d(t) = \frac{9}{2} \sqrt{16t^2 + 1}$
12. $d(t) = \frac{1}{2} \sqrt{t^2 - 16}$

CAPÍTULO 2

EJERCICIO 15

1. -1
2. $0.16666 = \frac{1}{6}$
3. -1
4. 0
5. 1
6. 4
7. No existe
8. No existe
9. 2
10. 3

EJERCICIO 16

11. $\delta = 0.01$
12. $\delta = 0.08$
13. $\delta = 0.025$
14. $\delta = 0.4$
15. $\varepsilon = 0.18$
16. $\varepsilon = 0.0098$
17. $\varepsilon = 0.25$
18. $\varepsilon = 0.002$

EJERCICIO 17

1. 3
2. 24
3. 18
4. 0
5. 7
6. 64
7. -3
8. $-\frac{4}{27}$
9. 15
10. $2\sqrt{3}$
11. 32
12. 1
13. 1
14. $-\frac{5}{8}$
15. 0
16. No existe
17. 1
18. $\frac{\sqrt{2}}{4}$
19. 1
20. h
21. $\frac{4\sqrt{3}}{3}$

EJERCICIO 18

1. $\frac{3}{5}$
2. No existe
3. 0
4. $\frac{a}{c}$
5. $-\frac{2}{3}$
6. $\frac{1}{2}$
7. 4
8. $\frac{1}{2h}$
9. $-\frac{1}{2}$
10. $2a$
11. 9
12. 0
13. $-\frac{1}{2}$
14. $\frac{5}{4}$
15. $\frac{1}{4}$
16. 2
17. 4
18. $-\frac{3}{7}$
19. 15
20. $\frac{9}{19}$

21. $\frac{7}{9}$

22. -6

23. $\frac{1}{3}$

24. -3

25. 3

26. $\frac{1}{12}$

27. 48

28. $\frac{1}{4}$

29. 2

30. No existe

31. $\frac{1}{2}$

32. $2\sqrt{5}$

33. $-\frac{1}{6}$

34. $\frac{24}{5}$

35. $\frac{b}{a}$

EJERCICIO 19

1. $\frac{7}{4}$

2. 2

3. 0

4. No existe

5. 3

6. $\frac{1}{2}$

7. $-\frac{1}{3}$

8. 0

9. 1

10. 0

36. $\frac{1}{n \cdot \sqrt[n]{p^{n-1}}}$

37. $-\frac{1}{2}$

38. $\frac{5}{12}$

39. 2

40. 0

41. $-\frac{3}{20}$

42. $\frac{1}{4}$

43. $\frac{1}{4}$

44. $\frac{1}{6a^3\sqrt{a^2}}$

45. $\frac{1}{3}$

46. $\frac{1}{8}$

47. $\frac{1}{2\sqrt[4]{x}}$

48. $\frac{1}{54}$

49. $\frac{1}{4}$

50. $\frac{2}{25}$

11. $-\frac{11}{6}$

12. -1

13. $\frac{9}{2}$

14. 1

15. 1

16. $\frac{a_m}{b_n}$ si $m = n$

0 si $m < n$

No existe si $m > n$

17. $\frac{a}{c}$

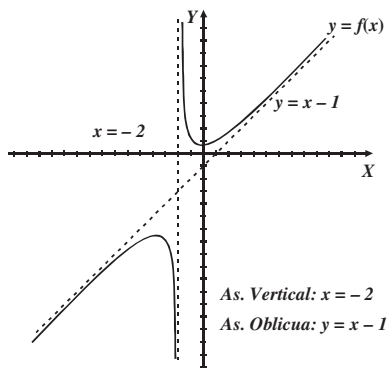
18. $\sqrt[n]{a}$

EJERCICIO 20

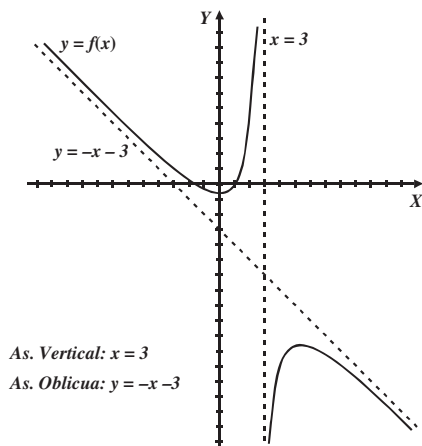
1. $y = \frac{1}{2}$
2. $y = 0$
3. No tiene asíntota horizontal
4. $y = 1, y = -1$
5. $y = 2$
6. $y = \frac{a}{c}$
7. $y = 0$
8. $y = -2$
9. No tiene asíntota horizontal
10. $y = \frac{a}{b}$

EJERCICIO 21

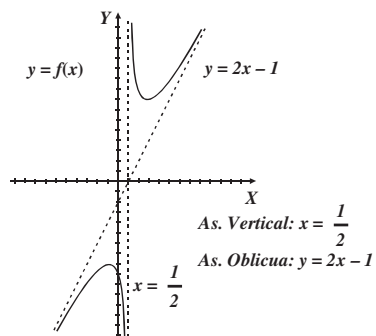
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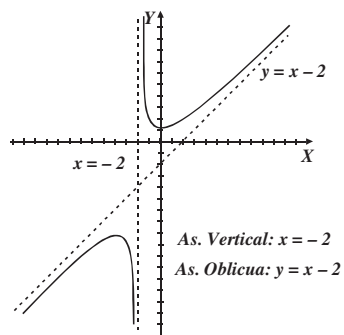
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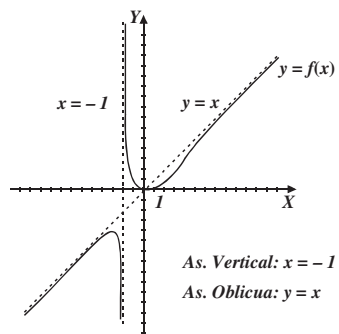
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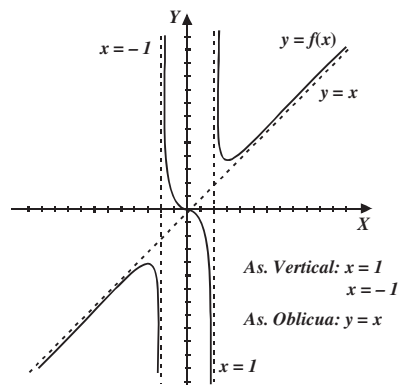
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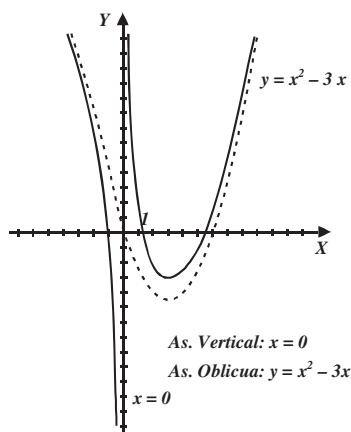
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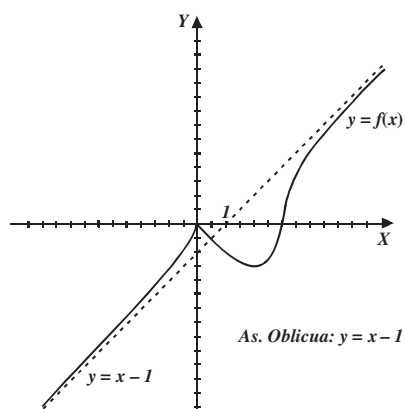
6.



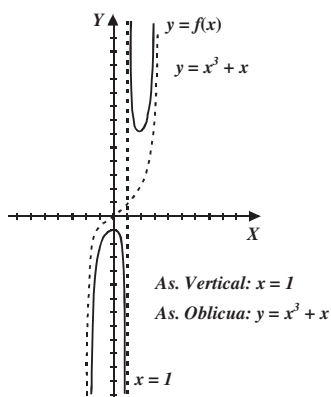
7.



8.



9.

**EJERCICIO 22**

1. a) 11, b) 9, c) No existe
2. a) -1, b) -1, c) -1, d) -6, e) -4, f) No existe
3. a) 0, b) 2, c) No existe, d) $-\frac{2}{3}$, e) $-\frac{2}{3}$, f) $-\frac{2}{3}$
4. a) 1, b) 4, c) 4, d) 4, e) 16
5. a) 4, b) 4, c) 4, d) 8, e) 3, f) No existe

6. -7

7. 1

8. 3

9. 2

10. No existe el límite

EJERCICIO 231. $\frac{1}{3}$

6. 2

2. $\frac{1+\sqrt{3}}{2}$ 7. $-4\sqrt{3}$

3. -2

8. 0

4. -1

9. 1

5. $-\frac{1}{2}$

10. No existe

EJERCICIO 24

1. -2

11. -2

2. $\frac{3}{4}$ 12. $\frac{1}{2^m}$ 3. $-\frac{1}{2}$

13. 1

4. 0

14. 0

5. $-\frac{1}{2}$ 15. $\frac{1}{2}$

6. 0

16. 0

7. 0

17. 9

8. $\sqrt{2}$

18. 0

9. -1

19. $\frac{n^2 - m^2}{4}$ 10. $-\sec^3(3)$

20. 0

CAPÍTULO 3**EJERCICIO 25**1. Es continua en $x = 0$ 2. No es continua en $x = 2$ 3. No es continua en $x = -\frac{3}{2}$ 4. Es continua en $x = 3$ 5. Continuidad removible en $x = 2$ 6. No es continua en $x = 2\pi$ 7. Es continua en $x = 2$ 8. No es continua en $x = 1$ 9. No es continua en $x = 0$

10. No es continua en $x = -2$; es continua en $x = 2$ 11. Es continua en $x = 1$; no es continua en $x = 2$ 12. Es continua en $x = \pi$ y $x = \frac{3}{2}\pi$ 13. No es continua en $x = -3$; es continua en $x = 3$ 14. Continuidad removible en $x = 3$ 15. Continuidad removible en $x = 1$ 16. Continuidad removible en $x = -2$ 17. Continuidad removible en $x = 8$ 18. No es continua en $x = \frac{1}{2}$ 19. $k = 1$ 20. $k = 0$ o $k = 2$ 21. $k = 1$ o $k = -\frac{2}{9}$ 22. $a = -\frac{9}{4}$, $b = -7$ 23. $a = -\frac{17}{2}$, $b = -2$ 24. $a = 4$, $b = 2$ **EJERCICIO 26**

1. sí

2. no

3. sí

4. no

5. sí

6. sí

7. sí

8. no

9. no

10. sí

EJERCICIO 27

1. 2

2. 0

3. 4

4. $\pm\sqrt{2}$, $\frac{1}{2}$

5. 5

6. -5

7. 5

8. $\frac{2}{5}$ 9. $\frac{\sqrt{2}}{2}$

10. 1

CAPÍTULO 4**EJERCICIO 28**1. $y' = 3$ 2. $y' = -b$ 3. $y' = 2x$ 4. $f'(x) = 6x - 5$ 5. $y' = 2ax + b$ 6. $y' = 3x^2$ 7. $y' = 3x^2 - 2x$ 8. $y' = 4$ 9. $y' = -\frac{2}{(x-1)^2}$ 10. $y' = 3x^2$

11. $f'(x) = -\frac{6}{x^3}$

12. $f'(x) = \frac{4x}{(x^2+1)^2}$

13. $f'(x) = \frac{1}{2\sqrt{x-2}}$

14. $f'(x) = \frac{x}{\sqrt{x^2-4}}$

15. $y' = \frac{2}{3\sqrt[3]{(2x+1)^2}}$

16. $y' = -\frac{1}{x\sqrt{x}}$

17. $y' = \frac{1}{3\sqrt[3]{x^2}}$

18. $y' = \frac{-2}{3(x-1)\sqrt[3]{x-1}}$

19. $y' = \frac{2}{(x-1)^{\frac{1}{2}}(x+3)^{\frac{3}{2}}}$

20. $y' = \frac{1}{n^q\sqrt[q]{x^{n-1}}}$

EJERCICIO 29

1. $y' = 0$

2. $y' = 0$

3. $f'(x) = 0$

4. $s'(t) = 0$

5. $y' = 6$

6. $y' = \frac{3}{4}$

7. $f'(x) = a$

8. $s'(t) = b^2$

9. $f'(x) = 5\sqrt{2}$

10. $y' = a\sqrt{b}$

11. $f'(x) = 5x^4$

12. $f'(x) = 12x^2$

13. $s'(t) = \frac{4}{5}t^3$

14. $y' = \frac{9}{2}x^{\frac{7}{2}}$

15. $f'(x) = \frac{4}{3}x^{\frac{1}{3}}$

16. $y' = 9x^{\frac{1}{2}}$

17. $f'(x) = \frac{2}{5\sqrt[5]{x^3}}$

18. $f'(x) = \frac{1}{\sqrt[4]{x^3}}$

19. $f'(x) = \frac{1}{2\sqrt{x}}$

20. $s'(t) = \frac{1}{4\sqrt[4]{t^3}}$

21. $f'(x) = \frac{1}{\sqrt[3]{x^4}}$

22. $f'(x) = \frac{5x^4}{7}$

23. $f'(x) = \frac{4x^3}{9}$

24. $s'(t) = \frac{3t^2}{a}$

25. $f'(x) = -\frac{20}{x^5}$

26. $f'(x) = -\frac{12}{x^7}$

27. $f'(x) = \frac{1}{4\sqrt{x}}$

28. $s'(t) = \frac{1}{15\sqrt[3]{t^2}}$

29. $f'(x) = -\frac{2}{x\sqrt{x}}$

30. $s'(t) = -\frac{5}{4t\sqrt[3]{t}}$

$$31. f'(x) = -\frac{4}{3x\sqrt[3]{x}}$$

$$32. f'(x) = 21x^2 - 6x + 3$$

$$33. f'(x) = 4x^3 - 15x^2 + 16x - 1$$

$$34. f'(x) = 10x + 4$$

$$35. f'(x) = 12ax^3 - 12ax^2 - 10bx + 7c$$

$$36. f'(x) = \frac{x^2}{2} - \frac{6x}{5} - \frac{4}{9}$$

$$37. s'(t) = \frac{5t^4}{6} - \frac{4t^3}{5} + \frac{3t^2}{4} - \frac{2t}{7} + \frac{1}{9}$$

$$38. f'(x) = \frac{2x}{\sqrt{a^2 + b^2}} - \frac{1}{a}$$

$$39. s'(t) = -\frac{8}{t^3} + \frac{5}{t^2}$$

$$40. f'(x) = -\frac{20}{x^5} + \frac{18}{x^4} + \frac{14}{x^3} + \frac{3}{x^2}$$

$$41. s(t) = \frac{3t^2}{5} + \frac{4}{t^3} - \frac{6}{t^2}$$

$$42. f'(x) = \frac{1}{15\sqrt[3]{x^2}} + \frac{1}{x\sqrt[3]{x}}$$

$$43. f'(x) = 3x^2 - 6x - 6 - \frac{2}{x^2}$$

$$44. f'(x) = 3\sqrt{x} + \frac{5}{4\sqrt{x}} + \frac{2}{x^2\sqrt{x}}$$

$$45. f'(x) = \frac{4}{\sqrt{x}} + \frac{6}{\sqrt[3]{x}} + 6\sqrt{x}$$

$$46. f'(x) = anx^{n-1} + b(n-1)x^{n-2}$$

$$47. f'(x) = \frac{2}{3}x + \frac{5}{7}$$

$$48. f'(x) = \frac{a}{n\sqrt[n]{x^{n-1}}} + \frac{b}{3\sqrt[3]{x^2}}$$

$$49. y' = \frac{1}{3\sqrt[3]{x}} + \frac{5\sqrt[4]{x}}{12}$$

$$50. f'(x) = -\frac{5}{2x^2\sqrt[3]{x}} + \frac{1}{2x\sqrt{x}} + 3$$

$$51. f'(x) = 14x + 15x^2$$

$$52. f'(x) = -\frac{6}{x^3} - \frac{5}{x^2} - 2$$

$$53. f'(x) = 5\sqrt[3]{x^2} + \frac{10}{3\sqrt[3]{x}} - \frac{8}{3x\sqrt[3]{x}}$$

$$54. y' = \frac{3}{x^2}$$

$$55. y' = 15(3x - 4)^4$$

$$56. y' = -12(2 - 4x)^2$$

$$57. y' = (72x^5 - 32x^3)(3x^6 - 2x^4)^3$$

$$58. y' = 12\sqrt{x}(2x - 1)^2(6x - 1)$$

$$59. y' = \frac{-3x}{\sqrt{5 - 3x^2}}$$

$$60. y' = \frac{x^2}{\sqrt[3]{(x^3 + 2)^2}}$$

$$61. y' = \frac{1 - x^2}{(x^2 + 1)^2}$$

$$62. y' = \frac{4x + 6}{3\sqrt{2x^2 + 6x}}$$

$$63. y' = \left(1 + \frac{9}{\sqrt{x}}\right)\left(\frac{x}{3} + 6\sqrt{x}\right)^2$$

$$64. f'(x) = \frac{x^3}{\sqrt[4]{(x^4 - 2)^3}}$$

$$65. f'(x) = (6x + 15)(x^2 + 5x - 3)^2$$

$$66. y' = \frac{4}{3\sqrt[3]{2x - 3}}$$

$$67. y' = \frac{1}{(4x + 3)^{\frac{3}{4}}}$$

$$68. f'(x) = \left(\frac{1}{3}x + 2\right)^2$$

$$69. y' = \left(\frac{2}{x} - \frac{1}{x^2}\right)^{-\frac{1}{2}}\left(-\frac{1}{x^2} + \frac{1}{x^3}\right)$$

$$70. f'(z) = \frac{z}{\sqrt{z^2 - 4}}$$

$$71. y' = \frac{2x^5 + 1}{\sqrt[3]{(x^6 + 3x)^2}}$$

$$72. y' = 108x^2 + 55x - 4$$

$$73. y' = 40x - 12 - \frac{9}{x^2}$$

$$74. y' = 12x^3 + 3x^2$$

$$75. f'(x) = \frac{3x + 1}{\sqrt{2x + 1}}$$

$$76. y' = \frac{1}{3}(2x+1)^2(8x+1)$$

$$77. y' = \frac{5x^2 - 4x}{2\sqrt{x-1}}$$

$$78. f'(x) = 12x(3x^2 - 5)^3(2x^2 + 1)^2(7x^2 - 3)$$

$$79. f'(\theta) = (6\theta^4 - 12\theta)(\theta^2 + 1)^2(2\theta^3 + \theta - 2)$$

$$80. s' = \frac{8-9t}{2\sqrt{4-3t}}$$

$$81. s'(t) = (2t+3)\left(\frac{2}{t} - \frac{3}{t^2}\right) = 4 - \frac{9}{t^2}$$

$$82. f'(x) = \frac{6}{(1-2x)^2}$$

$$83. f'(t) = \frac{-bt}{a\sqrt{a^2 - t^2}}$$

$$84. f'(r) = \frac{r^3 - 5r}{(r^2 - 4)^{\frac{3}{2}}}$$

$$85. f'(t) = \frac{63}{(5t+8)^2}$$

$$86. f'(z) = \frac{21}{(5-6z)^2}$$

$$87. f'(x) = -\frac{2ab}{(ax-b)^2}$$

$$88. f'(x) = -\frac{1}{x\sqrt{x}} - \frac{1}{2\sqrt{3x}}$$

$$89. f'(t) = \frac{-2}{(1+2t)\sqrt{1-4t^2}}$$

$$90. f'(w) = \frac{10(w-3)}{(w+2)^3}$$

$$91. f'(\theta) = \frac{24\theta^3 - 54\theta^2 + 24}{(3-2\theta)^2}$$

$$92. f'(s) = \frac{-6s^2 + 4s - 12}{(s^2 - 6s)^2}$$

$$93. f'(x) = \frac{10b^2x + 5x^3}{2(b^2 + x^2)^{\frac{3}{2}}}$$

$$94. f'(t) = \frac{(693-27t)(9t-6)^2}{(27-3t)^3}$$

$$95. f'(x) = \frac{2ab}{(a-3x)^2}$$

$$96. f'(x) = \frac{8-4x^2}{\sqrt{4-x^2}}$$

$$97. y' = \frac{-4x^3}{(x^4 - a^4)^{\frac{3}{2}}}$$

$$98. y' = \frac{7x^4 + 27x^2}{3(x^2 + 3)^{\frac{4}{3}}}$$

$$99. y' = \frac{8x^2 + 24x + 9}{2\sqrt{x^2 + 3x}}$$

$$100. y' = \frac{x+2}{2(x+1)^{\frac{3}{2}}}$$

$$101. y' = \frac{3x-3}{2\sqrt{x-3}}$$

$$102. y' = \frac{-2x^2}{(x^6 - 1)^{\frac{2}{3}}(x^3 - 1)^{\frac{2}{3}}}$$

$$103. y' = \frac{nx^{n-1}}{(1-x^n)\sqrt{x^{2n}-1}}$$

$$104. y' = \frac{-n\sqrt[n]{x^{n-m}}}{m\left(\sqrt[n]{x^n}-1\right)^2}$$

$$105. y' = \frac{-20x^2 + 19x + 8}{2(4-5x)^{\frac{3}{2}}\sqrt{2x+1}}$$

$$106. y' = \frac{8x^6 - 8x^3 - 2}{\sqrt[3]{(4x^6 - 1)^2(2x^3 - 1)^2}}$$

EJERCICIO 30

$$1. \frac{dy}{dx} = \frac{1-2u}{x^2}$$

$$2. \frac{dy}{dx} = \frac{1}{2\sqrt{x}\sqrt{u^2-1}(1+u)}$$

$$3. \frac{dy}{dx} = \frac{x(6u^2-3)}{\sqrt{2u^3-3u}}$$

$$4. \frac{dy}{dx} = \frac{4}{u^3} - \frac{9}{u^4}$$

$$5. \frac{dy}{dx} = \frac{(8+12x-3x^2)(u^2+1)}{(u^2-1)^2}$$

$$6. \frac{dy}{dx} = \frac{5u^3+3u}{\sqrt{u^3+u}}$$

$$7. \frac{dy}{dx} = \frac{3x^2}{\sqrt{u}(x^3+1)^2}$$

$$8. \frac{dy}{dx} = \frac{8x}{(u-1)^2(v-2)^2\sqrt{x^2-1}}$$

$$9. \frac{dy}{dx} = -\frac{v}{2\sqrt{x}\sqrt{u-1}(v^2-1)^2}$$

$$10. \frac{dy}{dx} = -\frac{2x(x^2+3)}{(v+1)\sqrt{v^2-1}\sqrt{u^3}}$$

$$11. \frac{dy}{dx} = -\frac{2u}{\sqrt{v}(x-1)^2}$$

EJERCICIO 31

$$1. y' = 8 \cos 8x$$

$$2. f'(x) = -6x \operatorname{sen} 3x^2$$

$$3. f'(x) = 3x^2 \sec^2 x^3$$

$$4. s'(t) = 6 \sec 6t \tan 6t$$

$$5. f'(x) = -12x^2 \csc^2 4x^3$$

$$6. f'(x) = -9 \csc 9x \cot 9x$$

$$7. f'(x) = -a \operatorname{sen} ax$$

$$8. s'(t) = 2bt \sec^2 bt^2$$

$$9. f'(x) = 12x \sec x^2 \tan x^2$$

$$10. f'(x) = -\frac{1}{8} \csc \frac{x}{4} \cot \frac{x}{4}$$

$$11. f'(x) = -3a \operatorname{sen} 3x$$

$$12. f'(x) = -3 \csc^2(3x-5)$$

$$13. f'(x) = \cos \frac{x}{2}$$

$$14. f'(x) = -5 \operatorname{sen} \left(5x - \frac{\pi}{2} \right)$$

$$15. s'(t) = a \sec^2(at + \pi)$$

$$16. f'(x) = \cos x - \operatorname{sen} x$$

$$17. s'(t) = \frac{1}{2\sqrt{t}} \cos \sqrt{t}$$

$$18. f'(x) = -\frac{1}{3\sqrt[3]{x^2}} \csc^2 \sqrt[3]{x}$$

$$19. f'(x) = -\frac{1}{x^2} \cos \frac{1}{x}$$

$$20. s'(t) = \frac{3}{t^4} \operatorname{sen} \frac{1}{t^3}$$

$$21. f'(x) = -\frac{\sec \frac{1}{\sqrt{x}} \tan \frac{1}{\sqrt{x}}}{2x\sqrt{x}}$$

$$22. f'(x) = 3 \sec^2 3x - 3 = 3 \tan^2 3x$$

$$23. f'(x) = a - a \csc^2 ax = -a \cot^2 ax$$

$$24. f'(x) = 2(x-1)\cos(x-1)^2$$

$$25. f'(x) = -18t(3t^2+2)^2 \operatorname{sen}(3t^2+2)^3$$

$$26. f'(x) = -\frac{2}{\sqrt{x-1}} \csc^2 \sqrt{x-1}$$

$$27. f'(x) = -\frac{2}{(x-1)^2} \sec^2 \left(\frac{x+1}{x-1} \right)$$

$$28. f'(x) = -\frac{2ab}{(ax-b)^2} \sec \left(\frac{ax+b}{ax-b} \right) \tan \left(\frac{ax+b}{ax-b} \right)$$

$$29. f'(x) = 10 \operatorname{sen} 5x \cos 5x = 5 \operatorname{sen} 10x$$

$$30. f'(x) = -3b \cos^2 bx \operatorname{sen} bx$$

$$31. f'(x) = 24x \tan^3 3x^2 \sec^2 3x^2$$

$$32. f'(x) = \frac{2 \cos 4x}{\sqrt{\operatorname{sen} 4x}} = 2 \sqrt{\operatorname{sen} 4x} \cot 4x$$

$$33. f'(x) = 5x \tan 5x^2 \sqrt{\sec 5x^2}$$

$$34. f'(x) = \frac{2x \sec^2 x^2}{\sqrt[3]{9 \tan^2 x^2}}$$

$$35. f'(x) = x \cos x + \operatorname{sen} x$$

$$36. f'(x) = 2x \cos x^2 - 2x^3 \operatorname{sen} x^2$$

$$37. f'(x) = \frac{3x \cos 3x - \operatorname{sen} 3x}{x^2}$$

$$38. f'(x) = -\frac{10t^2 \operatorname{sen} 5t^2 + 2 \cos 5t^2}{t^3}$$

$$39. y' = 2ax \cos(ax^2)$$

$$40. y' = -3a \operatorname{sen}(3x)$$

$$41. y' = \frac{\sec^2 \sqrt{x}}{2\sqrt{x}}$$

$$42. y' = x \sec 3x^2 \tan 3x^2$$

$$43. y' = -\frac{1}{3} \csc \frac{2x}{3} \cot \frac{2x}{3}$$

$$44. y' = 2x + 3 + \frac{1}{x^2} \cos \left(\frac{1}{x} \right)$$

$$45. y' = -6x \csc^2(1-x^2)$$

$$46. y' = \frac{-4}{3(x-1)^2} \cdot \cos \left(\frac{x+1}{x-1} \right)$$

$$47. y' = 2b \operatorname{sen} 4bx$$

$$48. y' = 24(2x-1)^2 \tan^3(2x-1)^3 \sec^2(2x-1)^3$$

$$49. y' = \frac{\operatorname{sen} 2x}{\sqrt{\cos^3 2x}}$$

$$50. y' = \frac{2x \sec^2 x^2}{\sqrt[3]{9 \tan^2 x^2}}$$

$$51. y' = \cos 4x(\cos^2 4x - 6x \operatorname{sen} 8x)$$

$$52. y' = x \csc ax(2 - ax \cot ax)$$

$$53. y' = (1 - x \cot 2x) \sqrt{\csc 2x}$$

$$54. y' = \frac{-m \operatorname{sen} nx \operatorname{sen} mx - n \cos nx \cos mx}{\operatorname{sen}^2 nx}$$

$$55. y' = \frac{-\cos x}{(1 + \operatorname{sen} x)^2}$$

$$56. y' = -x \operatorname{sen} x$$

$$57. y' = \frac{\sec^2 x (\tan x - 1)}{\sqrt{(\tan^2 x - 1)^3}}$$

$$58. y' = (2x^2 - 6) \cos 2x + 10x \operatorname{sen} 2x$$

$$59. y' = -2 \cos(2x - 1)$$

$$60. y' = x \sec(\pi - x) \cdot [2 - x \tan(\pi - x)]$$

$$61. y' = \frac{81x^2 \operatorname{sen}^2 x [3x^2 \cos x + x \cos x + \operatorname{sen} x]}{(3x + 1)^4}$$

$$62. y' = \frac{1}{(x-1)\sqrt{x^2-1}} \operatorname{sen} \sqrt{\frac{x+1}{x-1}}$$

$$63. y' = \left(\frac{\sec x}{x} \right)^2 (x \operatorname{sen} x - \cos x)$$

$$64. y' = -x \operatorname{sen} x + \cos x$$

$$65. y' = 2 \cos 2x$$

$$66. y' = 2 \sec^2 x \tan x$$

$$67. y' = -\frac{1}{2} \operatorname{sen} \frac{x}{2}$$

$$68. y' = -6 \operatorname{sen}(6x + 2)$$

$$69. y' = -\frac{x \operatorname{sen} x + 2 \cos x}{x^3}$$

$$70. y' = (\operatorname{sen} x + \cos x)(\tan^2 x - \tan x + 2)$$

$$71. y' = -\cos^3 x$$

$$72. y' = x^2 \operatorname{sen} x$$

$$73. y' = \operatorname{sen}^4 2x$$

EJERCICIO 32

$$1. y' = \frac{5}{\sqrt{1-25x^2}}$$

$$2. f'(x) = \frac{-8x}{\sqrt{1-16x^4}}$$

$$3. f'(x) = \frac{3}{1+9x^2}$$

$$4. y' = -\frac{3x^2}{1+x^6}$$

$$5. f'(x) = \frac{2}{x\sqrt{x^4-1}}$$

$$6. f'(x) = -\frac{2}{x\sqrt{9x^4-1}}$$

$$7. f'(x) = -\frac{1}{\sqrt{b^2-x^2}}$$

$$8. f'(x) = \frac{1}{\sqrt{16-x^2}}$$

$$9. f'(x) = \frac{a}{a^2+x^2}$$

$$10. f'(x) = \frac{1}{x\sqrt{x-1}}$$

$$11. y' = -\frac{2x}{\sqrt{-8+6x^2-x^4}}$$

$$12. y' = \frac{1}{\sqrt{1-x^2}}$$

$$13. y' = \frac{x^2}{x^2+1} + 2x \operatorname{arc} \tan x$$

$$14. y' = \operatorname{arc} \operatorname{sen} x$$

$$15. y' = \frac{x^2}{\sqrt{16-x^2}}$$

$$16. y' = \operatorname{arc} \csc\left(\frac{1}{x}\right) + \sqrt{\frac{1+x}{1-x}}$$

$$17. y' = x \operatorname{arc} \tan x$$

$$18. \varphi' = \frac{\theta}{(1-\theta^2)\sqrt{\theta^2-2}}$$

$$19. y' = \sqrt{1-4x^2}$$

$$20. y' = x^2 \operatorname{arc} \operatorname{sen} x$$

$$21. f'(r) = \sqrt{\frac{b-r}{b+r}}$$

$$22. y' = \frac{x^2}{x^2+1}$$

$$23. y' = \frac{3+6x^2}{(1+4x^2)(1+x^2)}$$

$$24. y' = \frac{1}{2\sqrt{x-x^2}}$$

$$25. y' = \frac{1}{\sqrt{x^2-1}} + \operatorname{arc} \cos\left(\frac{1}{x}\right)$$

$$26. y' = \frac{4a-8x}{\sqrt{1-(4ax-4x^2)^2}}$$

$$27. f'(r) = \frac{1}{\sqrt{-r^2 + 4r - 3}}$$

$$28. y' = \frac{1}{4x^2 + 4x + 5}$$

$$29. y' = \frac{x+2}{\sqrt{4x-x^2}}$$

$$30. y' = \frac{x^2}{\sqrt{4x-x^2}}$$

$$31. y' = \sqrt{2x-x^2}$$

$$32. s'(t) = \frac{2-3t}{\sqrt{9-t^2}}$$

$$33. y' = \sqrt{25-9x^2}$$

$$34. w' = \frac{\theta+5}{(\theta+4)\sqrt{\theta+2}}$$

$$35. y' = \frac{1}{5+4\cos x}$$

$$36. y' = \frac{\cos x}{5-3\cos x}$$

$$37. y' = -\frac{1}{3}$$

$$38. y' = \arccot(\tan x) - x$$

$$39. y' = \frac{1}{(4x^2-1)} \left(\frac{1}{x} - \frac{4x \arccsc 2x}{\sqrt{4x^2-1}} \right)$$

$$40. y' = \frac{\sqrt{7-8\cos x + \cos^2 x}}{14-2\cos x}$$

$$41. y' = \frac{64}{(3x+2)\sqrt{5x^2+28x-12}}$$

$$42. s' = \frac{t^2}{\sqrt{2t-t^2}} + 2t \arccos(1-t) + 2$$

$$43. y' = -\frac{1}{\sqrt{1-(a+x)^2}}$$

EJERCICIO 33

$$1. y' = \frac{3}{x}$$

$$2. f'(x) = \frac{2}{x}$$

$$3. f'(x) = \frac{6x-5}{3x^2-5x+2}$$

$$4. f'(x) = \frac{1}{2x}$$

$$5. f'(x) = \frac{6 \log e}{x}$$

$$6. f'(x) = \frac{3 \log e}{x}$$

$$7. f'(x) = \frac{\log_3 e}{x}$$

$$8. f'(x) = \frac{\log_4 e}{3x}$$

$$9. f'(x) = \frac{4 \ln^3 x}{x}$$

$$10. f'(x) = \frac{3 \ln^2 5x}{x}$$

$$11. y' = x(1 + \ln x^2)$$

$$12. y' = 2(1 + \ln x)$$

$$13. y' = \frac{1 - \ln x}{x^2}$$

$$14. f'(x) = \frac{2 - \ln x^2}{x^2}$$

$$15. y' = -\frac{a}{2(b-ax)} = \frac{a}{2(ax-b)}$$

$$16. f'(x) = \frac{9x^2-2}{x(3x^2-1)}$$

$$17. f'(x) = \frac{2ax^2-b}{x(ax^2-b)}$$

$$18. y' = \frac{13}{(3x-5)(2x+1)}$$

$$19. f'(x) = \frac{bc}{c^2 x^2 - b^2}$$

$$20. y' = \cot x$$

$$21. y' = -5 \tan 5x$$

$$22. y' = \frac{2x}{x^2-4}$$

$$23. y' = \frac{3}{2(3x+4)}$$

$$24. y' = \frac{12}{4x^2-9}$$

$$25. y' = \frac{x^2}{x^3+8}$$

$$26. y' = \frac{\ln x}{2x}$$

$$27. y' = \frac{27x^2 + 12x + 6}{(3x + 2)(3x^2 + 2)}$$

$$28. y' = \frac{2 \log_3 e}{4x^2 - 1}$$

$$29. y' = \frac{(30bx^2\sqrt{x} - 3)\log e}{10bx^3\sqrt{x} - 6x}$$

$$30. y' = \tan x$$

$$31. y' = 2 \cot x$$

$$32. y' = 1 + \ln x$$

$$33. y' = -2 \tan 2x$$

$$34. y' = \frac{1}{2x\sqrt{\ln x}}$$

$$35. y' = \sec x$$

$$36. y' = -\frac{1 + \operatorname{sen} 2x}{\cos 2x}$$

$$37. y' = \frac{x + \tan x}{x \tan x}$$

$$38. y' = 2x^2(1 + \ln x^3)$$

$$39. y' = \frac{3 \sec^2 \sqrt{x}}{2\sqrt{x} \tan x} = \frac{3 \sec \sqrt{x} \csc \sqrt{x}}{2\sqrt{x}}$$

$$40. y' = \frac{\log e}{2x}$$

$$41. y' = 2^{x^2+5x} \cdot \ln 2 \cdot (2x + 5)$$

$$42. f'(x) = \frac{b^{\sqrt{x}} \ln b}{2\sqrt{x}}$$

$$43. y' = \frac{3^{\ln x} \ln 3}{x}$$

$$44. y' = (x \cos x + \operatorname{sen} x) \cdot 5^{x \operatorname{sen} x} \cdot \ln 5$$

$$45. y' = 2^{\ln x}(1 + \ln 2)$$

$$46. y' = 5^x(\ln 5^x + 1)$$

$$47. y' = 2xe^{x^2} = 2xy$$

$$48. y' = (6x - 2)e^{3x^2-2x+1} = (6x - 2)y$$

$$49. y' = \frac{3xe^{\sqrt{3x^2-1}}}{\sqrt{3x^2-1}} = \frac{3xy}{\sqrt{3x^2-1}}$$

$$50. y' = (x \sec^2 x + \tan x)y$$

$$51. y' = e^{\frac{2x}{b}} + e^{-\frac{2x}{b}}$$

$$52. y' = \frac{8}{(e^{2x} + e^{-2x})^2}$$

$$53. f'(x) = 4e^{4x}$$

$$54. f'(x) = 10xe^{5x^2}$$

$$55. f'(x) = 3e^{3x-1}$$

$$56. f'(x) = \frac{1}{5}e^{\frac{x}{5}}$$

$$57. f'(x) = \frac{1}{3}\sqrt[3]{e^x}$$

$$58. f'(x) = \frac{1}{4}\sqrt[4]{e^x}$$

$$59. f'(x) = -\frac{2}{x^3}e^{\frac{1}{x^2}}$$

$$60. f'(x) = \frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

$$61. f'(\theta) = \operatorname{sen} 2\theta \cdot e^{\operatorname{sen}^2 \theta}$$

$$62. f(x) = -2 \operatorname{sen} 2x \cdot e^{\cos 2x}$$

$$63. y' = (\operatorname{sen} x + \cos x)e^{x+e^x \operatorname{sen} x}$$

$$64. f'(x) = (3 \ln 5)5^{3x}$$

$$65. f'(x) = (2 \ln 7)7^{2x}$$

$$66. f'(x) = (2x \ln 5)5^{x^2}$$

$$67. y' = 2x^{2x}(1 + \ln x)$$

$$68. y' = x^{\cos x-1}(\cos x - x \ln x^{\operatorname{sen} x})$$

$$69. y' = \frac{\sqrt[3]{x}}{x^2}(1 - \ln x)$$

$$70. y' = \frac{e^{\arctan x}}{1+x^2} = \frac{y}{1+x^2}$$

$$71. y' = \frac{1+2x}{2x}$$

$$72. y' = 3e^{\ln x^2} = 3x^2$$

$$73. y' = \frac{xe^x}{(x+1)^2}$$

$$74. y' = \frac{e^x(x \ln x + \ln x - 1)}{2 \ln^2 x}$$

$$75. y' = \frac{-1}{x(\ln x - 1)\sqrt{\ln^2 x - 1}}$$

$$76. y' = \frac{e^{\operatorname{sen} x} \cos x}{(1 - e^{\operatorname{sen} x})\sqrt{e^{2 \operatorname{sen} x} - 1}}$$

$$77. y' = \frac{2a \cot ax}{e^y}$$

$$78. y' = \frac{e^{\ln \sqrt{e^x \operatorname{sen} x}}(1 + \cot x)}{2} = \frac{y(1 + \cot x)}{2}$$

79. $y' = xe^{\sin x}(x \cos x + 2)$

80. $y' = \cot x$

81. $y' = \frac{3x^2 - 8}{x(x^2 - 4)}$

82. $y' = \frac{x+3}{\sqrt{x^2+9}}$

83. $y' = \sec^3 2x$

84. $y' = \arctan x$

85. $y' = \sqrt{x^2 - 4}$

86. $y' = \arcsin x$

87. $y' = \frac{1}{4x^2 - 9}$

88. $y' = \arccot x$

89. $y' = \operatorname{arccsc} \frac{x}{2}$

EJERCICIO 34

1. $y' = -\frac{x}{y}$

2. $y' = -\frac{y}{x}$

3. $y' = \frac{4}{y}$

4. $y' = -\frac{2x+5}{4y-2}$

5. $y' = -\frac{3x+y}{x-6y}$

6. $y' = \frac{x+1}{1-y}$

7. $y' = \frac{(x-y)^2 + 2y}{2x}$

8. $y' = \frac{b^2 x}{a^2 y}$

9. $y' = -\frac{y}{x}$

10. $y' = \frac{2y^2 + 3x^2 y + 10xy^2}{3y^2 - 4xy - x^3 - 10x^2 y + 1}$

11. $y' = \frac{4xy - 9x^2 - 5y - 3}{5x - 2x^2 - 1}$

12. $y' = \frac{2\sqrt{x+y} - y}{2x+3y}$

13. $y' = \frac{2y\sqrt{x+y} - 1}{1 - 2x\sqrt{x+y}}$

14. $y' = \frac{2-4x-3y}{3x+3}$

15. $y' = (1-4\sqrt{x})\sqrt{\frac{y}{x}}$

16. $y' = \frac{y}{x(2y-1)}$

17. $y' = -\frac{y}{x}$

18. $y' = \frac{\tan e^y}{e^y}$

19. $y' = 3e^{x-y}$

20. $y' = \frac{2xy}{x^2+1}$

21. $y' = \frac{1-x+y}{1-y+x}$

22. $y' = \frac{x(1-e^{x^2})}{y(e^{y^2}-1)}$

23. $y' = -x$

24. $y' = -\frac{y}{x \ln x}$

25. $y' = \frac{y}{x^2 + y^2 + x}$

26. $y' = \frac{y}{x} \left(\frac{x \ln y - y}{y \ln x - x} \right)$

27. $y' = \frac{1}{x \ln x}$

28. $y' = \frac{e^x(1+e^y)}{e^y} = (e^{-y} + 1) \ln(1 + e^y)$

29. $y' = \frac{y(x - \ln y)}{x \ln x}$

30. $y' = \frac{e^x}{1+x e^y} = \frac{e^y}{1-y}$

31. $y' = \frac{y^2}{e^{\ln y} - xy} = \frac{y}{1-x}$

32. $y' = \frac{1}{e^{x+y}[\cos(e^{x+y}) - 1]} - 1$

33. $y' = \frac{e^{x \cos y} \cos y - 3}{x e^{x \cos y} \sin y} = \frac{1}{x} \left(\frac{1}{\tan y} - \frac{1}{x \sin y} \right)$

$$34. y' = -\frac{\cos(x+a)}{\operatorname{sen}(y-b)}$$

$$35. y' = \frac{\operatorname{sen} x}{\cos y - 1} = -\frac{\operatorname{sen} x(\csc y + \cot y)}{\operatorname{sen} y}$$

$$36. y' = \frac{\operatorname{sen}(4x) \cos(2x)}{\operatorname{sen}(2y) \cos(2y)} = \frac{\operatorname{sen}(8x)}{\operatorname{sen}(8y)}$$

$$37. y' = -\frac{\operatorname{sen} x e^{\cos x}}{\cos y(1 + e^{\operatorname{sen} y})}$$

$$38. y' = \frac{2 - y \cos(xy)}{x \cos(xy)} = \frac{2 \sec(xy) - y}{x}$$

$$39. y' = -\frac{\cos x}{\operatorname{sen} y}$$

$$40. y' = \frac{\operatorname{sen} x}{\operatorname{sen} y \cdot e^{\cos y}}$$

$$41. y' = \frac{\cos x - \operatorname{sen} y - 1}{x \cos y}$$

$$42. y' = \left(\frac{y^2 + 1}{y^2 + 1 - x} \right) \arctan y$$

$$43. y' = \frac{\cot(x+y)}{1 - \cot(x+y)}$$

$$44. y' = \frac{1}{2^y \ln 2} = \frac{1}{\ln 2^x + \ln 8}$$

$$45. y' = -\frac{y}{e^{\operatorname{sen} y} \cos y + x - 2}$$

$$46. y' = \frac{y}{x} \left(\frac{x \ln y - y}{y \ln x - x} \right) = \frac{y^2}{x^2} \left(\frac{1 - \ln x}{1 - \ln y} \right)$$

$$47. y' = \frac{\operatorname{sen}(x+y) + \cos(x+y)}{1 - \operatorname{sen}(x+y) - \cos(x+y)}$$

$$48. y' = \frac{\tan^2(xy) + y^2 \sec^2(xy)}{\tan(xy) - xy \sec^2(xy)}$$

$$49. y' = \frac{x^2 + y + 1}{(x^2 + 1) \arccot x}$$

$$50. y' = \frac{y \cdot e^x}{\sqrt{1 - e^{2x}} (\operatorname{sen} y + \arccos(e^x))}$$

EJERCICIO 35

$$1. \frac{d^4 y}{dx^4} = 24$$

$$2. \frac{d^3 y}{dx^3} = 0$$

$$3. \frac{d^2 y}{dx^2} = -\frac{170}{(5x+3)^3}$$

$$4. \frac{d^2 y}{dx^2} = \frac{4ab^2}{(ax-b)^2}$$

$$5. \frac{d^3 y}{dx^3} = 24a^3(ax+b)$$

$$6. \frac{d^4 y}{dx^4} = \operatorname{sen} x + \cos x$$

$$7. y'' = -\csc^2 x$$

$$8. y''' = -\frac{72}{(x-1)^5}$$

$$9. y'' = e^x \sec^2 e^x (2e^x \tan e^x + 1) = e^x \sec^2 e^x (2e^x y + 1)$$

$$10. \frac{d^2 y}{dx^2} = \frac{18y-6}{(2-3x)^2}$$

$$11. \frac{d^2 y}{dx^2} = -\frac{9}{(9-x^2)^{\frac{3}{2}}}$$

$$12. \frac{d^2 y}{dx^2} = -\frac{x^2 + y^2}{y^3} = -\frac{16}{y^3}$$

$$13. \frac{d^4 y}{dx^4} = \frac{2}{x^3}$$

$$14. \frac{d^2 y}{dx^2} = -\frac{\operatorname{sen} x \operatorname{sen}^2 y + \cos^2 x \cos y}{\operatorname{sen}^3 y} = -\csc y \left(\operatorname{sen} x + \frac{\cos^2 x \cos y}{\operatorname{sen}^2 y} \right)$$

$$15. \frac{d^3 y}{dx^3} = -x^2 \cos x - 6x \operatorname{sen} x + 6 \cos x$$

$$16. \frac{d^3 y}{dx^3} = \frac{12}{(x+1)^4}; \frac{d^n y}{dx^n} = \frac{2 \cdot n! \cdot (-1)^{n+1}}{(x+1)^{n+1}}$$

$$17. y'' = \frac{2y}{(x+1)^2}$$

$$18. \frac{d^3 y}{dx^3} = -2 \sec^2 x \tan x$$

$$19. \frac{d^2 y}{dx^2} = \frac{2 \cos^2 x}{(1 + \operatorname{sen} x)^3} + \frac{\operatorname{sen} x}{(1 + \operatorname{sen} x)^2} = \frac{2 - \operatorname{sen} x}{(1 + \operatorname{sen} x)^2}$$

$$20. y'' = \frac{-12}{(x+2y)^3}; y''' = \frac{-108x}{(x+2y)^5}$$

EJERCICIO 36

1. $\frac{dy}{dx} = \frac{2 \operatorname{sen} 2\theta + 3 \cos 2\theta}{2 \cos 2\theta - 3 \operatorname{sen} 2\theta}$
2. $\frac{dy}{dx} = 0$
3. $\frac{dy}{dx} = \frac{5 \operatorname{sen} \theta \cos 5\theta + \operatorname{sen} 5\theta \cos \theta}{5 \cos \theta \cos 5\theta - \operatorname{sen} 5\theta \operatorname{sen} \theta}$
4. $\frac{dy}{dx} = \frac{\cos 2\theta \operatorname{sen} \theta + \operatorname{sen} 2\theta \cos \theta}{\cos 2\theta \cos \theta - \operatorname{sen} 2\theta \operatorname{sen} \theta}$
5. $\frac{dy}{dx} = -\frac{\cos 2\theta + \cos \theta}{\operatorname{sen} 2\theta + \operatorname{sen} \theta} = -\cot\left(\frac{3\theta}{2}\right)$
6. $\frac{dy}{dx} = \csc \theta - \cot \theta$
7. $\frac{dy}{dx} = \frac{a \operatorname{sen} \theta + \cos \theta}{a \cos \theta - \operatorname{sen} \theta}$
8. $\frac{dy}{dx} = \frac{\operatorname{sen} \theta \tan \frac{\theta}{2} + \cos \theta}{\cos \theta \tan \frac{\theta}{2} - \operatorname{sen} \theta}$
9. $\frac{dy}{dx} = \frac{3 \cos \theta - 2 \cos 2\theta}{2 \operatorname{sen} 2\theta - 3 \operatorname{sen} \theta}$
10. $\frac{dy}{dx} = \frac{\operatorname{sen} \theta + 2\theta \cos \theta}{\cos \theta - 2\theta \operatorname{sen} \theta}$
11. -1
12. $2 + \sqrt{3}$
13. 3
14. $-a$
15. $-\frac{1}{4}$

EJERCICIO 37

1. $\frac{dy}{dx} = 4t\sqrt{t}$
2. $\frac{dy}{dx} = \frac{1}{36t^2}$
3. $\frac{dy}{dx} = \frac{2\sqrt{t^2 - t}}{2t - 1}$
4. $\frac{dy}{dx} = \frac{a}{b} \csc \theta$
5. $\frac{dy}{dx} = \frac{2t^2 - 3t^3}{2\sqrt{t-1}}$
6. $\frac{dy}{dx} = \frac{\cos \theta + 4 \operatorname{sen} \theta}{5 \cos \theta + \operatorname{sen} \theta}$
7. $\frac{dy}{dx} = \frac{(t-1)^2(1-2t-t^2)}{(t^2+1)}$
8. $\frac{dy}{dx} = \csc \frac{\theta}{2} (4 - 8 \cos^2 \theta)$
9. $\frac{dy}{dx} = -\frac{4}{5t^4}$
10. $\frac{dy}{dx} = -\frac{1}{2} \cos \theta \cot^2 \theta$

11. $\sqrt{3}$
12. $\frac{1}{4m^2}$
13. $-\frac{1}{3}$
14. $-\frac{1}{3}$
15. 1
16. $-\frac{1}{10}$

CAPÍTULO 5**EJERCICIO 38**

1. Sub-tangente = 1
Sub-normal = 1
Tangente = $\sqrt{2}$
Normal = $-\sqrt{2}$
2. Sub-tangente = $-\frac{12}{7}$
Sub-normal = -84
Tangente = $-\frac{60}{7}\sqrt{2}$
Normal = $-60\sqrt{2}$
3. Sub-tangente = $-\frac{12}{7}$
Sub-normal = -84
Tangente = $-\frac{60}{7}\sqrt{2}$
Normal = $60\sqrt{2}$
4. Sub-tangente = $\frac{7}{4}$
Sub-normal = 28
Tangente = $\frac{7\sqrt{17}}{4}$
Normal = $7\sqrt{17}$
5. Sub-tangente = $-\frac{3}{2}$
Sub-normal = -6
Tangente = $-\frac{3}{2}\sqrt{5}$
Normal = $3\sqrt{5}$
6. Sub-tangente = -18
Sub-normal = $-\frac{1}{2}$
Tangente = $-3\sqrt{37}$
Normal = $\frac{\sqrt{37}}{2}$

7. Sub-tangente = 8

Sub-normal = $\frac{1}{2}$

Tangente = $2\sqrt{17}$

Normal = $\frac{\sqrt{17}}{2}$

8. Sub-tangente = -2

Sub-normal = $-\frac{1}{8}$

Tangente = $-\frac{\sqrt{17}}{2}$

Normal = $\frac{\sqrt{17}}{8}$

9. Sub-tangente = $-\frac{3}{2}$

Sub-normal = -6

Tangente = $-\frac{3\sqrt{5}}{2}$

Normal = $-3\sqrt{5}$

10. Sub-tangente = $-\frac{3}{4}$

Sub-normal = $-\frac{16}{27}$

Tangente = $-\frac{\sqrt{145}}{12}$

Normal = $\frac{2\sqrt{145}}{27}$

11. T: $4x + y - 1 = 0$

N: $x - 4y + 38 = 0$

12. T: $x + y - 1 = 0$

N: $x - y + 1 = 0$

13. T: $y = 0$

N: $2x - 1 = 0$

14. T: $8x - y - 12 = 0$

N: $x + 8y - 34 = 0$

15. T: $4x + y + 8 = 0$

N: $x - 4y - 15 = 0$

16. T: $\sqrt{5}x + 2y - 9 = 0$

N: $2x - \sqrt{5}y = 0$

17. T: $x + 2y - 7 = 0$

N: $2x - y - 4 = 0$

18. T: $2x + y - 2 = 0$

N: $x - 2y + 4 = 0$

19. T: $y - 1 = 0$

N: $2x - \pi = 0$

20. T: $3\sqrt{3}x + 6y - (3 + \sqrt{3}\pi) = 0$

N: $12x - 6\sqrt{3}y + (3\sqrt{3} - 4\pi) = 0$

21. T: $4x - 2y + (6 - \pi) = 0$

N: $4x + 8y - (24 + \pi) = 0$

22. T: $2x - y - 6 = 0$

N: $x + 2y - 8 = 0$

23. T: $x + y - 2 = 0$

N: $x - y = 0$

24. T: $6x + 2y - 9 = 0$

N: $2x - 6y + 7 = 0$

25. T: $x - 2 = 0$

N: $2y - 1 = 0$

26. T: $7x - y - 8 = 0$

N: $x + 7y - 94 = 0$

27. T: $x - ey = 0$

N: $ex + y - 1 - e^2 = 0$

28. T: $8x + y - 7 = 0$

N: $2x - 16y + 47 = 0$

EJERCICIO 39

1. Agudo $26^\circ 33'$, obtuso $153^\circ 27'$

2. Agudo $73^\circ 42'$, obtuso $106^\circ 18'$

3. Agudo $78^\circ 41'$, obtuso $101^\circ 19'$

4. Agudo $35^\circ 15'$, obtuso $144^\circ 44'$

5. Agudo $28^\circ 23'$, obtuso $151^\circ 36'$

6. Agudo $28^\circ 4'$, obtuso $151^\circ 55'$

7. Agudo $71^\circ 33'$, obtuso $104^\circ 28'$

8. $125^\circ 32'$

9. $\theta = 63^\circ 26'$

10. $\theta = 18^\circ 26'$

11. $\theta = 6^\circ 54', 57^\circ 25'$

12. $\theta = 33^\circ 41'$

13. $\theta = 54^\circ 44'$

EJERCICIO 40

1. $r = \frac{5\sqrt{5}}{3}, \frac{d\theta}{ds} = \frac{3\sqrt{5}}{25}$

2. $r = \frac{17\sqrt{17}}{8}, \frac{d\theta}{ds} = \frac{8\sqrt{17}}{289}$

3. $r = 4\sqrt{2}, \frac{d\theta}{ds} = \frac{\sqrt{2}}{8}$

4. $r = \frac{5\sqrt{10}}{3}, \frac{d\theta}{ds} = \frac{3\sqrt{10}}{50}$

5. $r = 1, \frac{d\theta}{ds} = 1$

$$6. \quad r = \frac{17\sqrt{17}}{2}, \frac{d\theta}{ds} = \frac{2\sqrt{17}}{289}$$

$$7. \quad r = \frac{1}{2}, \frac{d\theta}{ds} = 2$$

$$8. \quad r = \frac{\sqrt{11}}{4}, \frac{d\theta}{ds} = \frac{4\sqrt{11}}{11}$$

$$9. \quad C(-2, 5)$$

$$10. \quad C\left(-\frac{3}{4}, -\frac{3}{2}\right)$$

$$11. \quad C\left(\frac{\pi}{2}, 0\right)$$

$$12. \quad C(-2, 3)$$

$$13. \quad C\left(\frac{23}{2}, -32\right)$$

EJERCICIO 41

$$1. \quad \text{Punto m\u00ednimo } (3, -4)$$

Creciente en $(3, \infty)$

Decreciente en $(-\infty, 3)$

$$2. \quad \text{Punto m\u00e1ximo } \left(\frac{5}{6}, -\frac{23}{12}\right)$$

Creciente en $\left(-\infty, \frac{5}{6}\right)$

Decreciente en $\left(\frac{5}{6}, \infty\right)$

$$3. \quad \text{Punto m\u00e1ximo } (-1, 2)$$

Punto m\u00ednimo $(1, -2)$

Creciente en $(-\infty, -1) \cup (1, \infty)$

Decreciente en $(-1, 1)$

$$4. \quad \text{Punto m\u00e1ximo } (0, 0)$$

Punto m\u00ednimo $(4, -32)$

Creciente en $(-\infty, 0) \cup (4, \infty)$

Decreciente en $(0, 4)$

$$5. \quad \text{Punto m\u00e1ximo } (-1, 5)$$

Punto m\u00ednimo $\left(\frac{1}{2}, -\frac{7}{4}\right)$

Creciente en $(-\infty, -1) \cup \left(\frac{1}{2}, \infty\right)$

Decreciente en $\left(-1, \frac{1}{2}\right)$

$$6. \quad \text{Punto m\u00e1ximo } \left(-\frac{1}{2}, \frac{17}{4}\right)$$

Punto m\u00ednimo $\left(\frac{2}{3}, \frac{29}{27}\right)$

Creciente en $\left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{2}{3}, \infty\right)$

Decreciente en $\left(-\frac{1}{2}, \frac{2}{3}\right)$

$$7. \quad \text{Punto m\u00e1ximo } (2, 15)$$

Punto m\u00ednimo $(-1, -12)$

Creciente en $(-1, 2)$

Decreciente en $(-\infty, -1) \cup (2, \infty)$

$$8. \quad \text{Punto m\u00e1ximo } \left(-1, \frac{8}{3}\right)$$

Punto m\u00ednimo $(3, -8)$

Creciente en $(-\infty, -1) \cup (3, \infty)$

Decreciente en $(-1, 3)$

$$9. \quad \text{Punto m\u00e1ximo } \left(-2, \frac{34}{3}\right)$$

Punto m\u00ednimo $\left(3, -\frac{19}{2}\right)$

Creciente en $(-\infty, -2) \cup (3, \infty)$

Decreciente en $(-2, 3)$

$$10. \quad \text{Punto m\u00e1ximo } \left(1, \frac{17}{12}\right)$$

Punto m\u00ednimo $(0, 1) \left(3, -\frac{5}{4}\right)$

Creciente en $(0, 1) \cup (3, \infty)$

Decreciente en $(-\infty, 0) \cup (1, 3)$

$$11. \quad \text{Punto m\u00e1ximo } (1, -3)$$

Creciente $(-\infty, 0) \cup (0, 1)$

Decreciente $(1, 2) \cup (2, \infty)$

$$12. \quad \text{No tiene m\u00e1ximos y m\u00ednimos}$$

Decreciente en $(-\infty, 3) \cup (3, \infty)$

$$13. \quad \text{Punto m\u00e1ximo } \left(2, \frac{1}{2}\right)$$

Punto m\u00ednimo $\left(-2, -\frac{1}{2}\right)$

Creciente en $(-2, 2)$

Decreciente en $(-\infty, -2) \cup (2, \infty)$

$$14. \quad \text{Punto m\u00ednimo } \left(0, -\frac{1}{4}\right)$$

Creciente en $(0, 2) \cup (2, \infty)$

Decreciente en $(-\infty, -2) \cup (-2, 0)$

$$15. \quad \text{Punto m\u00e1ximo } (-6, -12)$$

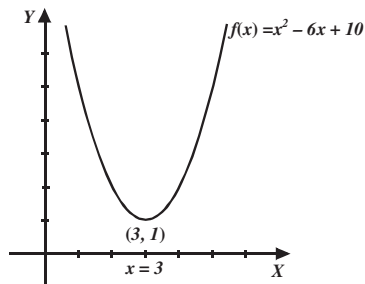
Punto m\u00ednimo $(0, 0)$

Creciente en $(-\infty, -6) \cup (0, \infty)$

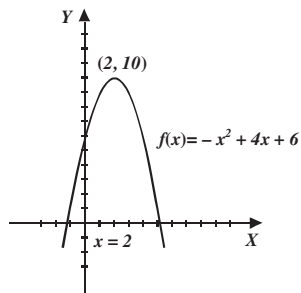
Decreciente en $(-6, -3) \cup (-3, 0)$

EJERCICIO 42

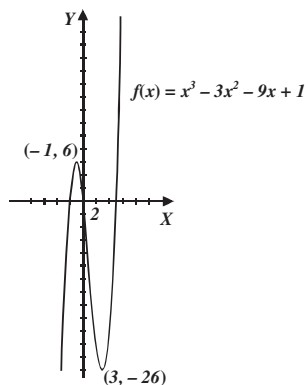
1.

Punto mínimo $(3, 1)$ Crece $(3, \infty)$ Decrece $(-\infty, 3)$ Concavidad hacia arriba $(-\infty, \infty)$

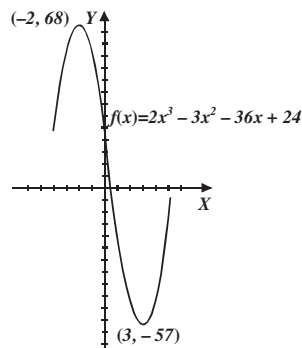
2.

Punto máximo $(2, 10)$ Crece $(-\infty, 2)$ Decrece $(2, \infty)$ Concavidad hacia abajo $(-\infty, \infty)$

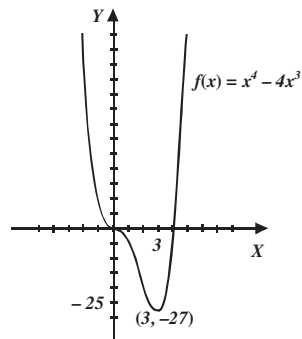
3.

Punto máximo $(-1, 6)$, Punto mínimo $(3, -26)$ Crece $(-\infty, -1) \cup (3, \infty)$ Decrece $(-1, 3)$ Concavidad hacia abajo $(-\infty, 1)$ Concavidad hacia arriba $(1, \infty)$ Punto de inflexión $(1, -10)$

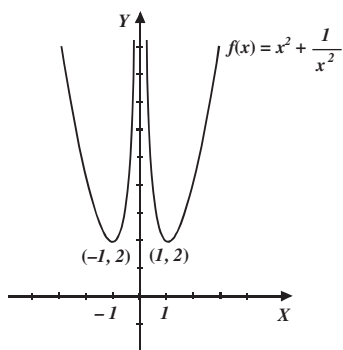
4.

Punto máximo $(-2, 68)$ Punto mínimo $(3, -57)$ Crece $(-\infty, -2) \cup (3, \infty)$ Decrece $(-2, 3)$ Concavidad hacia abajo $(-\infty, \frac{1}{2})$ Concavidad hacia arriba $(\frac{1}{2}, \infty)$ Punto de inflexión $(\frac{1}{2}, \frac{11}{2})$

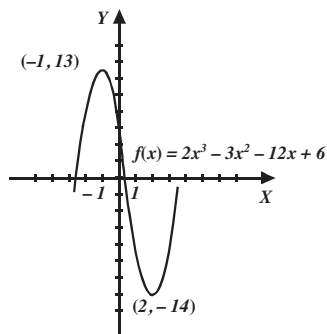
5.

Punto mínimo $(3, -27)$ Puntos de inflexión $(0, 0)$, $(2, -16)$ Crece $(3, \infty)$ Decrece $(-\infty, 0) \cup (0, 3)$ Concavidad hacia abajo $(0, 2)$ Concavidad hacia arriba $(-\infty, 0) \cup (2, \infty)$

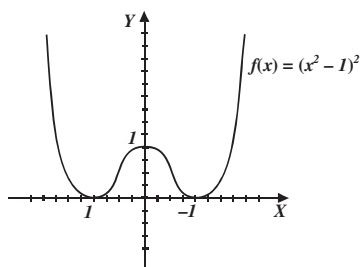
6.

Puntos mínimos $(-1, 2), (1, 2)$ Crece $(-1, 0) \cup (1, \infty)$ Decrece $(-\infty, -1) \cup (0, 1)$ Concavidad hacia arriba $(-\infty, 0) \cup (0, \infty)$

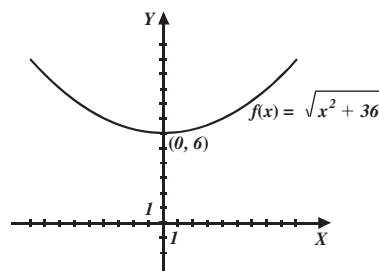
7.

Punto máximo $(-1, 13)$ Punto mínimo $(2, -14)$ Crece $(-\infty, -1) \cup (2, \infty)$ Decrece $(-1, 2)$ Concavidad hacia abajo $(-\infty, \frac{1}{2})$ Concavidad hacia arriba $(\frac{1}{2}, \infty)$, Punto de inflexión $(\frac{1}{2}, -\frac{1}{2})$

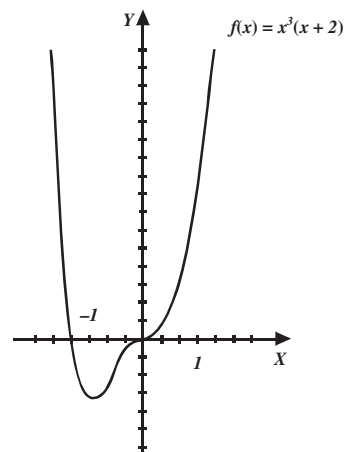
8.

Punto máximo $(0, 1)$ Puntos mínimos $(-1, 0), (1, 0)$ Crece $(-1, 0) \cup (1, \infty)$ Decrece $(-\infty, -1) \cup (0, 1)$ Concavidad hacia abajo $(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$ Concavidad hacia arriba $(-\infty, -\frac{1}{\sqrt{3}}) \cup (\frac{1}{\sqrt{3}}, \infty)$ Punto de inflexión $(-\frac{1}{\sqrt{3}}, \frac{4}{9}), (\frac{1}{\sqrt{3}}, \frac{4}{9})$

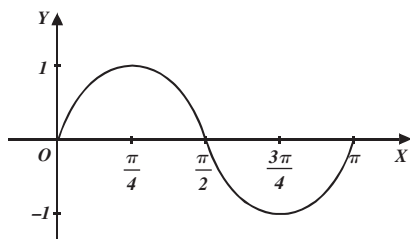
9.

Punto mínimo $(0, 6)$ Crece $(0, \infty)$ Decrece $(-\infty, 0)$ Concavidad hacia arriba $(-\infty, \infty)$

10.

Punto mínimo $(-\frac{3}{2}, -\frac{27}{16})$ Puntos de inflexión $(0, 0), (-1, -1)$ Crece $(-\frac{3}{2}, 0) \cup (0, \infty)$ Decrece $(-\infty, -\frac{3}{2})$ Concavidad hacia abajo $(-1, 0)$ Concavidad hacia arriba $(-\infty, -1) \cup (0, \infty)$

11.

Punto mínimo $\left(\frac{3\pi}{4}, -1\right)$ Punto máximo $\left(\frac{\pi}{4}, 1\right)$ Punto de inflexión $\left(\frac{\pi}{2}, 0\right)$ Decrece $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ Crece $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{3\pi}{4}, \pi\right)$ Concavidad hacia arriba $\left(\frac{\pi}{2}, \pi\right)$ Concavidad hacia abajo $\left(0, \frac{\pi}{2}\right)$ **EJERCICIO 43**

1. 20 y 20
2. -25 y 25
3. 2 pulgadas por lado y el volumen de $128 \ln^3$
4. $V = 6144\pi \text{ cm}^3$
5. $V = \frac{500\sqrt{6}\pi}{9\pi} \text{ in}^3$
6. $r = \sqrt[4]{\frac{100}{3}}$
 $h = \sqrt[4]{\frac{400}{3}}$
7. $A = 54 \text{ cm}^2$
8. $A = 24u^2$
9. Número = 1
10. Base = $\sqrt{3} r$
altura = $\frac{3}{2} r$
11. (1, 1), (-1, -1)
12. $A = \frac{32\sqrt{3}}{9} u^2$
13. Base = $2\sqrt{2}$
altura = $\sqrt{2}$
14. $2\sqrt{6}$ y $2\sqrt{3}$ ft
15. $d = 2\sqrt{5}$
16. 8 y 8
17. $2\sqrt{5}$ y $\sqrt{5}$ unidades
18. $A = 6u^2$
19. $h = 2 \text{ cm}$
20. Cada lado mide $2u$
21. 20 y 20
22. $4\sqrt{2}$ y 8

23. $P\left(\frac{3}{2}, \frac{1}{2}\right)$

24. $A = 15u^2$

25. $A = 2ab u^2$

26. $4x + 3y - 24 = 0$

27. Radio $4\sqrt{2}$
altura = $8\sqrt{2}$ pulgadas

28. $\left(\frac{1}{2}\sqrt{4a^2 + k^2}, \frac{k}{2}\right)$

29. Números 4 y 4

30. $\frac{2P}{4 + \pi}; \frac{P}{4 + \pi}$

31. 400 m, 800 m

32. $\frac{100\sqrt{3}\pi}{9 + \sqrt{3}\pi} \text{ cm}, \frac{900}{9 + \sqrt{3}\pi} \text{ cm}$

33. $\frac{2000\sqrt{3}\pi}{27} \text{ cm}^3$

34. $25\sqrt{2} \text{ cm} \times 25\sqrt{2} \text{ cm}$

35. $r = \frac{5}{\sqrt{\pi}} \text{ in} = \frac{10}{\sqrt{\pi}} \text{ in}$

36. $5\sqrt{5} \text{ m}$

37. $r = \sqrt[3]{\frac{3V}{5\pi}}$

38. $P(2, 0)$

$$P\left(-\frac{\sqrt{6} + 2}{2}, \sqrt{6} - \frac{3}{2}\right)$$

EJERCICIO 44

1. $6 \frac{\text{m}}{\text{s}}, \frac{17 \text{ m}}{2 \text{ s}}, 10 \frac{\text{m}}{\text{s}}$
2. $0 < t < 4$ y $6 < t$
3. a) $s = 22$ si $t = 2, s = 18$ si $t = 4$
 $a = -6$ si $t = 2, a = 6$ si $t = 4$
b) $s = 20, v = -3$ si $t = 3$
c) "s" crece cuando $0 < t < 2$ o $t > 4$
d) "v" crece cuando $0 < t < 2$ o $t > 4$
4. a) $t = 18 \text{ s}, v = -54 \frac{\text{m}}{\text{s}}$
b) $t = 9 \text{ s}, s = 243 \text{ m}$
5. $v = -2 \frac{\text{m}}{\text{s}}, s = 35 \text{ m}$

EJERCICIO 45

1. $\frac{5}{2}\sqrt{10} \text{ cm}$
2. $\frac{3}{5}\sqrt{449} \frac{\text{m}}{\text{min}}$
3. $\frac{360 \text{ m}}{\sqrt{17} \text{ s}}$
4. $\frac{4}{75}\sqrt{\frac{5}{\pi}} \text{ m}^3$
5. $-\frac{25 \text{ m}}{12 \text{ s}}$
6. $10\pi \frac{\text{cm}^2}{\text{s}}$
7. $-\frac{3 \text{ m}}{49\pi \text{ min}}$
8. $-\frac{405 \text{ km}}{8\sqrt{14} \text{ h}}$

9. a) $100 \frac{\text{km}}{\text{h}}$
 b) $\frac{820 \text{ km}}{\sqrt{73} \text{ h}}$
10. $-\frac{4 \text{ m}}{9\pi \text{ min}}$
11. 90.58 km/h
12. $4.8 \frac{\text{m}}{\text{s}}$
13. $\frac{7 \text{ pies}}{4\pi \text{ min}}$
14. $\frac{\sqrt[4]{27} \text{ cm}}{5 \text{ min}}$
15. $\frac{14 \text{ u}}{3 \text{ s}}$
16. $\frac{7}{50} \sqrt{29} \frac{\text{m}}{\text{s}}$
17. $1.95 \frac{\text{m}}{\text{s}}$
- .4342 $\frac{\text{m}}{\text{s}}$

EJERCICIO 46

1. a) $I = \$15\,275.00$, $U = \$8\,370.00$, $Q = \$47.90$
 b) $I = \$9\,424.00$, $U = \$7\,208.00$, $Q = \$26.93$
 $Q = \$16.00$ por artículo
2. Costo promedio mínimo = \$14.80 por artículo
 Se deben producir 1 225 artículos para un costo mínimo
3. Ingreso real: $I(31) - I(30) = \$156.00$
 Ingreso aproximado: \$160.00
4. 59 metros
5. $p(x) = 100 - \frac{1}{800}x$
 \$50.00 por boleto

EJERCICIO 47

1. $\frac{15}{2}$
 2. 2
 3. -1
 4. $\frac{1}{3} \ln 2$
 5. 1
 6. $-\frac{9}{4}$
 7. 1
8. $\frac{1}{3}$
 9. e
 10. 0
 11. 1
 12. $-\frac{1}{2}$
 13. $e^{-\frac{1}{2}}$
 14. 0
15. e
 16. ∞
 17. -1
 18. 1
 19. 1
 20. 0

EJERCICIO 48

1. $c = 0$
 2. $c = \frac{3}{4}$
 3. $c = \frac{5}{2}$
4. $c = \frac{3}{4}$
 5. $c = \pm \sqrt{3}$
 6. $c = 0.36$

$$7. c = \pm \sqrt{\frac{13}{3}}$$

$$8. c = 0$$

$$9. c = 0$$

$$10. c = \pi$$

$$11. \text{ No es continua en } x = 1$$

$$12. c = 1$$

EJERCICIO 49

1. $c = \frac{3}{2}$
 2. $c = \frac{1}{2}$
 3. $c = \sqrt{3}$
 4. $c = 0$
 5. $c = 3$
6. $c = -1$
 7. $c = 1.7613$
 8. $c = 2.1750$
 9. $c = 0.5413$
 10. $c = 1.3204$

EJERCICIO 50

1. $dy = adx$
 2. $dy = (2ax + b)dx$
 3. $df(x) = (3x^2 - 4x)dx$
 4. $ds = \left(\frac{1}{2\sqrt{t}} - \frac{1}{3\sqrt[3]{t^2}} \right) dt$
 5. $dh(t) = -36t(5 - 3t^2)^5 dt$
 6. $dy = -\frac{6x}{(x^2 - 2)^5}$
 7. $dy = -\frac{1}{x^2} \sqrt[3]{\left(\frac{x}{2x+3} \right)^2} dx$
 8. $dy = \frac{2(x^2 + 1)}{\sqrt{x^2 + 2}} dx$
 9. $df(x) = (7x + 5)(x - 1)^2(x + 3)^3 dx$
 10. $dh(s) = \frac{8}{(2s + 3)^2} ds$
 11. $dg(x) = \frac{-2x}{(x^2 - 1)^2} dx$
 12. $dy = \frac{x + 8}{2(x + 3)^{\frac{3}{2}}} dx$
 13. $dy = \frac{-2abx}{(ax^2 - b)\sqrt{a^2x^4 - b^2}} dx$
 14. $df(x) = (1 + 2 \sin 2x)dx$
 15. $df(t) = 6 \tan^2 2t \sec^2 2t dt$
 16. $dy = -2 \tan x (\sec x - \sec^2 x) dx$
 17. $dg(x) = \frac{-2 \cos x}{(1 + \sin x)^2} dx$

$$18. ds(t) = \frac{-t \operatorname{sen} t - 2 \cos t}{2t^2 \sqrt{\cos t}} dt$$

$$19. df(x) = \frac{\sec x}{\sec x + 1} dx = \frac{1}{1 + \cos x} dx$$

$$20. dy = \frac{2x}{x^2 + 5} \log e \, dx$$

$$21. dy = \frac{x}{x^2 - 3} dx$$

$$22. dy = \frac{3}{2x^2x + 32x - 4} dx$$

$$23. dy = \frac{3}{2} \sqrt{x} e^{\sqrt{x^3}} dx$$

$$24. dy = 2^{x^3+5} (3x^2 \ln 2) dx$$

$$25. dh(t) = -\frac{2}{(e^t - e^{-t})^2} dt$$

$$26. df(x) = x(\ln x^2 + 1) dx$$

$$27. df(x) = -\frac{2}{\sqrt{1-4x^2}} dx$$

$$28. dy = -\frac{2}{x^2 + 4} dx$$

$$29. dy = \frac{1}{2x\sqrt{x-1}} dx$$

$$30. dy = -\frac{3}{x\sqrt{9x^6-1}} dx$$

EJERCICIO 51

$$1. \approx \frac{167}{18} = 9.277$$

$$2. \approx \frac{89}{27} = 3.296$$

$$3. \approx \frac{17}{8} = 2.125$$

$$4. \approx \frac{45 + 2\sqrt{3}\pi}{90} = 0.620$$

$$5. \approx \frac{108 + \sqrt{3}\pi}{36} = 3.151$$

$$6. \approx \frac{76 + 30\sqrt{3} + 2\pi}{15} = 8.949$$

$$7. \approx \frac{5489}{4} = 1372.25$$

$$8. \approx \frac{180(\sqrt{3} - \sqrt{2}) - \pi(7 + \sqrt{2})}{360} = 0.8549$$

$$9. \approx \frac{45 - 3\pi}{720} = 0.054$$

$$10. \approx \frac{9 - \sqrt{3}\pi}{9\sqrt{3}} = 0.228$$

$$11. dA = 0.286 \text{ cm}^2$$

$$12. dA = 2.265 \text{ cm}^2, dV = 3.341 \text{ cm}^3$$

$$13. dV = 1.8\pi \text{ cm}^3$$

$$14. \text{Lado} = 8 \text{ cm}$$

$$15. \text{Error relativo} = \frac{dA}{A} = 0.00249,$$

$$\text{Error porcentual} = 0.249\%$$

$$\text{Error relativo} = \frac{dV}{V} = 0.00374,$$

$$\text{Error porcentual} = 0.374\%$$

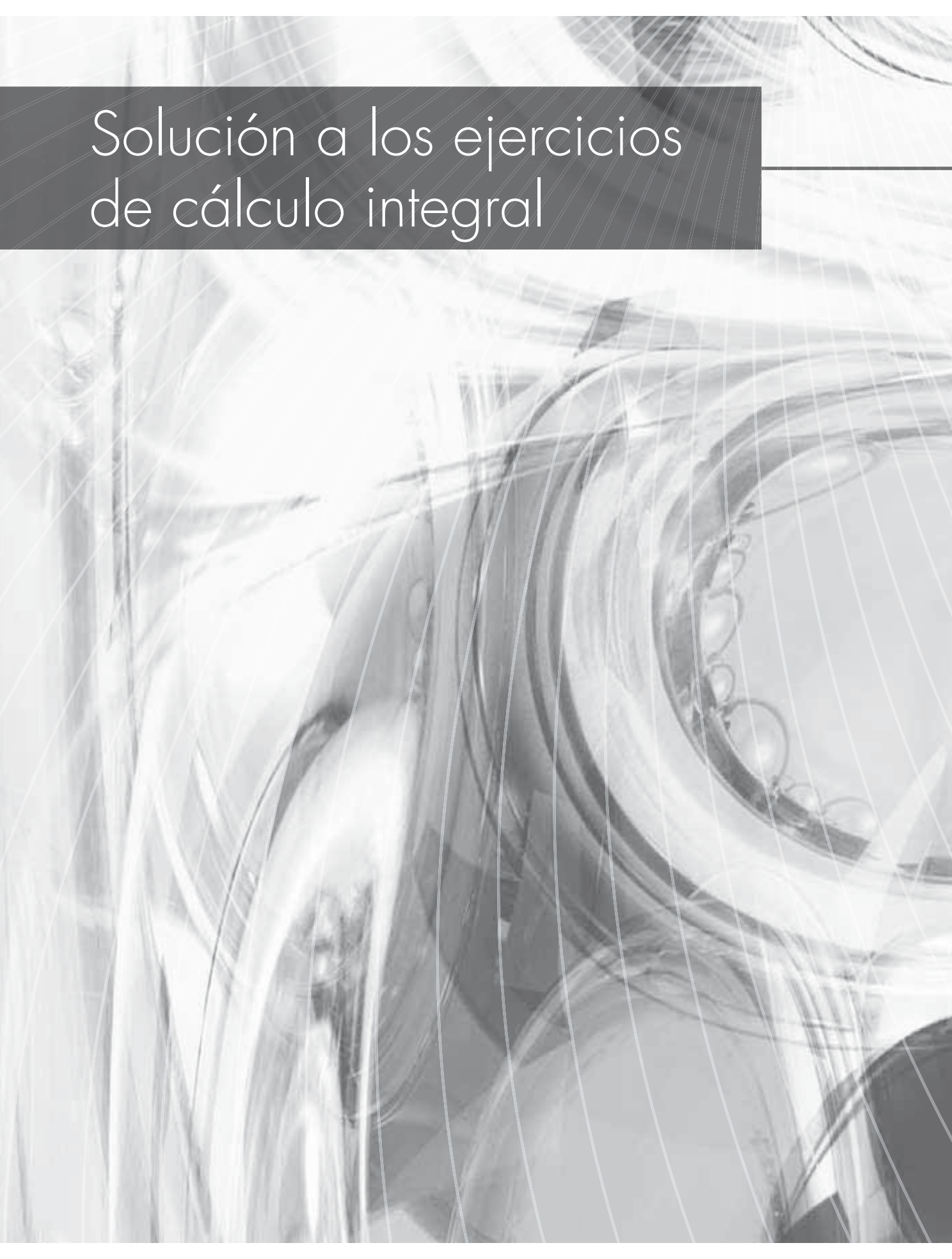
$$16. \text{Lado} = \frac{1}{9} \text{ cm}$$

$$17. d\phi = 0.02 \text{ cm}$$

$$18. \text{Error relativo} = \frac{dA}{A} = 0.00088,$$

$$\text{Error porcentual} = 0.088\%$$

Solución a los ejercicios de cálculo integral



CAPÍTULO 1

EJERCICIO 1

- | | | |
|---------------------|-------------------------------------|----------------------|
| 1. 354 | 5. 14 560 | 9. 63 |
| 2. -40 | 6. 17 | 10. $\frac{853}{70}$ |
| 3. $-\frac{5}{4}$ | 7. $-\frac{322}{3}$ | 11. 414 |
| 4. $\frac{223}{70}$ | 8. $3\left(1 + \frac{b}{2a}\right)$ | 12. 81 |

EJERCICIO 2

- | | | |
|----------------------|----------------------|---------------------|
| 1. $57u^2$ | 5. $\frac{10}{3}u^2$ | 9. $\frac{9}{4}u^2$ |
| 2. $5u^2$ | 6. $\frac{bh}{2}u^2$ | 10. $242u^2$ |
| 3. $\frac{32}{3}u^2$ | 7. $12u^2$ | |
| 4. $\frac{7}{2}u^2$ | 8. $2u^2$ | |

CAPÍTULO 2

EJERCICIO 3

- | | | |
|----------------------------------|---|--|
| 1. $\frac{x^7}{7} + C$ | 12. $4 \ln x + C$ | 23. $\frac{4x^6}{3} - x^5 - x^4 - 2x^3 - x^2 - 3x + C$ |
| 2. $x^5 + C$ | 13. $\frac{4\sqrt[4]{x^3}}{3} + C$ | 24. $\frac{ax^4}{4} - \frac{bx^3}{3} - \frac{cx^2}{2} + dx + C$ |
| 3. $\frac{bx^4}{4} + C$ | 14. $9\sqrt[3]{x^2} + C$ | 25. $\frac{x^3}{3\sqrt{a^2+b^2}} - \frac{3x^2}{2\sqrt{a}} - 5\sqrt{bx} + C$ |
| 4. $\frac{\sqrt{3}x^3}{3} + C$ | 15. $\frac{5x^5\sqrt[3]{x^3}}{8} + C$ | 26. $\frac{x^4}{4} - 2x^3 - 7x + C$ |
| 5. $ax + C$ | 16. $3a\sqrt[3]{x} + C$ | 27. $5x^{\frac{3}{5}} - \frac{5x^{\frac{4}{5}}}{2} + C$ |
| 6. $\frac{3}{4}x + C$ | 17. $\frac{5}{2}\ln x + C$ | 28. $6\sqrt[3]{x^2} - \frac{20\sqrt[4]{x^3}}{3} + C$ |
| 7. $\frac{1}{3}x + C$ | 18. $\frac{2x\sqrt{bx}}{3} + C$ | 29. $\frac{2y^{\frac{7}{2}}}{7} - \frac{15y^{\frac{7}{3}}}{7} - \frac{8y^{\frac{5}{4}}}{5} - \frac{2y^{\frac{3}{2}}}{3} + C$ |
| 8. $\frac{3x\sqrt[3]{x}}{4} + C$ | 19. $\frac{15\sqrt[3]{x^2}}{2} - 3x\sqrt[3]{x} + C$ | 30. $\frac{2y^{\frac{5}{2}}}{5} - \frac{3y^{\frac{2}{3}}}{2} + \frac{4}{3y^{\frac{3}{4}}} + C$ |
| 9. $4x\sqrt[4]{x} + C$ | 20. $-\frac{3}{4x^4} + \frac{2}{x} - 6 \ln x + C$ | 31. $\frac{3t^3\sqrt[3]{t}}{2} - \frac{9t^2\sqrt[3]{t}}{7} + \frac{3t\sqrt[3]{t}}{2} + C$ |
| 10. $-\frac{1}{2x^2} + C$ | 21. $\frac{3t\sqrt[3]{at}}{4} + C$ | 32. $\frac{3t\sqrt[3]{7t}}{4} + C$ |
| 11. $-\frac{5}{3x^3} + C$ | 22. $\frac{2t\sqrt{6t}}{3} + C$ | 33. $\frac{(3x+4)^7}{21} + C$ |
| | | 34. $\frac{(ax^2-b)^6}{12a} + C$ |
| | | 35. $\frac{(t^3-4)^3}{9} + C$ |
| | | 36. $-\frac{(a-by)^5}{5b} + C$ |
| | | 37. $\frac{t^5}{5} - 4t^3 + 36t + C$ |
| | | 38. $\frac{x^4}{4} + \frac{8x^3}{3} + 8x^2 + C$ |
| | | 39. $\frac{x^6}{6} + \frac{3x^5}{5} + \frac{3x^4}{4} + \frac{x^3}{3} + C$ |
| | | 40. $\frac{2(m+ny)\sqrt{m+ny}}{3n} + C$ |
| | | 41. $\frac{2(5x-3)^{\frac{3}{2}}}{15} + C$ |
| | | 42. $\frac{1}{a}\sqrt{at^2+b} + C$ |

43. $\frac{1}{6}\sqrt[3]{(9x-1)^2} + C$
44. $\frac{x^2}{2} - \frac{16x\sqrt{x}}{3} + 16x + C$
45. $-\frac{1}{18(3x^2-4)^3} + C$
46. $-\frac{5}{3(3x-4)} + C$
47. $-\frac{2}{3(2x^2+5)^3} + C$
48. $\frac{2(\sqrt{x}-b)^3}{3} + C$
49. $\frac{1}{a}\ln(at+b) + C$
50. $\frac{1}{6}\ln|3x^2-4| + C$
51. $\ln|x+3| + C$
52. $\ln|x^2-3| + C$
53. $-\frac{1}{(x^2-3x+6)} + C$
54. $\frac{2(x^3-6x+3)^{\frac{3}{2}}}{9} + C$
55. $-\frac{1}{an(m-1)(ay^n+b)^{m-1}} + C \quad \forall m \neq 1$
56. $-\frac{(1-e^{3x})^3}{9} + C$
57. $-\frac{(4-\ln|x+3|)^4}{4} + C$
58. $-\frac{(1-\sin 4x)^4}{16} + C$
59. $-\frac{2(3+\cot x)^{\frac{3}{2}}}{3} + C$
60. $-\sqrt{1-\sec 2x} + C$
61. $-\frac{1}{a}\ln|1-\sin ax| + C$
62. $\frac{4}{3}(e^{\sqrt{x}}-1)^{\frac{3}{2}} + C$
63. $\frac{(2+\ln|\sin x|)^2}{2} + C$
64. $-\frac{1}{2(1-\cos^2 x)^2} + C$
65. $\frac{1}{3b}\sin^3 bx + C$
66. $-\frac{\cot^2 mx}{2m} + C$
67. $-\frac{\cos^3 4x}{12} + C$
68. $\frac{2\sqrt{\sin 5x+4}}{5} + C$
69. $4x - \ln(x+2)^6 + C$
70. $\frac{3x^2}{2} + 3x + \ln|(x-1)^5| + C$
71. $-\frac{1}{\ln y} + C$
72. $\frac{1}{2}\ln|\ln 3x| + C$
73. $\frac{2(ax^{n+1}+b)^{\frac{3}{2}}}{3a(n+1)} + C$
74. $\frac{1}{3}\sqrt[3]{\left(1-\frac{1}{x^2}\right)^3} + C$
75. $-\frac{1}{3}\csc 3x + C$
76. $-\frac{1}{(x+1)^2} + \frac{3}{x+1} + 4\ln(x+1) + C$
77. $\ln\left|\frac{(x+2)^3}{(x+5)^4}\right| + C$
78. $\ln\left|(2x-1)^{\frac{3}{2}}(3x-4)^{\frac{5}{3}}\right| + C$
79. $-3\sqrt[3]{\cos x} + C$
80. $\frac{2}{5}\sin^5 x + C$
81. $2\sqrt{1-\cot w} + C$
82. $-\frac{3}{2}\sqrt{2\cos^2 y-1} + C$
83. $2\sqrt{1-\cos \alpha} + C$
84. $\frac{4}{7}\tan^{\frac{7}{4}} x + C$

EJERCICIO 4

1. $\frac{1}{4} e^{4x} + C$
2. $16 e^{\frac{x}{2}} + C$
3. $\frac{1}{a} e^{ax+b} + C$
4. $\frac{2}{3} e^{\sqrt{3x}} + C$
5. $\frac{1}{3} e^{3x} + C$
6. $-\frac{1}{4} e^{\cos 4x} + C$
7. $\frac{2}{3} e^{x^3} + C$
8. $\frac{b^{4x}}{4 \ln b} + C$
9. $\frac{3^{2x}}{2 \ln 3} + C$
10. $\frac{2^x e^x}{1 + \ln 2} + C$
11. $3 \sqrt[3]{e^x} + C$
12. $\frac{2}{3} \sqrt{e^{3x}} + C$
13. $-\frac{1}{5^{4x} \ln 625} + C$
14. $-\frac{1}{2} e^{\frac{1}{x^2}} + C$
15. $\frac{3}{4} \sqrt[3]{e^{4x}} + C$
16. $x^3 - \frac{1}{3} e^{x^3} + C$
17. $e^{x^2-3x+1} + C$
18. $-\frac{5}{2\sqrt[3]{e^{2t}}} + C$
19. $-\frac{1}{2} e^{\frac{1}{\sec 2x}} + C$
20. $-\frac{1}{8e^{2t}} + C$
21. $\frac{4^x \cdot e^{2x}}{2 + \ln 4} + C$
22. $2 \left(e^{\frac{x}{2}} - 2e^{-\frac{x}{4}} \right) + C$
23. $\frac{1}{6} e^{6x} - \frac{4}{3} e^{3x} + 4x + C$
24. $\frac{5^{x^2}}{\ln 25} + C$
25. $\frac{1}{4} (e^{4x} - e^{-4x}) - 2x + C$
26. $\frac{1}{3} e^{\tan 3x} + C$
27. $\frac{5^{x^3}}{3 \ln 5} + C$
28. $\frac{10^{3x}}{3 \ln 10} - \frac{2^x}{\ln 2} + C$
29. $n \left(e^{\frac{x}{n}} - \frac{1}{\ln a} a^{\frac{x}{n}} \right) + C$
30. $\frac{1}{2} (e^{2x} + 5e^{-2x}) + C$
31. $-\frac{1}{ae^{ax}} - x + C$
32. $-e^{\cos^2 x} + C$
33. $\frac{1}{2} e^{\arcsin 2x} + C$
34. $e^{\arctan x} + C$
35. $\frac{3^{4x}(3 \cdot 3^{4x} + 8 \cdot 3^{2x} + 6)}{24 \ln 3} + C$

EJERCICIO 5

1. $-\frac{1}{5} \cos 5x + C$
2. $\frac{1}{6} \sin 6x + C$
3. $-4 \cos \frac{x}{4} + C$
4. $\frac{1}{b} \ln |\sec bx| + C$
5. $a \tan \frac{x}{a} + C$
6. $-\frac{1}{a} \cot ax + C$
7. $\frac{1}{b} \tan bx + C$
8. $\sec x + C$
9. $-4 \csc \frac{t}{4} + C$
10. $-\frac{1}{8} \cos 4x^2 + C$
11. $\frac{5}{3} \sin \frac{x^3}{5} + C$
12. $-\frac{1}{2} \csc^2 x + C$
13. $\frac{1}{a} \sec ax + C$
14. $\frac{3}{8} \tan 4x^2 + C$
15. $-\frac{1}{3} \cot(3x-1) + C$
16. $\frac{1}{a} \ln |\sin(ax-b)| + C$
17. $\frac{1}{a} \ln |\sec ax + \tan ax| + C$
18. $\frac{1}{8} \ln |\csc 4x^2 - \cot 4x^2| + C$
19. $-2\sqrt{\csc x} + C$
20. $\frac{1}{b} (\tan b\theta - \cot b\theta) + C$
21. $\frac{2}{3} (\csc 3x - \cot 3x) - x + C$
22. $\frac{2}{5} (\tan 5x + \sec 5x) - x + C$
23. $\frac{1}{2} [\sin^2 x - 2 \cos x] + C$
24. $-\frac{1}{2} \sin(2-x^2) + C$
25. $\ln |\csc x - \cot x| + \cos x + C$
26. $\sin x - \cos x + C$
27. $\ln |\cos x - 1| + C$
28. $\cot \left(\frac{\pi}{3} - x \right) + C$
29. $-\cot w + C$
30. $\frac{1}{2} (2 \tan x - x) + C$
31. $-\frac{1}{3} \cot^3 x + C$
32. $\frac{2\sqrt{\sin x}(\cos^2 x + 4)}{5} + C$
33. $\frac{1}{4} \ln |1 - 4 \cot w| + C$
34. $\frac{-[x \cos x - 2(\sin x - 1)]}{\cos x} + C$
35. $-2 \ln \left| \cot \left(\frac{y}{2} \right) \right| + C$
36. $\ln \sqrt{\sec 2\alpha} + C$
37. $2\sqrt{\sec^2 \theta + 1} + C$
38. $-\frac{1}{2} \cos(e^{2x}) + C$
39. $-\frac{1}{2} \cos(\ln x^2) + C$
40. $\frac{2}{3} \ln |\sec \sqrt{x} + \tan \sqrt{x}| + C$

EJERCICIO 6

1. $\frac{1}{9} \arctan\left(\frac{x}{9}\right) + C$
2. $\frac{1}{b^2} \arctan\left(\frac{y}{b}\right) + C$
3. $\frac{1}{8} \ln\left|\frac{y-4}{y+4}\right| + C$
4. $\frac{1}{20} \ln\left|\frac{5+2x}{5-2x}\right| + C$
5. $\frac{\sqrt{2}}{16} \ln\left|\frac{\sqrt{2}x-4}{\sqrt{2}x+4}\right| + C$
6. $\frac{1}{72} \ln\left|\frac{x-4}{x+4}\right| + C$
7. $\frac{1}{3} \arcsen\frac{3x}{5} + C$
8. $\frac{1}{2} \ln|2x + \sqrt{4x^2 - 7}| + C$
9. $\frac{1}{3} \arcsen\frac{2x}{3} + C$
10. $\frac{\sqrt{2}}{2} \ln|x + \sqrt{x^2 - 4}| + C$
11. $\frac{4}{b^2 m} \arctan\left(\frac{b^2 x}{m}\right) + C$
12. $\frac{1}{2b^2} \ln\left|\frac{v^2 - b^2}{v^2 + b^2}\right| + C$
25. $\frac{e^{2x}}{4} \sqrt{16 - e^{4x}} + 4 \arcsen\left(\frac{e^{2x}}{4}\right) + C$
26. $\frac{x}{2} \sqrt{1 - 2x^2} + \frac{\sqrt{2}}{4} \arcsen\sqrt{2}x + C$
27. $\sqrt{5} \arcsen\left(\frac{\sqrt{10} m}{20}\right) + C$
28. $\left(\frac{2x+1}{4}\right) \sqrt{4x^2 + 4x + 1 - a^2} - \frac{a^2}{4} \ln(2x+1 + \sqrt{4x^2 + 4x + 1 - a^2}) + C$
29. $\left(\frac{\sqrt{7}x^m}{2m}\right) \sqrt{49x^{2m} + 4} + \left(\frac{2\sqrt{7}}{7m}\right) \ln(7x^m + \sqrt{49x^{2m} + 4}) + C$
30. $\frac{\sqrt{14}}{14} \arctan\left(\frac{\sqrt{14} t}{7}\right) + C$
13. $\frac{1}{2} \arctan\left(\frac{\ln x}{2}\right) + C$
14. $\frac{1}{2} \ln\left|\frac{1 + \sen x}{1 - \sen x}\right| + C$
15. $\frac{1}{2} \arcsen\left(\frac{x^2}{3}\right) + C$
16. $\frac{5\sqrt{3}}{3} \arcsen(x) + C$
17. $2 \ln(\sqrt{e^x} + \sqrt{e^x + 4}) + C$
18. $\frac{1}{2} \arcsen\left(\frac{2\sqrt{5}y}{5}\right) + C$
19. $\frac{1}{10a^2} \ln\left|\frac{5+ay}{5-ay}\right| + C$
20. $\frac{\sqrt{3}}{3} \ln|\sqrt{3}t + \sqrt{3t^2 + 5}| + C$
21. $\frac{\sqrt{10}}{20} \ln\left|\frac{\sqrt{10}y+5}{\sqrt{10}y-5}\right| + C$
22. $\frac{\sqrt{3}}{3} \ln|\sqrt{3}x + \sqrt{3x^2 + 4}| + C$
23. $\frac{1}{b^2} \arctan\frac{x}{b^2} + C$
24. $\frac{1}{\sqrt{2}} \arcsen\frac{\sqrt{2}y}{2} + C$

31. $\left(\frac{3}{5}\right) \sqrt{5x^2 - 16} + \left(\frac{2\sqrt{5}}{5}\right) \ln(\sqrt{5}x + \sqrt{5x^2 - 16}) + C$
32. $-\frac{\sqrt{5}}{20} \ln\left(\frac{\sqrt{5} + \cos 2t}{\sqrt{5} - \cos 2t}\right) + C$
33. $-\arcsen(\cos x) + C$
34. $\frac{t \ln(3t)}{2} \sqrt{t^2 \ln^2(3t) + 4} + 2 \ln(t \ln(3t) + \sqrt{t^2 \ln^2(3t) + 4}) + C$

EJERCICIO 7

1. $\frac{1}{6} \ln\left|\frac{x}{x+6}\right| + C$
2. $\frac{1}{8} \ln\left|\frac{x}{x+8}\right| + C$
3. $\ln\left|\frac{x+2}{x+3}\right| + C$
4. $\ln\left|\frac{2x+1}{x+1}\right| + C$
5. $\frac{1}{9} \ln\left|\frac{x-2}{x+7}\right| + C$
6. $\frac{1}{7} \ln\left|\frac{2x+1}{x+4}\right| + C$
7. $\frac{1}{2a} \ln\left|\frac{ax+3}{ax+5}\right| + C$
15. $\frac{1}{2} \ln|2x - 1 + \sqrt{4x^2 - 4x + 3}| + C$
16. $\frac{\sqrt{3}}{3} \ln|3z + 2 + \sqrt{9z^2 + 12z}| + C$
17. $\frac{\sqrt{2}}{2} \ln|4x + 1 + 2\sqrt{4x^2 + 2x}| + C$
18. $\ln|2 \ln x + 7 + 2\sqrt{\ln^2 x + 7 \ln x + 6}| + C$
19. $\ln|2w - 9 + 2\sqrt{w^2 - 9w + 5}| + C$
20. $\frac{x+2}{2} \sqrt{x^2 + 4x - 3} - \frac{7}{2} \ln|x + 2 + \sqrt{x^2 + 4x - 3}| + C$
21. $\frac{41\sqrt{2}}{32} \arcsen\left(\frac{\sqrt{41}(4x+3)}{41}\right) + \left(\frac{4x+3}{8}\right) \sqrt{4 - 3x - 2x^2} + C$
8. $3 \ln\left|\frac{e^x + 4}{e^x + 5}\right| + C$
9. $\frac{1}{7} \ln\left|\frac{2w-3}{w-5}\right| + C$
10. $\frac{1}{11} \ln\left|\frac{2\alpha-1}{\alpha-5}\right| + C$
11. $\frac{\sqrt{7}}{7} \arctan\left(\frac{\sqrt{7}(x-1)}{7}\right) + C$
12. $x + \frac{1}{4} \ln\left|\frac{e^x - 1}{e^x + 3}\right| + C$
13. $\frac{\sqrt{3}}{6} \ln\left|\frac{\sen x - 3 - \sqrt{3}}{\sen x - 3 + \sqrt{3}}\right| + C$
14. $\frac{\sqrt{5}}{5} \arcsen\left(\frac{5w-11}{9}\right) + C$

$$22. \left(\frac{2x-3}{4} \right) \sqrt{3x-x^2} + \frac{9}{8} \arcsen \left(\frac{2x-3}{3} \right) + C$$

$$23. \frac{(3x-2)\sqrt{3x^2-4x}}{6} - \frac{2\sqrt{3}}{9} \ln |3x-2+\sqrt{9x^2-12x}| + C$$

$$24. \left(\frac{2x^2-1}{8} \right) \sqrt{x^4-x^2-20} - \frac{81}{16} \ln |2x^2-1+2\sqrt{x^4-x^2-20}| + C$$

$$25. \left(\frac{2x+5}{4} \right) \sqrt{24-5x-x^2} + \frac{121}{8} \arcsen \left(\frac{2x+5}{11} \right) + C$$

$$26. \left(\frac{2x+3}{8} \right) \sqrt{x^2+3x+8} + \frac{23}{16} \ln |2x+3+2\sqrt{x^2+3x+8}| + C$$

$$27. 2 \ln |\sqrt{x-2} + \sqrt{x-4\sqrt{x}-21}| + C$$

$$28. \frac{1}{n} \left[\left(\frac{e^{nx}-1}{2} \right) \sqrt{3+2e^{nx}-e^{2nx}} + 2 \arcsen \left(\frac{e^{nx}-1}{2} \right) \right] + C$$

$$29. \ln |2y+1+2\sqrt{y^2+y+1}| + C$$

$$30. \left(\frac{3x+2}{6} \right) \sqrt{3x^2+4x+1} - \frac{\sqrt{3}}{18} \ln |\sqrt{3(3x^2+4x+1)}+3x+2| + C$$

$$31. \frac{\sqrt{2}}{2} \arcsen \left(\frac{4w-5}{5} \right) + C$$

$$32. \frac{2\sqrt{a}}{a} \ln |2\sqrt{ax}+3+2\sqrt{ax+3\sqrt{ax}+2}| + C$$

$$33. \frac{\sqrt{3}}{3} \ln |\sqrt{3}(6y+13)+6\sqrt{3y^2+13y-10}| + C$$

$$34. \frac{1}{2} \ln \left| \frac{(x+1)^{11}}{(3x+1)^7} \right| + C$$

$$35. -\frac{1}{6} \ln |(3-x)^5(3+x)^{13}| + C$$

$$36. -\frac{1}{9} \ln |(3x+4)^5(3x-4)^2| + C$$

$$37. \ln \left| \frac{e^x(x-\sqrt{3}-2)^{\frac{6\sqrt{3}+7}{2}}}{(x+\sqrt{3}-2)^{\frac{6\sqrt{3}-7}{2}}} \right| + C$$

$$38. \frac{\sqrt{73}}{438} \ln |(6x+5+\sqrt{73})^{\sqrt{73}+17} (6x+5-\sqrt{73})^{\sqrt{73}-17}| + C$$

$$39. \frac{1}{5} \ln \left| \frac{(x-6)^{11}}{(x-1)^6} \right| + C$$

$$40. \frac{1}{3} \ln |x^2+9x-5| + \frac{4\sqrt{101}}{101} \ln \left| \frac{2x+9-\sqrt{101}}{2x+9+\sqrt{101}} \right| + C$$

$$41. 3\sqrt{x^2-4} + 2 \ln |x+\sqrt{x^2-4}| + C$$

$$42. -\frac{11}{3} \arcsen \frac{3x}{2} - \frac{1}{3} \sqrt{4-9x^2} + C$$

$$43. \frac{5}{4} \ln |4x+\sqrt{16x^2+25}| - \frac{1}{8} \sqrt{16x^2+25} + C$$

$$44. \frac{3}{2} \sqrt{7-2x^2} + 2\sqrt{2} \arcsen \left(\frac{\sqrt{14}x}{7} \right) + C$$

$$45. \frac{\sqrt{129}}{2580} \ln \left| \frac{(10x+\sqrt{129}-7)^{67-\sqrt{129}}}{(10x-\sqrt{129}-7)^{67+\sqrt{129}}} \right| + C$$

$$46. 5\sqrt{x^2+3x-5} - \frac{37}{2} \ln |2x+3+2\sqrt{x^2+3x-5}| + C$$

$$47. \frac{13\sqrt{2}}{8} \ln \left| \frac{4x+5+\sqrt{8(2x^2+5x-1)}}{e^{\frac{2\sqrt{4x^2+10x-2}}{13}}} \right| + C$$

$$48. -5\sqrt{4-2x-x^2} - 4 \arcsen \frac{\sqrt{5}(x+1)}{5} + C$$

$$49. \frac{7}{2} \ln \left| \frac{e^{\frac{(4x^2-2)\sqrt{x^2-3x+4}}{21}}}{(2x-3+\sqrt{x^2-3x+4})^{-1}} \right| + C$$

$$50. \frac{175}{16} \ln \left| \frac{e^{\frac{(16x^2+84x-2)\sqrt{x^2+7x+6}}{175}}}{(2x+7+2\sqrt{x^2+7x+6})^{-1}} \right| + C$$

CAPÍTULO 3

EJERCICIO 8

$$1. \frac{1}{16} \sen^4 4x + C$$

$$2. -\frac{1}{18} \cos^6 3x + C$$

$$3. \frac{1}{3a} \cos^3 ax - \frac{1}{a} \cos ax + C$$

$$4. \frac{1}{15} \cos^3 5x - \frac{1}{5} \cos 5x + C$$

$$5. \frac{4}{3} \cos^3 \frac{x}{4} - 4 \cos \frac{x}{4} + C$$

$$6. \sen x - \frac{\sen^3 x}{3} + C$$

$$7. \frac{1}{a} \sen ax - \frac{\sen^3 ax}{3a} + C$$

8. $\frac{1}{6} \sin 6x - \frac{\sin^3 6x}{18} + C$
9. $3 \sin \frac{x}{3} - \sin^3 \frac{x}{3} + C$
10. $-\cos x + \frac{2}{3} \cos^3 x - \frac{1}{5} \cos^5 x + C$
11. $-\frac{1}{a} \cos ax + \frac{2}{3a} \cos^3 ax - \frac{1}{5a} \cos^5 ax + C$
12. $-\frac{1}{4} \cos 4x + \frac{1}{6} \cos^3 4x - \frac{1}{20} \cos^5 4x + C$
13. $-2 \cos \frac{x}{2} + \frac{4}{3} \cos^3 \frac{x}{2} - \frac{2}{5} \cos^5 \frac{x}{2} + C$
14. $\sin y - \frac{2}{3} \sin^3 y + \frac{1}{5} \sin^5 y + C$
15. $\frac{1}{b} \sin bx - \frac{2}{3b} \sin^3 bx + \frac{1}{5b} \sin^5 bx + C$
16. $3 \sin \frac{x}{3} - 2 \sin^3 \frac{x}{3} + \frac{3}{5} \sin^5 \frac{x}{3} + C$
17. $\frac{1}{7} \cos^7 \theta - \frac{3}{5} \cos^5 \theta + \cos^3 \theta - \cos \theta + C$
18. $\frac{1}{21} \cos^7 3x - \frac{1}{5} \cos^5 3x + \frac{1}{3} \cos^3 3x - \frac{1}{3} \cos 3x + C$
19. $-\frac{1}{7} \sin^7 y + \frac{3}{5} \sin^5 y - \sin^3 y + \sin y + C$
20. $-\frac{1}{28} \sin^7 4x + \frac{3}{20} \sin^5 4x - \frac{1}{4} \sin^3 4x + \frac{1}{4} \sin 4x + C$
21. $-\frac{1}{24} \cos^6 4x + \frac{1}{32} \cos^8 4x + C$
22. $\frac{1}{6} \sin^6 x - \frac{1}{8} \sin^8 x + C$
23. $\sqrt{\sin 2x} \left(1 - \frac{2 \sin^2 2x}{5} + \frac{\sin^4 2x}{9} \right) + C$

EJERCICIO 9

1. $\frac{1}{10} \tan^2 5x + \frac{1}{5} \ln |\cos 5x| + C$
2. $\tan^2 \frac{x}{2} + 2 \ln \left| \cos \frac{x}{2} \right| + C$
3. $-\frac{1}{8} \cot^2 4x - \frac{1}{4} \ln |\sin 4x| + C$
4. $-\frac{3}{2} \cot^2 \frac{x}{3} - 3 \ln \left| \sin \frac{x}{3} \right| + C$

5. $-\frac{1}{6} \left(\frac{\cot^4 6x}{4} - \frac{\cot^2 6x}{2} - \ln |\sin 6x| \right) + C$
6. $-4 \left(\frac{1}{4} \cot^4 \frac{y}{4} - \frac{1}{2} \cot^2 \frac{y}{4} - \ln \left| \sin \frac{y}{4} \right| \right) + C$
7. $\frac{1}{20} \tan^4 5x - \frac{1}{10} \tan^2 5x + \frac{1}{5} \ln |\sec 5x| + C$
8. $-\frac{1}{15} \cot^3 5x + \frac{1}{5} \cot 5x + x + C$
9. $\frac{1}{18} \tan^3 6x - \frac{1}{6} \tan 6x + x + C$
10. $\frac{1}{6} \tan^2 3x + \frac{1}{6} \cot^2 3x + \frac{4}{3} \ln |\sin 3x| + \frac{4}{3} \ln |\cos 3x| + C =$
 $\frac{1}{6} \tan^2 3x + \frac{1}{6} \cot^2 3x + \frac{4}{3} \ln |\sin 6x| + C$
11. $\frac{1}{6} \tan^3 2y + C$
12. $-\frac{1}{9} \cot^3 3y + C$

EJERCICIO 10

1. $\frac{1}{3} \tan 3x + \frac{1}{9} \tan^3 3x + C$
2. $\frac{1}{a} \tan ax + \frac{1}{3a} \tan^3 ax + C$
3. $6 \tan \frac{x}{6} + 2 \tan^3 \frac{x}{6} + C$
4. $-\frac{1}{9} \cot 9x - \frac{1}{27} \cot^3 9x + C$
5. $-\frac{1}{b} \cot bx - \frac{1}{3b} \cot^3 bx + C$
6. $-7 \cot \frac{x}{7} - \frac{7}{3} \cot^3 \frac{x}{7} + C$
7. $\frac{3}{2} \tan \frac{2x}{3} + \frac{1}{2} \tan^3 \frac{2x}{3} + C$
8. $-\frac{4}{5} \cot \frac{5x}{4} - \frac{4}{15} \cot^3 \frac{5x}{4} + C$
9. $\frac{1}{24} \tan^3 8x + \frac{1}{40} \tan^5 8x + C$
10. $\frac{1}{3a} \tan^3 ax + \frac{1}{5a} \tan^5 ax + C$
11. $\frac{7}{3} \tan^3 \frac{x}{7} + \frac{7}{5} \tan^5 \frac{x}{7} + C$
12. $\frac{1}{5} \tan^3 \frac{5x}{3} + \frac{3}{25} \tan^5 \frac{5x}{3} + C$

$$13. \frac{1}{25} \sec^5 5x - \frac{1}{15} \sec^3 5x + C$$

$$14. \frac{1}{5b} \sec^5 bx - \frac{1}{3b} \sec^3 bx + C$$

$$15. \frac{6}{5} \sec^5 \frac{x}{6} - 2 \sec^3 \frac{x}{6} + C$$

$$16. \frac{7}{20} \sec^5 \frac{4x}{7} - \frac{7}{12} \sec^3 \frac{4x}{7} + C$$

$$17. -\frac{1}{5b} \csc^5 bx + \frac{1}{3b} \csc^3 bx + C$$

$$18. -\frac{1}{20} \csc^5 4x + \frac{1}{12} \csc^3 4x + C$$

$$19. 2 \tan \frac{x}{2} \left(1 + \frac{2}{3} \tan^2 \frac{x}{2} + \frac{1}{5} \tan^4 \frac{x}{2} \right) + C$$

$$20. -\frac{2}{3} \left[\cot \left(\frac{3\theta}{2} \right) + \frac{1}{3} \cot^3 \left(\frac{3\theta}{2} \right) \right] + C$$

$$21. \frac{2}{3} \left(\tan x^3 + \frac{1}{3} \tan^3 x^3 \right) + C$$

$$22. \tan x \left(1 + \frac{2}{3} \tan^2 x + \frac{1}{5} \tan^4 x \right) + C$$

$$23. \tan \alpha - \frac{1}{5} \tan^5 \alpha + C$$

$$24. \frac{1}{2} \tan 2t + \frac{1}{6} \tan^3 2t + C$$

$$25. -\frac{1}{3} \cot(3x-1) - \frac{1}{9} \cot^3(3x-1) + C$$

$$26. \frac{1}{5} \sec^5 x - \frac{2}{3} \sec^3 x + \sec x + C$$

$$27. \frac{1}{6} \sec^3 2x + C$$

$$28. -\frac{1}{7} \csc^7 x + \frac{2}{5} \csc^5 x - \frac{1}{3} \csc^3 x + C$$

$$29. \frac{1}{21} \sec^7 3x - \frac{2}{15} \sec^5 3x + \frac{1}{9} \sec^3 3x + C$$

$$30. 2\sqrt{\tan x} \left(\frac{1}{9} \tan^4 x + \frac{2}{5} \tan^2 x + 1 \right) + C$$

$$31. \frac{1}{3} \tan 3t + 2 \cot \left(\frac{t}{2} \right) + \frac{1}{9} \tan^3 3t + \frac{2}{3} \cot^3 \left(\frac{t}{2} \right) + C$$

$$32. -\frac{1}{2} \cot(2x-1) - \frac{1}{6} \cot^3(2x-1) + C$$

$$33. -\cot \left(\frac{\theta}{5} \right) \cdot \left[5 + \frac{10}{3} \cot^2 \left(\frac{\theta}{5} \right) + \cot^4 \left(\frac{\theta}{5} \right) \right] + C$$

$$34. -\cot x \left(\frac{1}{7} \cot^6 x + \frac{3}{5} \cot^4 x + \cot^2 x + 1 \right) + C$$

$$35. \frac{1}{2} \tan x^2 - \frac{1}{6} \tan^3 x^2 + C$$

EJERCICIO 11

$$1. \frac{1}{2} x - \frac{1}{12} \sin 6x + C$$

$$2. \frac{1}{2} x - \frac{1}{4a} \sin 2ax + C$$

$$3. \frac{1}{2} x - \frac{5}{4} \sin \frac{2}{5} x + C$$

$$4. \frac{1}{2} x - \frac{1}{3} \sin \frac{3}{2} x + C$$

$$5. \frac{1}{2} x + \frac{1}{20} \sin 10x + C$$

$$6. \frac{1}{2} x + \frac{1}{4b} \sin 2bx + C$$

$$7. \frac{1}{2} x + \frac{7}{4} \sin \frac{2}{7} x + C$$

$$8. \frac{1}{2} x + \frac{1}{14} \sin 7x + C$$

$$9. \frac{3}{8} x - \frac{1}{32} \sin 16x + \frac{1}{256} \sin 32x + C$$

$$10. \frac{3}{8} x - \frac{1}{4a} \sin 2ax + \frac{1}{32a} \sin 4ax + C$$

$$11. \frac{3}{8} x - \frac{7}{4} \sin \frac{2}{7} x + \frac{7}{32} \sin \frac{4}{7} x + C$$

$$12. \frac{3}{8} x - \frac{1}{3} \sin \frac{3}{2} x + \frac{1}{24} \sin 3x + C$$

$$13. \frac{3}{8} x + \frac{1}{36} \sin 18x + \frac{1}{288} \sin 36x + C$$

$$14. \frac{3}{8} x + \frac{1}{4b} \sin 2bx + \frac{1}{32b} \sin 4bx + C$$

$$15. \frac{3}{8} x + \frac{3}{4} \sin \frac{2}{3} x + \frac{3}{32} \sin \frac{4}{3} x + C$$

$$16. \frac{3}{8} x + \frac{3}{20} \sin \frac{10}{3} x + \frac{3}{160} \sin \frac{20}{3} x + C$$

$$17. \frac{5}{16} x - \frac{1}{4} \sin 2x + \frac{3}{64} \sin 4x + \frac{1}{48} \sin^3 2x + C$$

$$18. \frac{5}{16} x - \frac{1}{16} \sin 8x + \frac{3}{256} \sin 16x + \frac{1}{192} \sin^3 8x + C$$

$$19. \frac{5}{16}x - \frac{1}{4a}\sin 2ax + \frac{3}{64a}\sin 4ax + \frac{1}{48a}\sin^3 2ax + C$$

$$20. \frac{5}{16}x - \sin \frac{1}{2}x + \frac{3}{16}\sin x + \frac{1}{12}\sin^3 \frac{1}{2}x + C$$

$$21. \frac{5}{16}x - \frac{1}{10}\sin 5x + \frac{3}{160}\sin 10x + \frac{1}{120}\sin^3 5x + C$$

$$22. \frac{5}{16}x + \frac{1}{4}\sin 2x + \frac{3}{64}\sin 4x - \frac{1}{48}\sin^3 2x + C$$

$$23. \frac{5}{16}x + \frac{1}{12}\sin 6x + \frac{1}{64}\sin 12x - \frac{1}{144}\sin^3 6x + C$$

$$24. \frac{5}{16}x + \frac{1}{4b}\sin 2bx + \frac{3}{64b}\sin 4bx - \frac{1}{48b}\sin^3 2bx + C$$

$$25. \frac{5}{16}x + \frac{1}{2}\sin x + \frac{3}{32}\sin 2x - \frac{1}{24}\sin^3 x + C$$

$$26. \frac{5}{16}x + \frac{5}{8}\sin \frac{4}{5}x + \frac{15}{128}\sin \frac{8}{5}x - \frac{5}{96}\sin^3 \frac{4}{5}x + C$$

$$27. \frac{5}{16}x + \frac{1}{4}\sin 2x + \frac{3}{64}\sin 4x - \frac{1}{48}\sin^3 2x + C$$

$$28. \frac{3}{8}x - \frac{1}{12}\sin 6x + \frac{1}{96}\sin 12x + C$$

$$29. \frac{1}{2}y - \frac{1}{4}\sin 2y + C = \frac{3}{8}y - \frac{1}{2}\sin y + \frac{1}{16}\sin 2y + C$$

$$30. \frac{1}{2}(x - \sin x \cos x) + C$$

$$31. \frac{3}{8}x + \frac{1}{12}\sin 6x + \frac{1}{96}\sin 12x + C$$

$$32. x + \sin x + C$$

$$33. \frac{19}{2}\alpha - 6\sin \alpha + \frac{1}{4}\sin 2\alpha + C$$

$$34. \frac{5}{2}x - \frac{3}{4}\sin 2x - 4\cos x + \frac{1}{3}\cos^3 x + C$$

$$35. \frac{x}{8} - \frac{b}{32}\sin\left(\frac{4x}{b}\right) + C$$

$$36. \frac{\theta}{2} - \frac{3}{4}\sin\left(\frac{2\theta}{3}\right) + 4\cos^2\left(\frac{\theta}{3}\right) + 3\sin\left(\frac{\theta}{3}\right) + C$$

$$37. \frac{35}{128}x + \frac{1}{4}\sin 2x + \frac{7}{128}\sin 4x - \frac{1}{24}\sin^3 2x + \frac{1}{1024}\sin 8x + C$$

EJERCICIO 12

$$1. \frac{1}{10}[5\sin x - \sin 5x] + C = \frac{\sin x}{2} - \frac{\sin 5x}{10} + C$$

$$2. -\frac{1}{8}[\cos 4x - 2\cos 2x] + C = -\frac{\cos 4x}{8} + \frac{\cos 2x}{4} + C$$

$$3. \frac{1}{8}\sin 4x - \frac{1}{12}\sin 6x + C$$

$$4. \frac{1}{20}\sin 10y + \frac{1}{8}\sin 4y + C$$

$$5. -\frac{1}{14}\cos 7x + \frac{1}{6}\cos 3x + C$$

$$6. -\frac{3}{7}\cos\left(\frac{7\alpha}{6}\right) - 3\cos\left(\frac{\alpha}{6}\right) + C$$

$$7. \frac{10}{17}\sin\left(\frac{17}{20}w\right) + \frac{10}{7}\sin\left(\frac{7}{20}w\right) + C$$

$$8. \frac{x \cos 2b}{2} - \frac{\sin 2mx}{4m} + C$$

$$9. \frac{x \cos 8}{2} - \frac{\sin 6x}{12} + C$$

$$10. \frac{1}{24}\cos 6w - \frac{1}{16}\cos 4w - \frac{1}{8}\cos 2w + C$$

CAPÍTULO 4**EJERCICIO 13**

$$1. \frac{1}{6}\ln\left|\frac{\sqrt{x^2+36}-6}{x}\right| + C$$

$$2. \frac{w}{5\sqrt{w^2+5}} + C$$

$$3. -\frac{y}{\sqrt{y^2+3}} + \ln\left(\sqrt{y^2+3}+y\right) + C$$

$$4. 8\arcsin\left(\frac{x}{4}\right) - \frac{x\sqrt{16-x^2}}{2} + C$$

$$5. -\frac{\sqrt{y^2+25}}{25y} + C$$

$$6. -\frac{\sqrt{(36-25x^2)^5}}{180x^5} + C$$

$$7. 6\arcsin\left(\frac{\alpha-2}{2}\right) - \frac{(\alpha+6)\sqrt{4\alpha-\alpha^2}}{2} + C$$

$$8. \frac{1}{4}\ln\left|\frac{\sqrt{25x^2+16}-4}{5x}\right| + C$$

$$9. \frac{(x^2+32)\sqrt{x^2-16}}{3} + C$$

$$10. -\frac{\sqrt{5-\theta^2}}{\theta} - \arcsin\left(\frac{\sqrt{5}\theta}{5}\right) + C$$

$$11. \sqrt{x^2+16} + 4\ln\left|\frac{\sqrt{x^2+16}-4}{x}\right| + C$$